

ALGEBRA 2

Unit 7

The collage shows several worksheets with the following content:

- Worksheet 1 (Top Left):** Titled "EXPONENTIAL Parent Function" and "ASYMPTOTE". It includes a graph of an exponential function and a grid. The "Notes/Examples" section lists properties of exponential functions based on the value of b .
- Worksheet 2 (Top Middle):** Titled "Solving EXPONENTIAL EQUATIONS". It includes a "SET 1: WITH A COMMON BASE" section with the equation $1. 2^{x+1} = 2^9$.
- Worksheet 3 (Top Right):** Titled "What is a LOGARITHM?". It includes a definition of a logarithm, the formula $\log_b a = x$, and a direction to write equations in exponential form. Examples include $1. \log_9 9 = 2$ and $2. \log_6 216 = 3$.
- Worksheet 4 (Bottom Left):** Titled "PRODUCT Property" and "QUOTIENT Property". It includes the formulas $\log_b(m \cdot n) = \log_b m + \log_b n$ and $\log_b\left(\frac{m}{n}\right) = \log_b m - \log_b n$.
- Worksheet 5 (Bottom Middle):** Titled "EXPONENTIAL Growth". It includes a definition of exponential growth and a formula for the value of an investment over time.
- Worksheet 6 (Bottom Right):** Titled "Solving LOGARITHMIC EQUATIONS". It includes a "Type 1: LOG = LOG" section with steps: 1. CONDENSE each logarithm, 2. Use the One-to-One Property, and 3. SOLVE and CHECK FOR EXTRANEOUS SOLUTIONS. Examples include $1. \log_5(5x+9) = \log_5(6x)$ and $2. \log_2(1-4n) = \log_2(2n+43)$.

EXPONENTIAL & LOGARITHMIC FUNCTIONS

NOTES • HOMEWORK • QUIZZES • TEST

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Gina



Unit 7 - Exponential & Logarithmic Functions: Sample Unit Outline

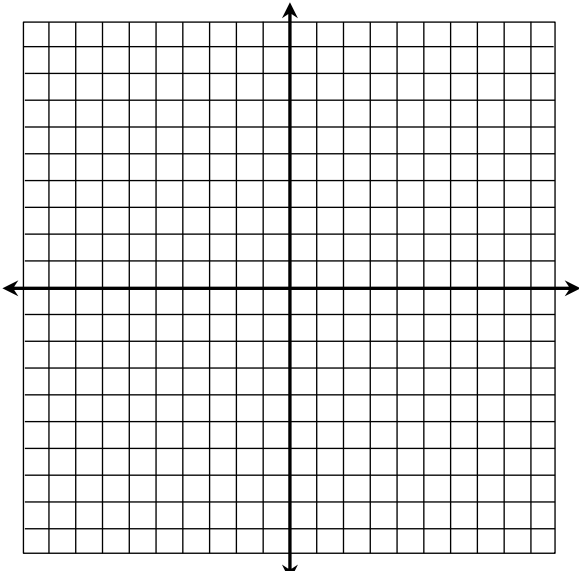
	TOPIC	HOMEWORK
DAY 1	Graphing Exponential Functions	HW #1
DAY 2	Solving Exponential Equations (using common bases)	HW #2
DAY 3	Quiz 7-1	None
DAY 4	Intro to Logarithms: Converting Exp/Log Forms, Base 10 Logs, Evaluating Simple Logs, Change of Base Formula	HW #3
DAY 5	Properties of Logarithms; Condensing & Expanding Logs	HW #4
DAY 6	Graphing Logarithmic Functions	HW #5
DAY 7	Quiz 7-2	None
DAY 8	Solving Logarithmic Equations	HW #6
DAY 9	Solving Exponential Equations using Logs	HW #7
DAY 10	Review: Solving Exponential/Logarithmic Equations	HW #8
DAY 11	Quiz 7-3	None
DAY 12	Applications of Exponential Functions: Exponential Growth & Decay/Compound Interest	HW #9
DAY 13	Base e & Natural Logarithms	HW #10
DAY 14	Modeling with Exponential/Logarithmic Functions; Choosing the Best Model (Lin, Quad, Cub, Quart, Exp, Ln)	HW #11
DAY 15	Unit 7 Review	Study for Test
DAY 16	UNIT 7 TEST	None

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
EXPONENTIAL Parent Function	- If $b > 1$, the function is an _____ and is _____. - If $b < 1$, the function is an _____ and is _____.
ASYMPTOTE	

Directions: Classify as an exponential growth or decay, graph, then identify its key characteristics.

1. $f(x) = 2^x$



Domain: _____

Range: _____

End Behavior:

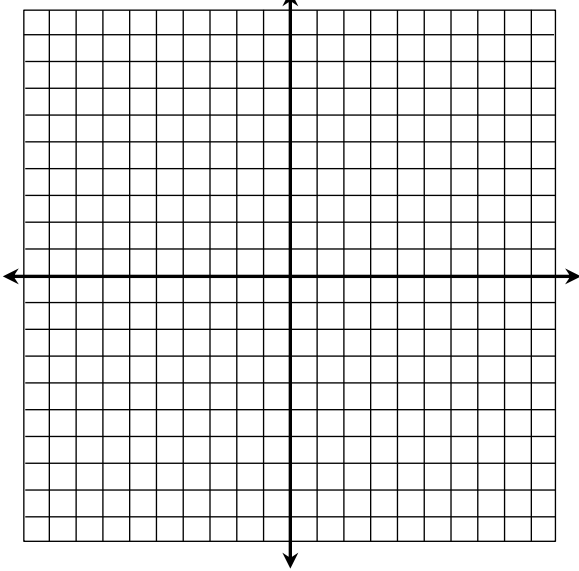
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

2. $f(x) = 3^x$



Domain: _____

Range: _____

End Behavior:

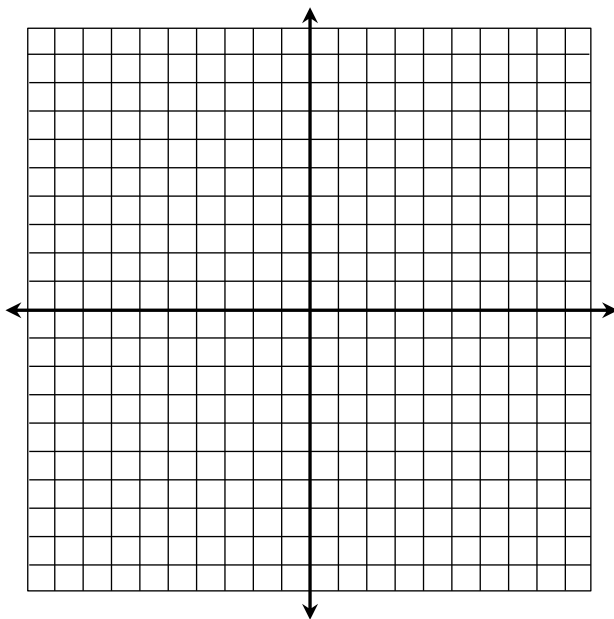
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

3. $f(x) = \left(\frac{1}{2}\right)^x$



Domain: _____

Range: _____

End Behavior:

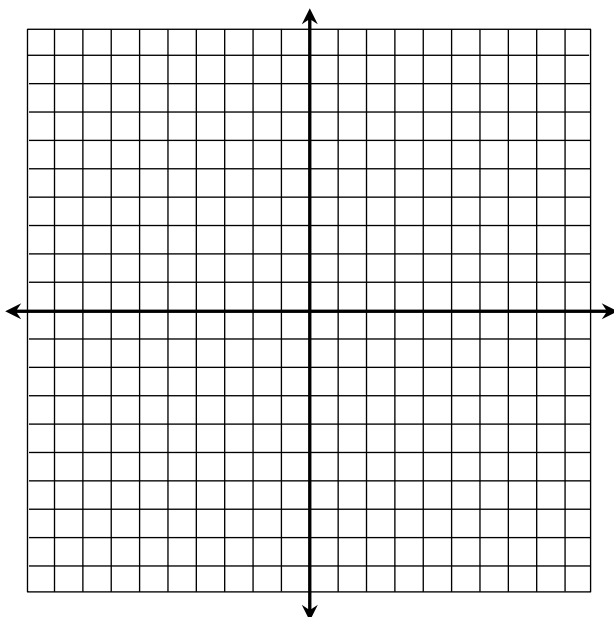
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

4. $f(x) = \left(\frac{2}{3}\right)^x$



Domain: _____

Range: _____

End Behavior:

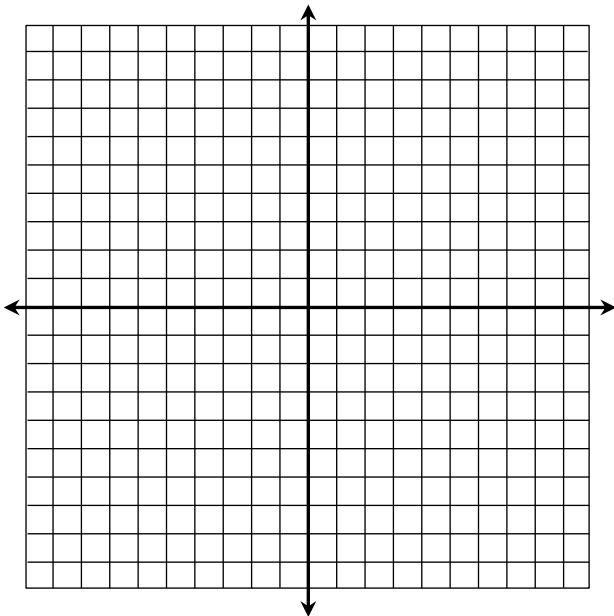
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

5. $f(x) = \left(\frac{5}{2}\right)^x$



Domain: _____

Range: _____

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

TRANSFORMATIONS

of Exponential
Functions

$$f(x) = a \cdot b^{x-h} + k$$

- h is the _____ shift. (+ shifts _____, - shifts _____)
- k is the _____ shift. (+ shifts _____, - shifts _____)
- If a is negative, the function is _____ across the ____ - _____
- $|a| > 1$ represents a vertical _____.
- $0 < |a| < 1$ represents a vertical _____.

Directions: (a) Identify the parent function, and (b) describe the transformations.

1. $f(x) = 3^x + 5$

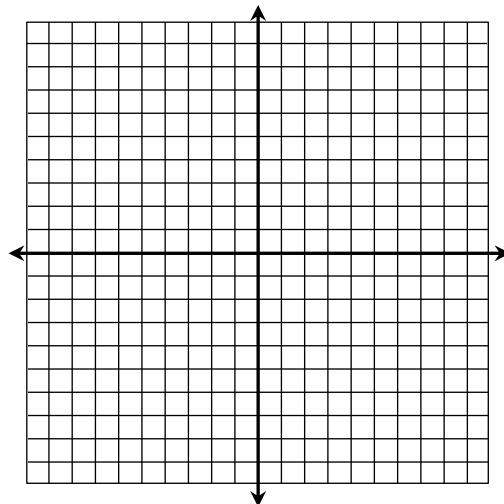
2. $f(x) = 2 \cdot \left(\frac{1}{4}\right)^{x-1}$

3. $f(x) = -\left(\frac{4}{3}\right)^{x+2} + 7$

4. $f(x) = \frac{1}{2} \cdot 5^{x-4} - 2$

Directions: Graph each function and identify its key characteristics.

5. $f(x) = 2^{x+5}$



Domain: _____

Range: _____

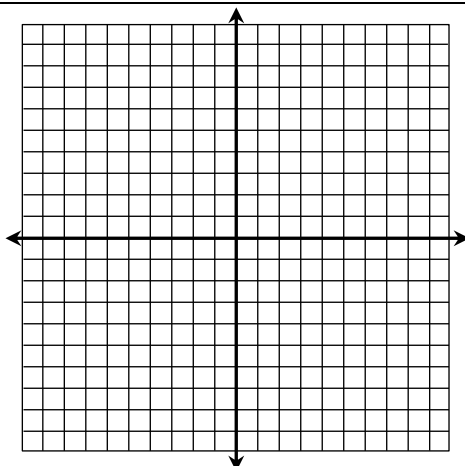
End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

6. $f(x) = \left(\frac{1}{3}\right)^x - 2$



Domain: _____

Range: _____

End Behavior:

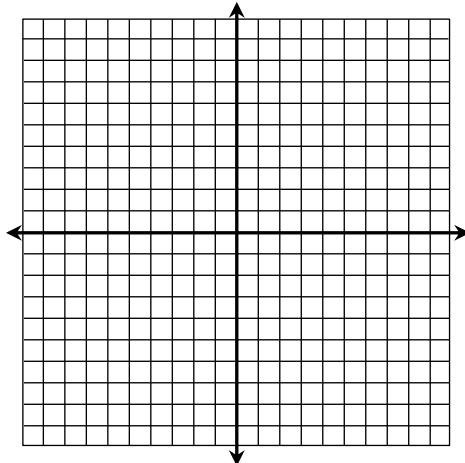
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

7. $f(x) = \frac{1}{2} \cdot 3^x + 1$



Domain: _____

Range: _____

End Behavior:

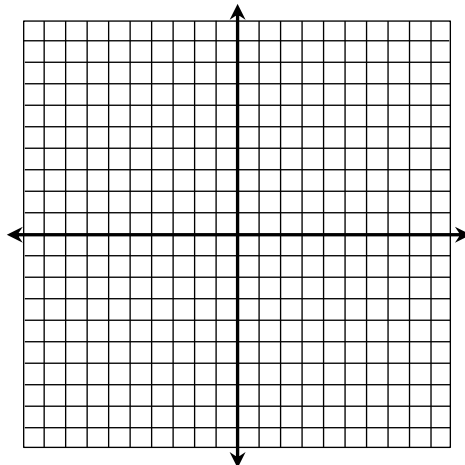
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

8. $f(x) = \left(\frac{3}{2}\right)^{x-4} - 5$



Domain: _____

Range: _____

End Behavior:

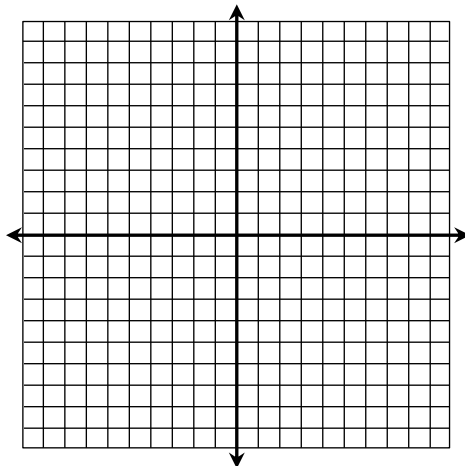
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

9. $f(x) = -2 \cdot 4^{x-2}$



Domain: _____

Range: _____

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

Name: _____

Unit 7: Exponential & Logarithmic Functions

Date: _____ Per: _____

Homework 1: Graphing Exponential Functions

**** This is a 2-page document! ****

Directions: Classify as an exponential growth or an exponential decay. Then, sketch the curve.

1. $f(x) = \frac{1}{2} \cdot 5^x$

2. $f(x) = \left(\frac{6}{5}\right)^x$

3. $f(x) = 4 \cdot \left(\frac{3}{8}\right)^x$

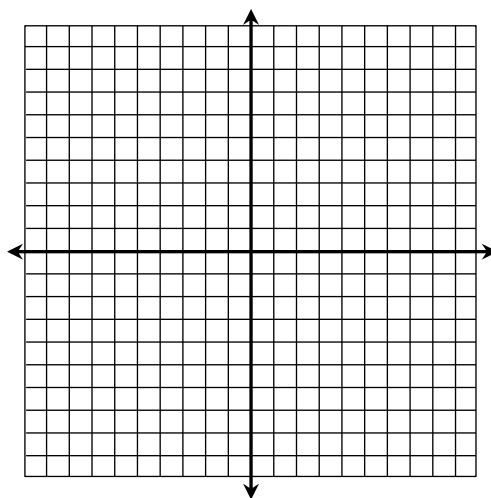
Directions: (a) Identify the parent function and **(b)** describe the transformations.

4. $f(x) = \left(\frac{4}{5}\right)^{x+2}$

5. $f(x) = -3 \cdot 2^{x-1} + 7$

Directions: Graph each function and identify its key characteristics.

6. $f(x) = 3^{x-2} - 7$



Domain: _____

Range: _____

End Behavior:

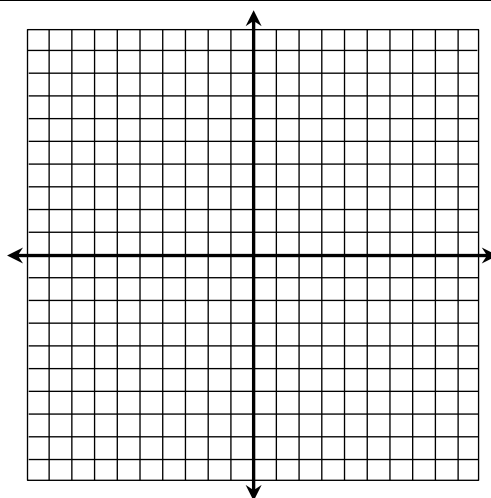
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

7. $f(x) = \left(\frac{1}{2}\right)^{x+4} - 3$



Domain: _____

Range: _____

End Behavior:

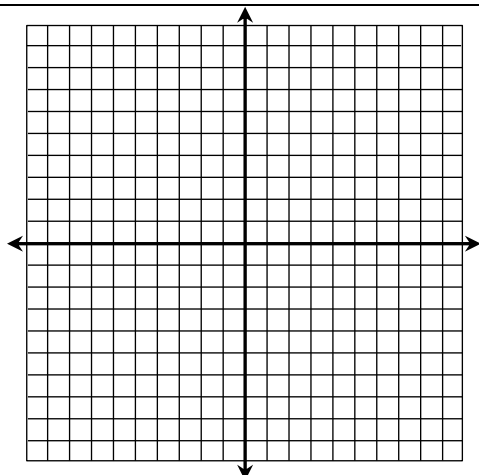
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

8. $f(x) = \frac{1}{2} \cdot 2^x - 6$



Domain: _____

Range: _____

End Behavior:

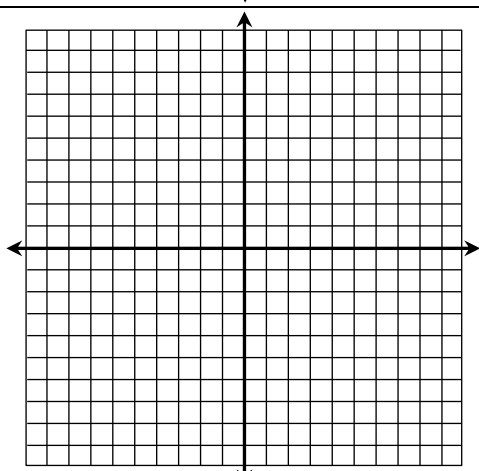
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

9. $f(x) = \frac{1}{8} \cdot 4^{x+1}$



Domain: _____

Range: _____

End Behavior:

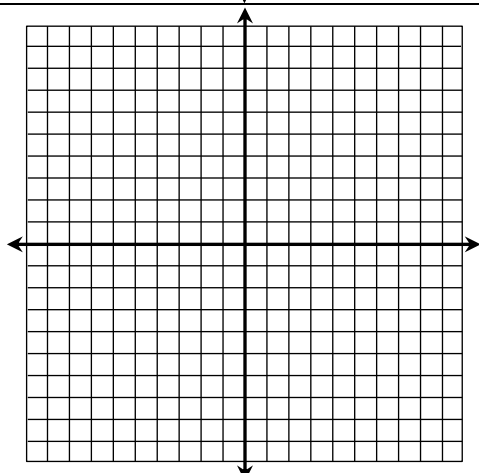
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

10. $f(x) = 4 \cdot \left(\frac{2}{3}\right)^x + 1$



Domain: _____

Range: _____

End Behavior:

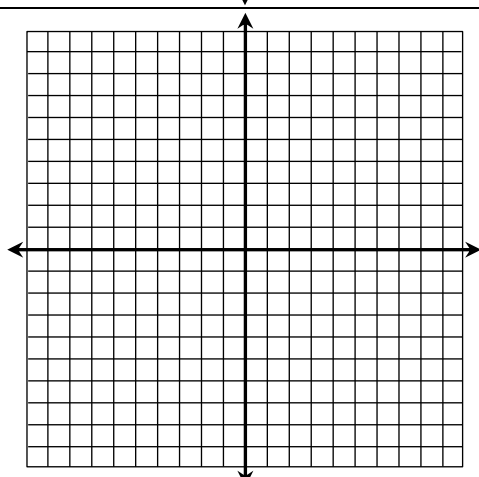
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

11. $f(x) = -\frac{1}{3} \cdot 6^x + 5$



Domain: _____

Range: _____

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
<i>Solving</i> EXPONENTIAL EQUATIONS	①	Use the properties of exponents to SIMPLIFY each side of the equation.
	②	Rewrite the equation so both sides have the SAME BASE .
	③	Drop the bases and SET THE EXPONENTS EQUAL TO EACH OTHER .
SET 1: WITH A COMMON BASE	Solve each equation.	
	1. $2^{x+1} = 2^9$	2. $5^{4n+5} = 5^{n-7}$
	3. $3^k \cdot 3^{k+2} = 3^{5k-1}$	4. $10^{-4} \cdot 10^9 = 10^{v+4} \cdot 10^{2v-11}$
SET 2: WITHOUT A COMMON BASE	Solve each equation.	
	5. $6^{2x-10} = 36$	6. $2^{p-7} = 8$
	7. $7^{4x+11} = \frac{1}{7}$	8. $32 = 2^{2m-9}$
	9. $27^{2x+6} = 3^{2x}$	10. $4^{y+2} = 16^{y-3}$

$$11. 125^y = 25$$

$$12. 16^{3x} = 8^{x+2}$$

$$13. 4^{3x} = 8^{x-1}$$

$$14. 81^{2x+5} = \left(\frac{1}{3}\right)^{2x}$$

$$15. 8^{2a-1} = 32^{2a+1}$$

$$16. 27^{2x} = 243^{x-2}$$

$$17. 64 = 4 \cdot 4^{4x}$$

$$18. 9^{2x+4} \cdot 9^{2x} = \frac{1}{81}$$

$$19. \frac{1}{7} = 49^{x-5} \cdot 7^{x-9}$$

$$20. 4^{2x} \cdot \frac{1}{16} = 4^{6x+18}$$

Name: _____

Unit 7: Exponential & Logarithmic Functions

Date: _____ Per: _____

Homework 2: Solving Exponential Equations

**** This is a 2-page document! ****

Directions: Solve each equation using a common base.

1. $9^{3x-7} = 9^{5-x}$

2. $2^{w+4} \cdot 2^{4w+6} = 2^{2w+1}$

3. $8^{6y+4} = 64$

4. $\frac{1}{5} = 5^{2c+3}$

5. $\frac{1}{27} = 3^{4m-1}$

6. $216 = 6^{2r-11}$

7. $2^{3k-1} \cdot 2^{5k-7} = 16$

8. $4^n \cdot 4^{2n-9} = 64$

9. $8^{x+2} = 4$

10. $125 = 25^{2h+1}$

11. $49^{p+1} = 343^{2p}$

12. $16^{r-2} = 64^{r+2}$

13. $27^{3n} = 81^{2n+1}$

14. $\left(\frac{1}{4}\right)^{2x} = 32^{4x-2}$

15. $16 \cdot 2^{6m} = 2^{3m-8}$

16. $\left(\frac{1}{9}\right)^a \cdot \left(\frac{1}{3}\right)^a = 3^{16-a}$

17. $256^y \cdot 16^{y-1} = 4^{2y-22}$

18. $36^{n-3} \cdot 216^n = 216^{2n+1}$

Name: _____ Algebra 2

Date: _____ Per: _____ Unit 7: Exponential & Logarithmic Functions

Quiz 7-1: Exponential Functions & Equations

For questions 1-3, indicate whether the function represents an exponential growth or decay.

1. $f(x) = \left(\frac{3}{4}\right)^x$

2. $f(x) = \frac{1}{6} \cdot 4^x$

3. $f(x) = \frac{1}{4} \cdot \left(\frac{5}{2}\right)^x$

1. _____

2. _____

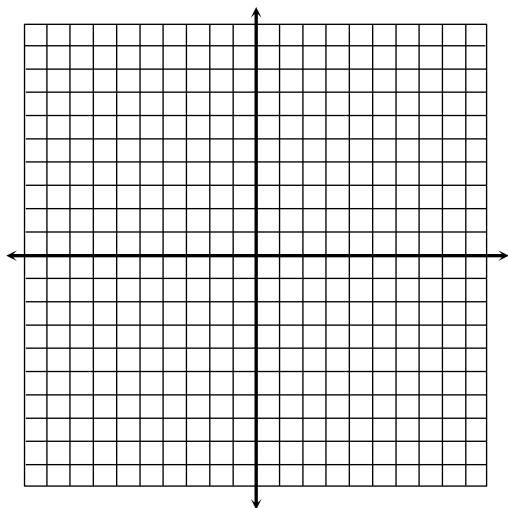
3. _____

4. _____

4. The parent function $f(x) = 2^x$ is reflected across the x-axis, then shifted 2 units left and 6 units down. Write an equation to represent the new function.

Graph each function and identify its key characteristics.

5. $f(x) = 3^{x-5} + 1$



Domain: _____

Range: _____

End Behavior:

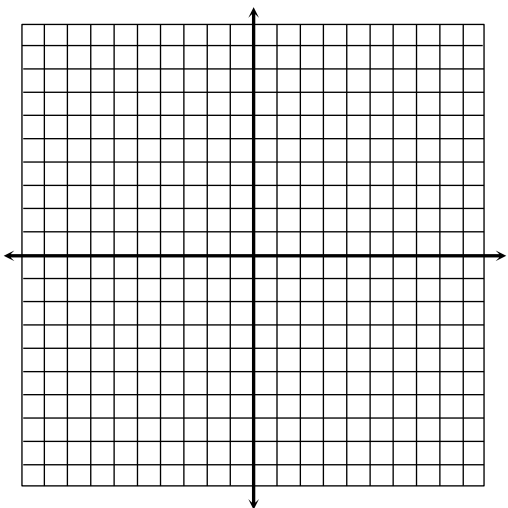
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

6. $f(x) = \frac{1}{4} \cdot 2^x - 7$



Domain: _____

Range: _____

End Behavior:

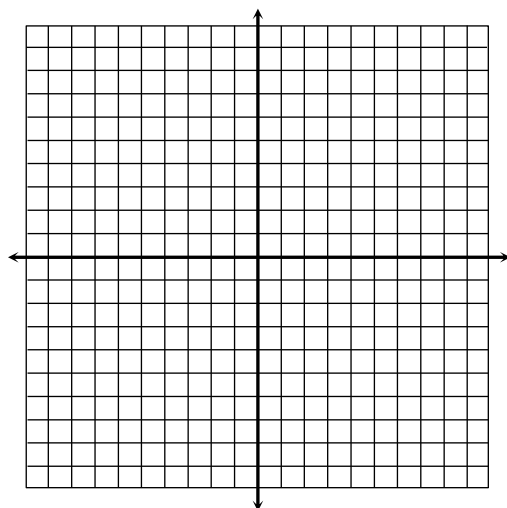
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

7. $f(x) = -\left(\frac{1}{3}\right)^{x+4}$



Domain: _____

Range: _____

End Behavior:

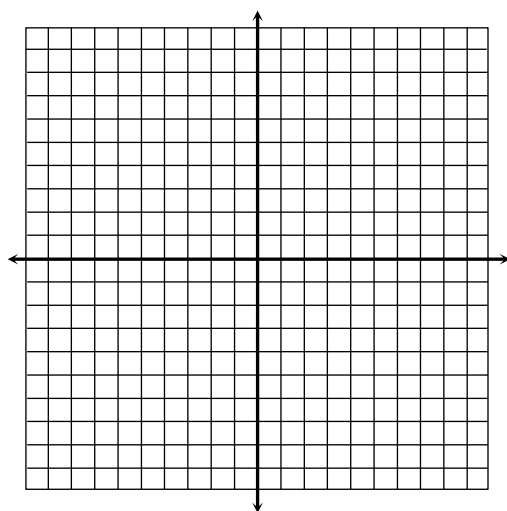
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

8. $f(x) = 2\left(\frac{1}{4}\right)^x - 9$



Domain: _____

Range: _____

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

Solve each equation.

9. $5^{3x-1} \cdot 5^{2x-5} = 5^{x+6}$

10. $64^x = 8^{3x+1}$

11. $\frac{1}{216} = 36^{3x+6}$

12. $32^{x+6} \cdot \frac{1}{2} = 8^{x-1}$

9. _____

10. _____

11. _____

12. _____

Name:		Date:	
Topic:		Class:	

Main Ideas/Questions	Notes/Examples	
What is a LOGARITHM?	A logarithm (log) is another way of writing exponents. Logarithmic Form $\log_b a = x$  Exponential Form  Read as "log base b of a equals x."	
Converting LOG $\Rightarrow$ EXP	Directions: Write each equation in exponential form .	
	1. $\log_3 9 = 2$	2. $\log_6 216 = 3$
	3. $\log_7 1 = 0$	4. $\log_2 16 = 4$
	5. $\log_4 \frac{1}{16} = -2$	6. $\log_9 27 = \frac{3}{2}$
Converting EXP $\Rightarrow$ LOG	Directions: Write each equation in logarithmic form .	
	7. $14^2 = 196$	8. $3^4 = 81$
	9. $12^1 = 12$	10. $36^{\frac{1}{2}} = 6$
	11. $2^{-3} = \frac{1}{8}$	12. $8^{\frac{4}{3}} = 16$

COMMON LOGARITHM	A logarithm with base 10 is called a common logarithm and can be written without the base. $\log_{10} x \rightarrow$	
EVALUATING LOGARITHMS	Directions: Use your knowledge of exponents to evaluate the following logarithms.	
	13. $\log_7 49$	14. $\log_3 27$
	15. $\log 100$	16. $\log_{12} 1$
	17. $\log_2 64$	18. $\log_3 243$
	19. $\log_9 \frac{1}{81}$	20. $\log_{64} 4$
CHANGE OF BASE FORMULA Choose BASE 10 because there is a calculator button for it!	Some logarithms are not as easy to evaluate as those above, and will require the change of base formula . $\log_b a =$	
	Directions: Evaluate each log using the change of base formula.	
	21. $\log_{16} 64$	22. $\log_8 32$
	23. $\log_2 54$	24. $\log_{10} 294$
	25. $\log_4 136$	26. $\log_6 \frac{1}{36}$

Name: _____

Unit 7: Exponential & Logarithmic Functions

Date: _____ Per: _____

Homework 3: Intro to Logarithms

Directions: Write each equation in exponential form.

1. $\log_2 128 = 7$

2. $\log_8 64 = 2$

3. $\log_3 \frac{1}{27} = -3$

Directions: Write each equation in logarithmic form.

4. $4^4 = 256$

5. $8^3 = 512$

6. $27^{\frac{2}{3}} = 9$

Directions: Evaluate each logarithm. Use the change of base formula when necessary.

7. $\log_6 36$

8. $\log_2 32$

9. $\log_4 64$

10. $\log_3 81$

11. $\log_{100} 10$

12. $\log_7 \frac{1}{7}$

13. $\log_{18} 1$

14. $\log_2 \frac{1}{16}$

15. $\log 1000$

16. $\log_{16} 8$

17. $\log_{243} 27$

18. $\log_3 92$

19. $\log_7 35$

20. $\log_2 260$

21. $\log_5 38$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples		
PRODUCT Property $\log_b(m \cdot n) =$	Condense into a single logarithm. Simplify if possible.		
	1. $\log_2 7 + \log_2 4$	2. $\log 25 + \log 4$	3. $\log_4 2x + \log_4 4x^2$
	Expand using the product property.		
	4. $\log 6$	5. $\log_7 45$	6. $\log_2 (5x)$
QUOTIENT Property $\log_b\left(\frac{m}{n}\right) =$	Condense into a single logarithm. Simplify if possible.		
	7. $\log_3 24 - \log_3 8$	8. $\log_2 15 - \log_2 15$	9. $\log_4 x^9 - \log_4 x^2$
	Expand using the quotient property.		
	10. $\log_8 4$	11. $\log_5 \frac{1}{3}$	12. $\log\left(\frac{m}{7}\right)$
POWER Property $\log_b m^n =$	Condense into a single logarithm. Simplify if possible.		
	13. $5 \cdot \log_4 2$	14. $7 \cdot \log_2 x$	15. $\frac{1}{3} \cdot \log 8$
	Expand using the power property. Simplify if possible.		
	16. $\log_2 8^7$	17. $3 \cdot \log 4^{x-1}$	18. $\log_7 \sqrt{w}$

PUTTING IT ALL TOGETHER!

CONDENSING LOGS

Rewrite as a single logarithm. Simplify if possible.

19. $2 \cdot \log_4 6 - \log_4 9$

20. $4 \cdot \log_4 a + 2 \cdot \log_4 b$

21. $7 \cdot \log_4 u - 3 \cdot \log_4 v^2$

22. $\log_2 15 + \log_2 4 - \log_2 6$

23. $\log_3 4 + \log_3 y + \frac{1}{2} \cdot \log_3 49$

24. $\frac{1}{3}(\log_5 8 + \log_5 27) - \log_5 3$

25. $3 \cdot \log_2 4 - \log_2 32$

26. $2 \cdot \log 6 - \frac{1}{4} \cdot \log 16 + \log 3$

EXPANDING LOGS

Expand each logarithm completely.

27. $\log_6 (xyz^4)$

28. $\log_4 \left(\frac{a^9}{b} \right)$

29. $\log_7 (q^4 r^2)^2$

30. $\log_2 \left(\frac{y}{z^5} \right)^2$

31. $\log \sqrt{7x^3}$

32. $\log_3 \sqrt[4]{m^5 n^2}$

PROPERTIES OF LOGARITHMS

Graphic Organizer

Name	Rule(s)	Example 1	Example 2
BASIC LOGARITHMS	$\log_b b =$; $\log_b 1 =$	Simplify: $\log_{14} 14 =$	Simplify: $\log_3 1 =$
PRODUCT RULE	$\log_b (m \cdot n) =$	Condense: $\log_5 6 + \log_5 7 =$	Expand: $\log_2 63 =$
QUOTIENT RULE	$\log_b \left(\frac{m}{n} \right) =$	Condense: $\log_4 84 - \log_4 12 =$	Expand: $\log 9 =$
POWER RULE	$\log_b m^n =$	Condense: $2 \cdot \log_3 8 =$	Expand: $\log_2 6^{x-1} =$
CHANGE OF BASE FORMULA	$\log_b a =$	Using a common base, evaluate the expression below. $\log_7 32 =$	

REMEMBER: BASE 10 LOGS ARE **COMMON LOGS** AND WRITTEN WITHOUT A BASE! (**log x**)

Name: _____

Unit 7: Exponential & Logarithmic Functions



Date: _____ Per: _____

Homework 4: Properties of Logarithms

**** This is a 2-page document! ****
Directions: Complete each rule.
PRODUCT RULE

$$\log_b(m \cdot n) =$$

QUOTIENT RULE

$$\log_b\left(\frac{m}{n}\right) =$$

POWER RULE

$$\log_b m^n =$$

Directions: Condense each expression into a single logarithm. Simplify if possible.

1. $\log_7 9 + \log_7 4$

2. $\log_2 80 - \log_2 5$

3. $\frac{1}{2} \cdot \log_3 81$

4. $3 \cdot \log_4 8 - 5 \cdot \log_4 2$

5. $7 \cdot \log_5 x + 3 \cdot \log_5 y^4$

6. $\log_3 a^7 + (\log_3 a^4 - 2 \cdot \log_3 b)$

7. $\log_4 x^7 - \log_4 x^2 + 4 \cdot \log_4 x^3$

8. $\frac{1}{2}(\log_6 45 - \log_6 5) + \log_6 12$

Directions: Condense, then use the **change of base formula** to evaluate the logarithm.

9. $\log_9 35 - \log_9 7$

10. $2 \cdot \log_3 8 - 4 \cdot \log_3 2$

11. $\frac{1}{3} \cdot \log_4 8 + \log_4 15$

12. $\log_2 27 + \log_2 4 - 2 \cdot \log_2 3$

Directions: Expand each logarithm completely.

13. $\log_8(mn^2)$

14. $\log\left(\frac{x^9}{y^4}\right)$

15. $\log_2(a^5b^2)^3$

16. $\log\left(\frac{p^4}{q^7}\right)^2$

17. $\log_5 \sqrt{a^7b}$

18. $\log_4 \sqrt[3]{c^2d^{15}}$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

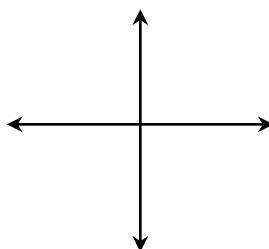
LOGARITHMIC

Parent Function

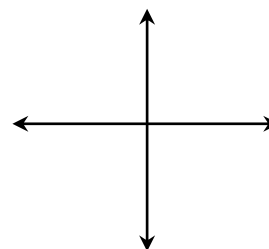


A logarithmic function is the **inverse** of an exponential function. Using your graphing calculator, sketch the following graphs:

$$f(x) = \log x$$



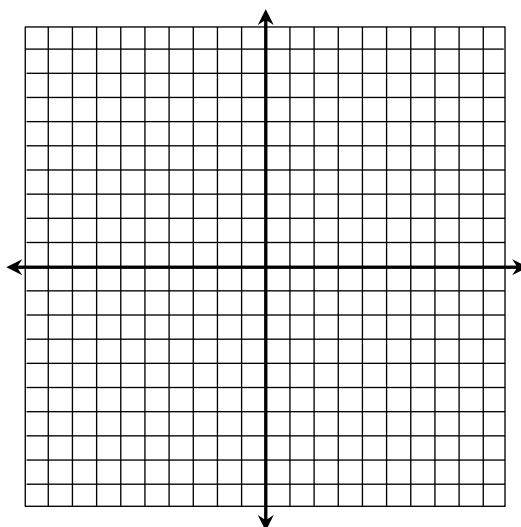
$$f(x) = 10^x$$



Because you can only graph base 10 logs on your calculator, you will need to use the inverse exponential function, then invert the values from the table to graph the logarithmic function.

Directions: Graph each function and identify its key characteristics.

1. $f(x) = \log_2 x$



Domain: _____

Range: _____

End Behavior:

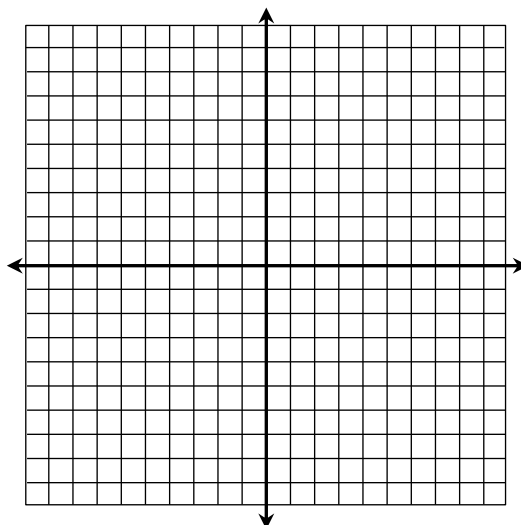
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

2. $f(x) = \log_{\frac{1}{3}} x$



Domain: _____

Range: _____

End Behavior:

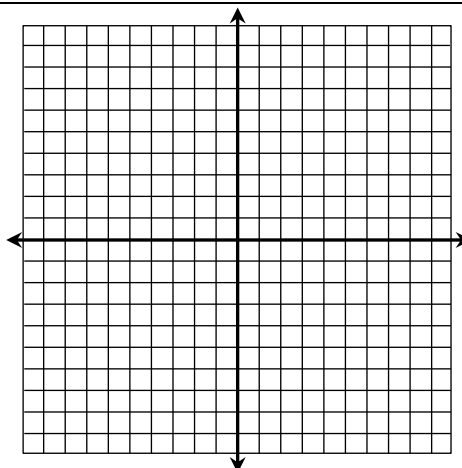
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

3. $f(x) = \log_4(x-1)$



Domain: _____

Range: _____

End Behavior:

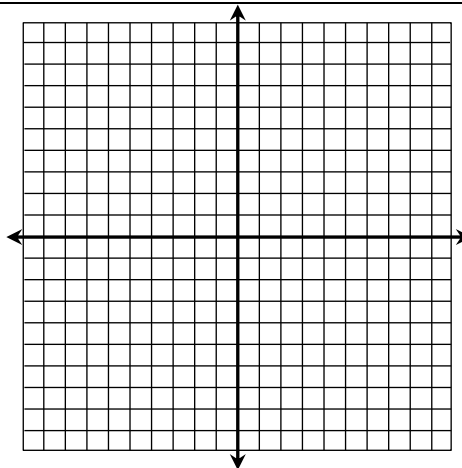
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

4. $f(x) = \log_3 x - 2$



Domain: _____

Range: _____

End Behavior:

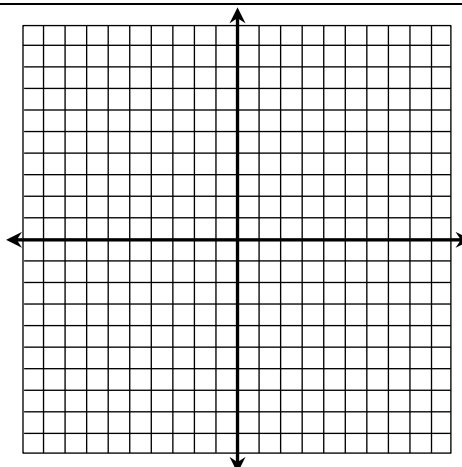
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

5. $f(x) = \log_{\frac{1}{2}}(x+9) + 4$



Domain: _____

Range: _____

End Behavior:

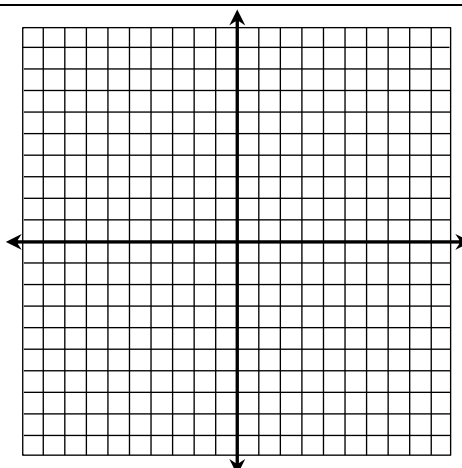
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

6. $f(x) = \log_5(x-2) + 1$



Domain: _____

Range: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____



Name: _____

Unit 7: Exponential & Logarithmic Functions

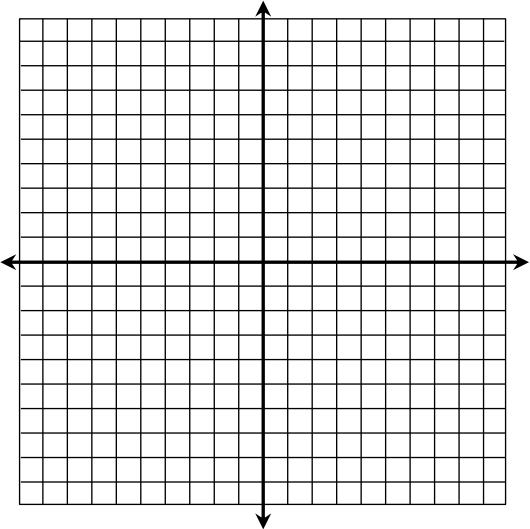
Date: _____ Per: _____

Homework 5: Graphing Logarithmic Functions

**** This is a 2-page document! ****

Directions: Graph each function and identify its key characteristics.

1. $f(x) = \log_3 x$



Domain: _____

Range: _____

End Behavior:

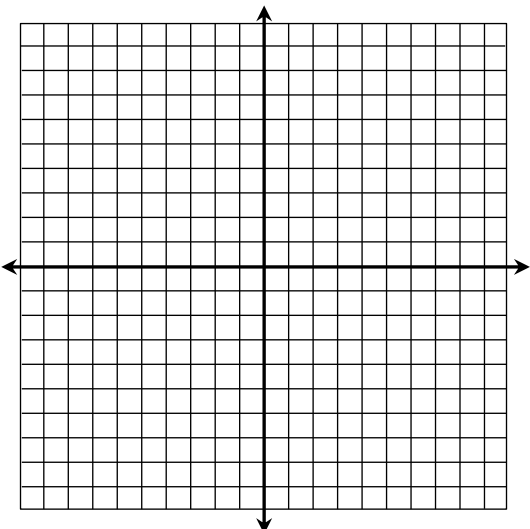
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

2. $f(x) = \log_{\frac{1}{2}} x + 3$



Domain: _____

Range: _____

End Behavior:

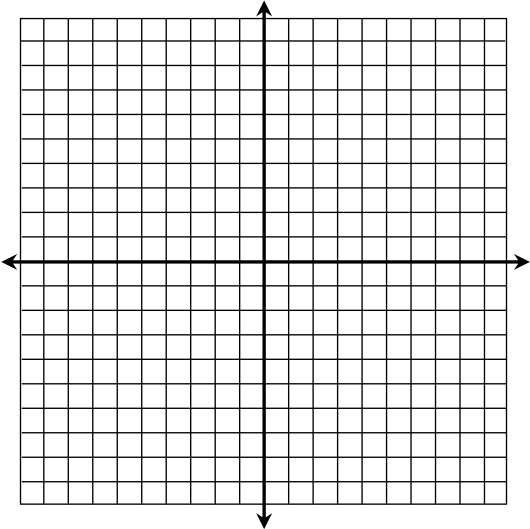
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

3. $f(x) = \log_4(x + 5)$



Domain: _____

Range: _____

End Behavior:

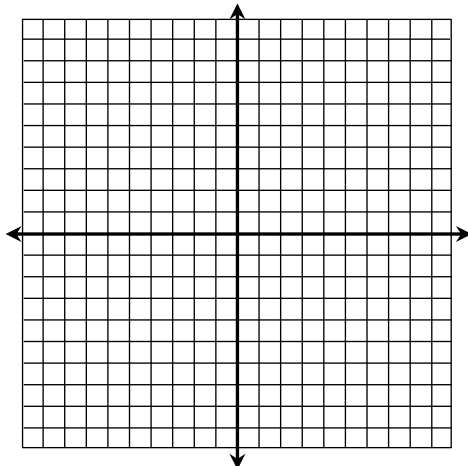
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

4. $f(x) = \log_2(x + 8) - 4$



Domain: _____

Range: _____

End Behavior:

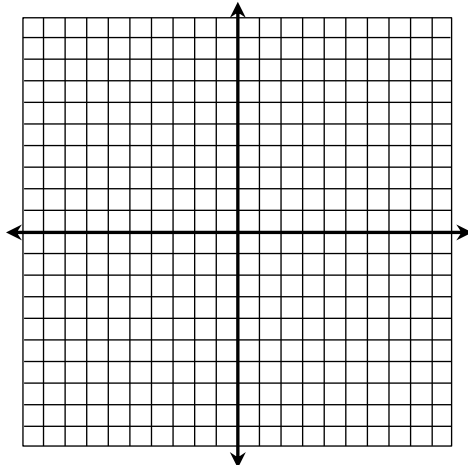
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

5. $f(x) = \log_{\frac{1}{3}}(x - 4)$



Domain: _____

Range: _____

End Behavior:

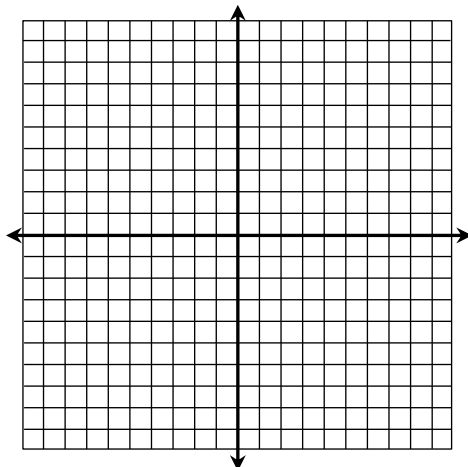
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

6. $f(x) = \log_5(x - 2) - 1$



Domain: _____

Range: _____

End Behavior:

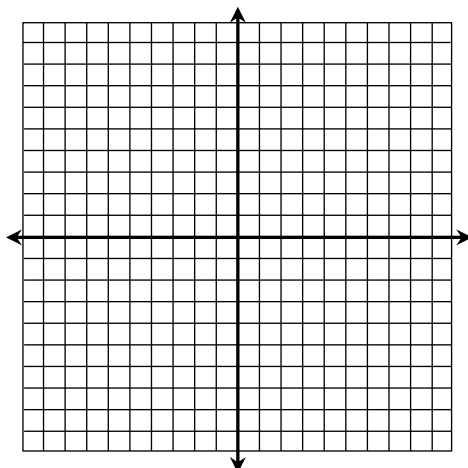
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

7. $f(x) = \log_{\frac{1}{4}}(x - 1) + 1$



Domain: _____

Range: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

Name: _____

Algebra 2

Date: _____ Per: _____

Unit 7: Exponential & Logarithmic Functions

Quiz 7-2: Logarithmic Expressions & Functions

Write each expression in exponential form.

1. $\log_3 81 = 4$

2. $\log_x 7 = \frac{1}{2}$

Write each expression in logarithmic form.

3. $2^{-3} = \frac{1}{8}$

4. $9^x = 36$

Evaluate. Use the change of base formula when necessary.

5. $\log_7 343$

6. $\log_3 \frac{1}{9}$

7. $\log_5 1$

8. $\log_2 64$

9. $\log_4 25$

10. $\log_{12} 90$

Condense each log into a single logarithm. DO NOT EVALUATE.

11. $\log_{12} 18 + 3 \cdot \log_{12} 2$

12. $\log_2 288 - \frac{1}{2} \cdot \log_2 81$

13. $5 \cdot \log x - 3 \cdot \log y$

14. $\frac{1}{4} \log_3 a^{20} + 3 \cdot \log_3 a^2$

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

Expand each logarithm.

15. $\log_4(2w^4)^3$

16. $\log\left(\frac{m^5}{n}\right)^2$

15. _____

16. _____

17. _____

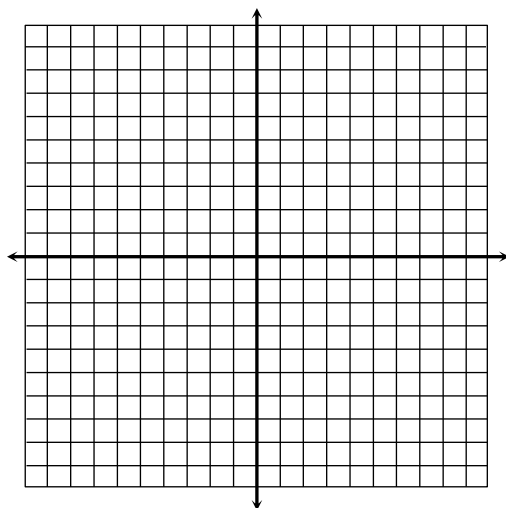
18. _____

17. $\log_5\sqrt{ab^7}$

18. $\log_2\sqrt[4]{p^5q^{12}}$

Graph each function and identify its key characteristics.

19. $f(x) = \log_2(x + 6) - 1$



Domain: _____

Range: _____

End Behavior:

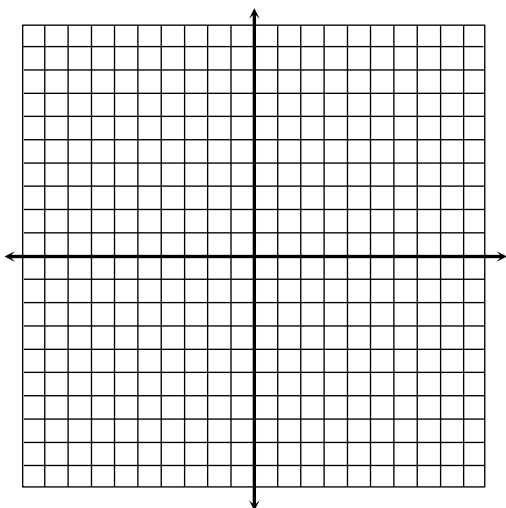
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

20. $f(x) = \log_{\frac{1}{3}}(x - 2)$



Domain: _____

Range: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
<i>Solving</i> LOGARITHMIC EQUATIONS Type 1: LOG = LOG	❶	CONDENSE each logarithm.
	❷	Use the One-to-One Property : If $\log_b m = \log_b n$, then
	❸	SOLVE and CHECK FOR EXTRANEOUS SOLUTIONS.
	1. $\log_5(5x + 9) = \log_5(6x)$	2. $\log_2(1 - 4n) = \log_2(2n + 43)$
	3. $\log_9(6 - 3w) = \log_9(-2w)$	4. $\log(y + 5) + \log 4 = \log 72$
	5. $3 \cdot \log_7 4 = \log_7(4a - 8)$	6. $\log_4 68 - \log_4 4 = \log_4(3n + 11)$
	7. $\frac{1}{2} \cdot \log_6 25 = \log_6(23 - 4w)$	8. $\log_3(2p - 5) = 2 \cdot \log_3 6 - \log_3 4$

	9. $\log_4(m^2) = \log_4(18 - 7m)$	10. $\log 2 + \log(k^2) = \log(k^2 + 16)$
<i>Solving</i> LOGARITHMIC EQUATIONS Type 2: LOG = NUMBER	❶ CONDENSE and ISOLATE the logarithm.	
	❷ Write the equation in EXPONENTIAL FORM .	
	❸ SOLVE and CHECK FOR EXTRANEIOUS SOLUTIONS .	
	11. $\log_2(x - 4) = 6$	12. $\log_3(4x + 8) - 7 = -3$
	13. $\log(2x) + \log(x - 5) = 2$	14. $2 \cdot \log x - \log 4 = 2$
	15. $\log_6(x + 9) + \log_6 x = 2$	16. $\log(x - 3) + \log x = 1$

Name: _____

Unit 7: Exponential & Logarithmic Functions

Date: _____ Per: _____

Homework 6: Solving Logarithmic Equations

**** This is a 2-page document! ****

Directions: Solve each equation. Check for extraneous solutions.

1. $\log_3(3x - 11) = \log_3(25 - x)$

2. $\log_7(4n - 7) = \log_7(-3n)$

3. $\log_2 75 = \log_2 3 + \log_2(2y - 1)$

4. $2 \cdot \log m = \log 36$

5. $\log_4 108 - \log_4 9 = \log_4(7a - 9)$

6. $\frac{1}{3} \cdot \log_5 64 = \log_5 8 + \log_5 p$

7. $\log(w^2 + 21) = \log(10w)$

8. $\log_2(2x) + \log_2(x - 7) = \log_2(4x)$

9. $\log_4(2m^3 - 14m^2) - \log_4(2m) = \log_4 8$

10. $2 \cdot \log(x - 3) = \log 25$

11. $\log_3(2x - 7) = 4$

12. $\log_8(28k - 20) + 15 = 18$

12. $\log_9(15 - 4n) = \frac{1}{2}$

14. $\log_2 4 + \log_2(c - 9) = 5$

15. $2 \cdot \log_4 k = 4$

16. $\log_8(p^2 + 15) = 2$

Name:		Date:	
Topic:		Class:	
Main Ideas/Questions REVIEW: Using a common base to solve an exponential equation.	Notes/Examples Directions: Solve the equations below using a common base.		
	1. $5^{n+10} = 25$	2. $9^{a+2} = 27^{4a-2}$	
What if a common base is NOT POSSIBLE? There are a couple ways to solve problems like these. Solve using both methods, then choose what works best for you!	METHOD 1 Take the Log of Both Sides and Solve. $2^x = 61$	METHOD 2 Write in Logarithmic Form and Solve. $2^x = 61$	
	Solve each equation. Round to the nearest ten-thousandth.		
Solving Exponential Equations: USING LOGS	3. $12^x = 8$	4. $8^{m-7} = 92$	
	5. $4 \cdot 7^n = 148$	6. $4^{3w} - 5 = 3$	

7. $7 - 4^{x+1} = 18$

8. $10 \cdot 5^{3k-3} = 40$

9. $4 \cdot 3^n + 15 = 359$

10. $-2 \cdot 5^p + 7 = -63$

11. $5 \cdot 9^{v-1} + 1 = 181$

12. $8 \cdot 11^{7k} - 3 = 213$

13. $6 \cdot 16^{7y+2} - 2 = 82$

14. $3 \cdot 8^{3-7n} + 10 = 94$

Name: _____

Unit 7: Exponential & Logarithmic Functions

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Homework 7: Solving Exponential Equations
(using logs)

**** This is a 2-page document! ****

Directions: Solve each exponential equation using logarithms.

1. $3^x = 18$

2. $7^y = 24$

3. $12^{n-3} = 60$

4. $2^{3a} = 142$

5. $15^{3v-5} = 87$

6. $4^{8n-2} = 84$

7. $4 \cdot 10^k = 60$

8. $16^n - 6 = 45$

9. $13^{c-8} - 9 = 17$

10. $2 \cdot 8^{5r} = 28$

11. $10^{2x-7} - 3 = 57$

12. $8^{6-4x} + 6 = 22$

13. $6 \cdot 4^m - 14 = 88$

14. $9 \cdot 12^{r+4} - 8 = 127$

15. $-5 \cdot 4^{6x} + 5 = -30$

16. $8 \cdot 11^{3p-9} + 10 = 194$

REVIEW: *Logarithmic & Exponential Equations*

LOGARITHMIC EQUATIONS

Solve. Check for extraneous solutions.

1. $\log_7(9x - 4) = \log_7(x + 20)$

2. $\log_5(m^2 - 12) = \log_5 m$

3. $\log_3 4 + \log_3(a + 5) = \log_3 56$

4. $\log(2y - 10) = 7 \cdot \log 2 - \log 8$

5. $\log_4(5m + 9) = 3$

6. $\log_{36}(20 - 4p) = \frac{1}{2}$

7. $\log_6(7k - 1) = 3$

8. $\log(n + 8) + \log 4 = 2$

EXPONENTIAL EQUATIONS

Solve. Check for extraneous solutions. Round to the nearest ten-thousandth when necessary.

9. $25^{v-2} = 625$

10. $\frac{1}{16} = 8^{4x-2}$

11. $8^k = 78$

12. $9^{m-6} = 78$

13. $15^{3a} + 7 = 67$

14. $14^{3-8x} + 9 = 77$

15. $8 \cdot 3^{n-1} - 21 = 51$

16. $2 \cdot 18^{10r-3} - 1 = 73$

Name: _____

Unit 7: Exponential & Logarithmic Functions



Date: _____ Per: _____

Homework 8: Logarithmic & Exponential Equations

**** This is a 2-page document! ******Directions:** Solve each equation. Check all answers for extraneous solutions.

1. $\log_4(25 - 2x) = \log_4(6x + 1)$

2. $\log_9(8y - 9) = \log_9 108 - \log_9 4$

3. $6 \cdot \log_2 2 = \log_2 8 + \log_2(a - 2)$

4. $\log_6(5w + 14) = 2 \cdot \log_6 w$

5. $\log_7(3x + 5) = 2$

6. $\log_{27}(11 - 2k) = \frac{1}{3}$

7. $\log(24x + 64) = 3$

8. $5 = \log_3 8 + \log_3(r + 6)$

9. $5^{x-4} = 25^{x-6}$

10. $36^{5v+2} = \left(\frac{1}{6}\right)^{11-v}$

11. $5^m = 220$

12. $14^{p-8} = 62$

13. $3 \cdot 4^{n+2} = 78$

14. $5^{8-2y} - 10 = 45$

15. $2 \cdot 10^{6c} + 9 = 17$

16. $9 \cdot 14^{5a+9} + 8 = 107$

Name: _____ Algebra 2

Date: _____ Per: _____ Unit 7: Exponential & Logarithmic Functions

Quiz 7-3: Logarithmic & Exponential Equations

Part I: Logarithmic Equations

1. $\log_2(3x - 10) = \log_2(14 - x)$

2. $\log 12 + \log 5 = \log(4m + 8)$

3. $\log_7(9p - 5) = 4 \cdot \log_7 2$

4. $\log_4(k^2 + 2k) = \log_4 60 - \log_4 4$

5. $\log_3(2a - 15) = 2$

6. $\frac{1}{4} = \log_{16}(17 - 5n)$

7. $\log_3 9 + \log_3(c - 2) = 3$

8. $\frac{1}{2} \cdot \log_2 36 = \log_2(7w + 1)$

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

Part II: Exponential Equations

9. $\left(\frac{1}{8}\right)^{x+3} = 64$

10. $36^{2m} = 216$

11. $4^w = 10$

12. $9^{a-5} = 12$

13. $7^{10n} + 2 = 81$

14. $12^{3p+8} + 10 = 97$

15. $-2 \cdot 10^x - 1 = -33$

16. $4 \cdot 18^{k-8} + 3 = 59$

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

15. _____

16. _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
EXPONENTIAL <i>Growth</i>	Exponential growth occurs when a quantity exponentially increases by a fixed percent each year (or other other time period).	
	Formula:	$a =$ _____ $r =$ _____ $t =$ _____
EXAMPLES	1. The original value of an investment is \$1,800. If the value has increased by 7% each year, write an exponential function to model the situation. Then, find the value of the investment after 15 years.	2. In 2002, there were 972 students enrolled at Oakview High School. Since then, the number of students has increased by 1.5% each year. Write an exponential function to model the situation, then find the number of students enrolled in 2014.
EXPONENTIAL <i>Decay</i>	Exponential decay occurs when a quantity exponentially decreases by a fixed percent each year (or other other time period).	
	Formula:	$a =$ _____ $r =$ _____ $t =$ _____
EXAMPLES	3. An investment of \$12,000 is losing value at a rate of 4% each year. Write an exponential function to model the situation, then find the value of the investment after 9 years.	4. Mark bought a brand new car for \$35,000 in 2008. If the value of the car depreciates approximately 8% each year, write an exponential function to model the situation. Then, find the value of the car in 2015.

COMPOUND <i>Interest</i>	Compound interest occurs when interest is calculated on both the principal amount AND the accrued interest thus far.	
	Formula:	$P =$ _____ $r =$ _____ $n =$ _____ $t =$ _____
EXAMPLES	5. Laura deposited \$12,000 into an account that earns 8% interest. How much money will she have in 5 years if the interest is compounded quarterly?	
	6. Jack moved \$25,000 of his savings to a new investment account that earns 4.5% interest compounded monthly. Find the total amount in the account after 6 years.	
	7. An investment account pays 3.9% interest compounded semi-annually. If \$4,000 is invested in this account, what will the balance be after 12 years?	
	8. A savings account offers 0.8% interest compounded bimonthly. If Bob deposited \$300 into this account, how much interest will he earn after 10 years?	
	9. Suppose you invest \$750 into an account that pays 3% interest compounded weekly. How much interest will you have earned after 20 years?	



Name: _____ Unit 7: Exponential & Logarithmic Functions

Date: _____ Per: _____ Homework 9: Applications of Exponential Functions

**** This is a 2-page document! ****

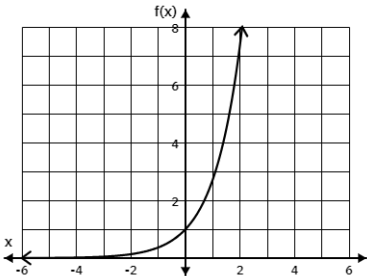
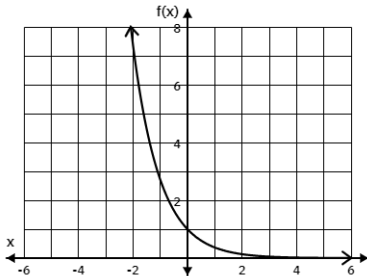
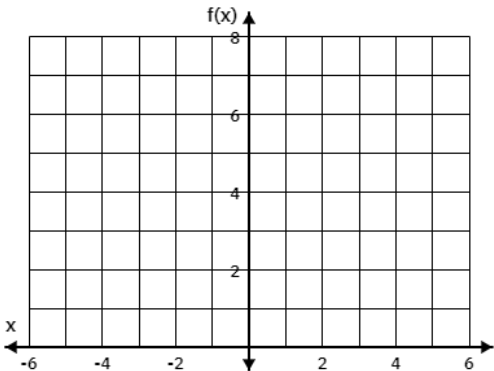
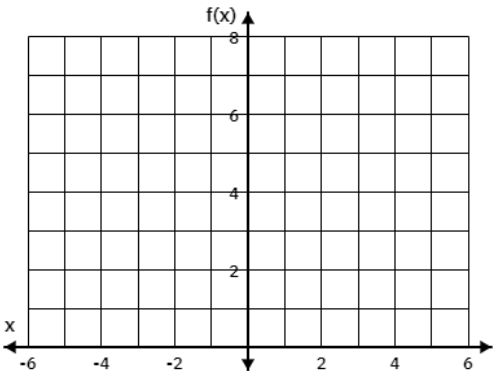
Exponential Growth & Decay

1. Vanessa invested \$2,500 into an account that will increase in value by 3.5% each year. Write an exponential function to model this situation, then find the value of the investment after 20 years.
2. The average price of a movie ticket in 1990 was \$4.22. Since then, the price has increased by approximately 3.1% each year. Write an exponential function to model this situation, then find the price of a ticket in 2016.
3. A virus has infected 400 people in the town and is spreading to 25% more people each day. Write an exponential function to model this situation, then find the number of people that will be infected in 10 days.
4. The population of a small town was 10,800 in 2002. Since then, the population has decreased at a rate of 2.5% each year. Write an exponential function to model the situation, then find the population of the town in 2020.
5. Manny bought a brand-new car in 2012 for \$28,750. If the car depreciates by 12% each year, write an exponential function to model the situation, then find the value of the car in 2018.

Compound Interest

6. Anisha invested \$8,000 in an account that earns 10% interest. How much money will she have in 15 years if the interest is compounded quarterly?
7. A realtor took \$32,500 made on the sale of a home and placed it in a new account that earns 6% compounded annually. Find the total amount in the account after 5 years.
8. Scott invested \$1,600 into a retirement account that earns 2.4% interest compounded monthly. What will the balance of the account be after 30 years?
9. Kaylee used her graduation money to set up a savings account that earns 3.4% interest compounded weekly. If the original amount deposited was \$500, how much interest will she have earned after 10 years?
10. Ryan inherited land from his grandfather. He decided to sell the land for \$160,000, then place the money in a new account that earns 1.2% interest compounded bimonthly. How much interest will he have earned after 30 years?

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
What is the NATURAL BASE, "<i>e</i>"?	- Like π, e is special number discovered in mathematics and occurs in real-world applications of <i>continuous</i> exponential growth and decay. e is an _____ with an approximate value of _____.	
Natural Base EXPONENTIAL Functions	The function _____ is used to model continuous exponential growth. 	The function _____ is used to model continuous exponential decay. 
	Classify as an exponential growth or decay, then graph the function.	
	1. $f(x) = e^{0.5x}$ 	2. $f(x) = e^{-2x}$ 
APPLICATIONS	3. A person is given 800 milligrams of a certain medicine. The amount of medicine, in miligrams, in their system after t hours is modeled by the function $f(t) = 800e^{-0.6t}$. How many milligrams of the medicine remains in the system after 2 hours?	4. The balance in an investment account is modeled by the function $f(t) = 2000e^{0.05t}$ where t is the years since the initial investment. Find the balance in the account after 15 years.

<i>Natural</i> LOGARITHMS	• A _____ is a logarithm with base e and and written as _____, but usually as _____.		
CONVERTING <i>Between Forms</i>	Write each equation in logarithmic form.		
	5. $e^x = 24$	6. $e^9 = x$	7. $e^{x+5} = 72$
	Write each equation in exponential form.		
	8. $\ln x = 58$	9. $\ln 6 = x$	10. $\ln(x - 9) = 32$
<i>Simplifying</i> EXPRESSIONS	Condense each expression into a single logarithm.		
	11. $\ln 3 + \ln 16$	12. $\ln 63 - 2 \cdot \ln 3$	13. $\frac{1}{3} \cdot \ln 64 + 2 \cdot \ln x$
	Expand each logarithm.		
	14. $\ln 5x$	15. $\ln\left(\frac{a^3}{b}\right)^2$	16. $\ln \sqrt[3]{m^2 n}$
<i>Solving</i> EQUATIONS	Solve each equation. Check for extraneous solutions.		
	17. $\ln(4x - 27) = \ln(15 - 2x)$	18. $2 \cdot \ln k = \ln(2k + 15)$	

19. $\ln 72 - \ln 4 = \ln 6 + \ln(a - 2)$

20. $2 \cdot \ln(m + 4) = \ln 4$

21. $\ln 8x = 2$

22. $\ln x - \ln 9 = 7$

23. $e^x = 57$

24. $e^{y+3} - 6 = 24$

25. $5e^{4n} = 95$

26. $2e^{c-9} + 3 = 87$

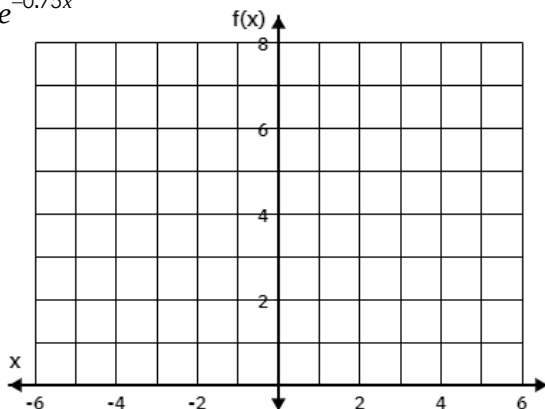
Name: _____

Unit 7: Exponential & Logarithmic Functions

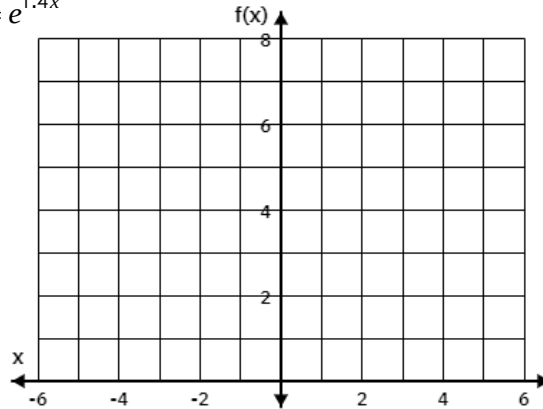
Date: _____ Per: _____

Homework 10: Base e and Natural Logarithms**** This is a 2-page document! ******Classify as an exponential growth or decay, then graph.**

1. $f(x) = e^{-0.75x}$



2. $f(x) = e^{1.4x}$



3. The amount of a bacteria in a culture can be modeled by the function $f(t) = 400e^{0.15t}$, where t is the number of hours since 7 AM. Find the bacteria count at 3:30 PM on the same day.

4. The value of a certain boat depreciates according to the model $f(t) = 30000e^{-0.2t}$, where t is the number of years after the boat was purchased new. Find the value of the boat in 18 months.

Write each equation in logarithmic form.

5. $e^3 = x$

6. $e^x = 36$

7. $e^{x-9} = 74$

Write each equation in exponential form.

8. $\ln 53 = x$

9. $\ln x = 18$

10. $\ln 87 = x + 4$

Condense each expression as a single logarithm.

11. $\ln 4 + \ln 3x$

12. $\frac{1}{2} \cdot \ln 256 - 3 \cdot \ln 2$

13. $7 \cdot \ln a - 4 \cdot \ln b$

Expand each logarithmic expression.

14. $\ln(2m^8)$

15. $\ln\left(\frac{m^5}{n^2}\right)^3$

16. $\ln\sqrt{r^8s^5}$

Solve each equation. Be sure to check for extraneous solutions.

17. $\ln(2x^2 - 15) = \ln(x^2 + 34)$

18. $\ln 8 + \ln(n - 9) = 5 \cdot \ln 2$

19. $\ln(4w + 9) = 5$

20. $\ln k - \ln 14 = 2$

21. $e^x = 21$

22. $e^{y-1} - 27 = 54$

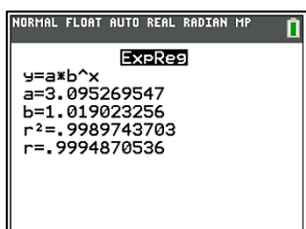
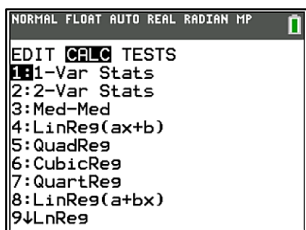
23. $4e^{3k} + 1 = 85$

24. $3e^{4m-7} - 8 = 106$

Name:	Date:
Topic:	Class:

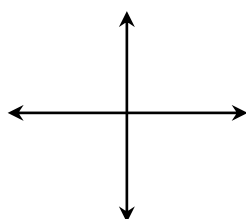
Main Ideas/Questions	Notes/Examples																																																
<table border="1" style="width: 100%; border-collapse: collapse; margin-bottom: 20px;"> <thead> <tr> <th style="width: 15%; padding: 5px;">Year</th><th style="width: 15%; padding: 5px;">Population</th></tr> </thead> <tbody> <tr><td style="text-align: center;">1900</td><td style="text-align: center;">3.1</td></tr> <tr><td style="text-align: center;">1920</td><td style="text-align: center;">4.6</td></tr> <tr><td style="text-align: center;">1940</td><td style="text-align: center;">6.4</td></tr> <tr><td style="text-align: center;">1960</td><td style="text-align: center;">9.5</td></tr> <tr><td style="text-align: center;">1980</td><td style="text-align: center;">14.2</td></tr> </tbody> </table> <table border="1" style="width: 100%; border-collapse: collapse; margin-bottom: 20px;"> <thead> <tr> <th style="width: 15%; padding: 5px;">Year</th><th style="width: 15%; padding: 5px;">Closing Price (\$)</th></tr> </thead> <tbody> <tr><td style="text-align: center;">1992</td><td style="text-align: center;">633</td></tr> <tr><td style="text-align: center;">1994</td><td style="text-align: center;">793</td></tr> <tr><td style="text-align: center;">1996</td><td style="text-align: center;">1,100</td></tr> <tr><td style="text-align: center;">1998</td><td style="text-align: center;">1,771</td></tr> <tr><td style="text-align: center;">2000</td><td style="text-align: center;">4,696</td></tr> </tbody> </table> <table border="1" style="width: 100%; border-collapse: collapse; margin-bottom: 20px;"> <thead> <tr> <th style="width: 15%; padding: 5px;">Year</th><th style="width: 15%; padding: 5px;">Female Workers (%)</th></tr> </thead> <tbody> <tr><td style="text-align: center;">1970</td><td style="text-align: center;">30.0</td></tr> <tr><td style="text-align: center;">1980</td><td style="text-align: center;">35.3</td></tr> <tr><td style="text-align: center;">1985</td><td style="text-align: center;">37.9</td></tr> <tr><td style="text-align: center;">1990</td><td style="text-align: center;">37.2</td></tr> <tr><td style="text-align: center;">2006</td><td style="text-align: center;">41.5</td></tr> </tbody> </table> <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 15%; padding: 5px;">Age</th><th style="width: 15%; padding: 5px;">Height</th></tr> </thead> <tbody> <tr><td style="text-align: center;">1</td><td style="text-align: center;">4</td></tr> <tr><td style="text-align: center;">2</td><td style="text-align: center;">6.3</td></tr> <tr><td style="text-align: center;">3</td><td style="text-align: center;">7.9</td></tr> <tr><td style="text-align: center;">4</td><td style="text-align: center;">9.2</td></tr> <tr><td style="text-align: center;">5</td><td style="text-align: center;">10.3</td></tr> </tbody> </table>	Year	Population	1900	3.1	1920	4.6	1940	6.4	1960	9.5	1980	14.2	Year	Closing Price (\$)	1992	633	1994	793	1996	1,100	1998	1,771	2000	4,696	Year	Female Workers (%)	1970	30.0	1980	35.3	1985	37.9	1990	37.2	2006	41.5	Age	Height	1	4	2	6.3	3	7.9	4	9.2	5	10.3	1. The table to the left shows the year and population (in millions) of Texas. Using an exponential model, write an equation for the curve of best fit, then estimate the population of Texas in 2020. 2. The table to the left shows the closing prices for the NASDAQ Stock Index at the end of February during certain years. Using an exponential model, write an equation for the curve of best fit, then estimate the year the closing price will reach \$10,000. 3. The table to the left gives the percent of all workers who are female during selected years from 1970 to 2006. Using a logarithmic model, write an equation for the curve of best fit, then estimate the percent of female workers in 2016. 4. The table to the left shows the height (in feet) of a pine tree in the park. When it was originally planted, it was one year old and had a height of 4 feet. Using a logarithmic model, write an equation for the curve of best fit, then determine the age of the tree at 20 feet.
Year	Population																																																
1900	3.1																																																
1920	4.6																																																
1940	6.4																																																
1960	9.5																																																
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3	7.9																																																
4	9.2																																																
5	10.3																																																

Choosing the BEST MODEL

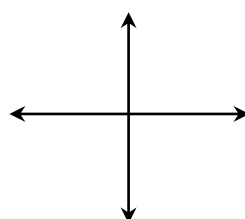


Sketch each parent function.

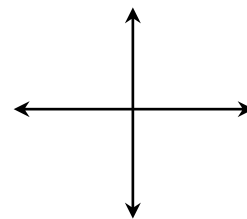
Linear (4:LinReg)



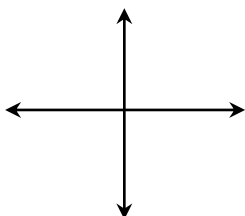
Quadratic (5:QuadReg)



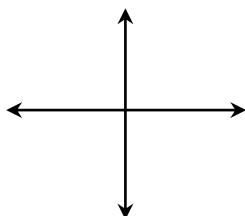
Cubic (6:CubicReg)



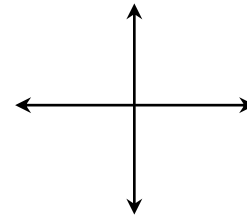
Quartic (7:QuartReg)



Exponential (0:ExpReg)



Logarithmic (9:LnReg)



To **determine the best model**, use the **correlation coefficient**, r .
The closer r is to -1 or 1, the better the fit. Make sure your diagnostics are turned ON to see this value!

5. The table below shows the operating costs (in thousands of dollars) of a small business from 2000 to 2005. Which model would fit this data best: quadratic, quartic, or logarithmic? Using the model, write a best-fit equation, then estimate the operation costs in 2008.

Year	Costs
2000	2.3
2001	2.6
2002	3.1
2003	3.4
2004	4
2005	5.6

6. The table below shows the balance of Marissa's savings account at the end of the year. Which model would fit this data best: linear, cubic, or exponential? Using the model, write a best-fit equation, then estimate the year the balance will reach \$1,500.

Year	Balance (\$)
2009	500
2010	540
2011	583.20
2012	629.86
2013	680.24
2014	734.66

Name: _____

Unit 7: Exponential & Logarithmic Functions



Date: _____ Per: _____

Homework 11: Exponential & Logarithmic Regression;
Choosing the Best Model**** This is a 2-page document! ****

1. The table below shows the value of a car during certain years. Using an **exponential model**, write an equation for the curve of best fit, then estimate the value of the car in 2016.

Year	Value (\$)
2008	27,500
2009	24,200
2010	21,296
2011	18,740
2012	16,492

2. The table below shows the number of students enrolled at Glennview High School during certain years. Using an **exponential model**, write an equation for the curve of best fit, then estimate the year in which enrollment reaches 2,000 students.

Year	Students
2004	1,250
2007	1,350
2010	1,458
2013	1,574

3. The table below shows the weights of five females of the same height along with their body mass index, BMI. Using a **logarithmic model**, write an equation for the curve of best fit, then estimate the BMI of a female that weighs 130 pounds.

Weight	BMI
112	17.5
125	19.1
139	22.7
156	26.4
173	30.2

4. The table below shows the airfare of a one-way ticket from New York City to Orlando as advertised by an airline for the first five days of April. Using a **logarithmic model**, write an equation for the curve of best fit. If Scott does not want to pay more than \$140, what day must he purchase his ticket by?

Date	Price
1	89
2	105
3	109
4	114
5	120

5. The table below shows the height, in feet, of a rock thrown from the top of a building each second after it was thrown. Which model would fit this data best: linear, quadratic, or cubic? Use the model to write a best-fit equation, then estimate the height of the rock at 5 seconds.

Time	Height
0	250
1	280
2	284
3	249
4	202

6. When the Morrison's adopted their six week old puppy, he was 8 pounds. The table below shows the weight of the puppy each week thereafter. Which model would best fit this data: exponential or logarithmic? Use the model to write a best-fit equation, then find the week at which the puppy will weigh 50 pounds.

Week	Weight
6	8
7	12
8	14
9	17
10	20

7. The table below shows the value of an investment during certain years. Which model would best fit this data: quadratic, cubic, or quartic? Use the model to write a best-fit equation, then estimate the value of the investment in 2022.

Year	Value (\$)
2002	1,200
2005	1,895
2008	2,399
2011	2,781
2014	3,500

8. The total yearly revenue of a small company, in thousands of dollars, is given in the table below. Which model would best fit this data: quadratic, cubic, or exponential? Use the model to write a best-fit equation, then estimate the revenue of the company in 2020.

Year	Revenue
2010	4.3
2011	8.3
2012	10.5
2013	11.4
2014	11.5

Unit 7 Test Study Guide

(Exponential & Logarithmic Functions)

Name: _____

Date: _____ Per: _____

Topic 1: Graphing Exponential & Logarithmic Functions

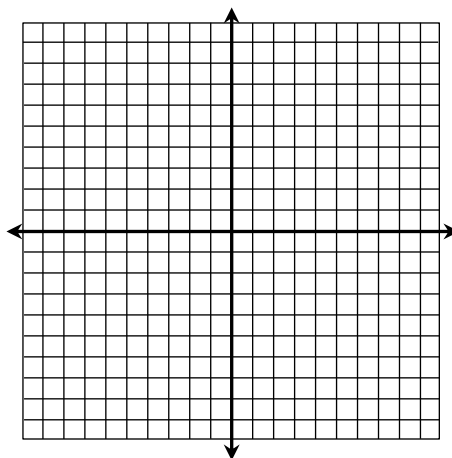
Describe as an exponential growth or decay.

1. $f(x) = 5\left(\frac{2}{3}\right)^x$

2. $f(x) = \frac{1}{3}\left(\frac{6}{5}\right)^x$

Graph each function, then identify its key characteristics.

3. $f(x) = 3^{x+1} - 6$



Domain: _____

Range: _____

End Behavior:

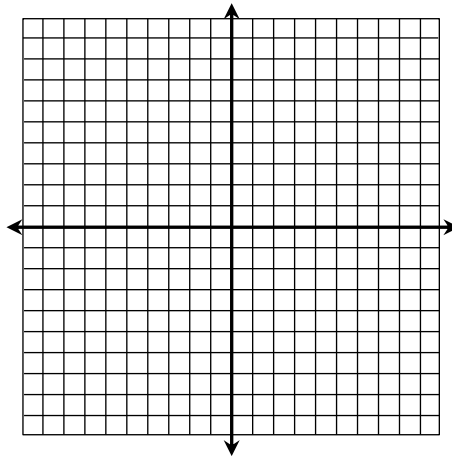
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

4. $f(x) = \left(\frac{1}{2}\right)^{x-5} + 2$



Domain: _____

Range: _____

End Behavior:

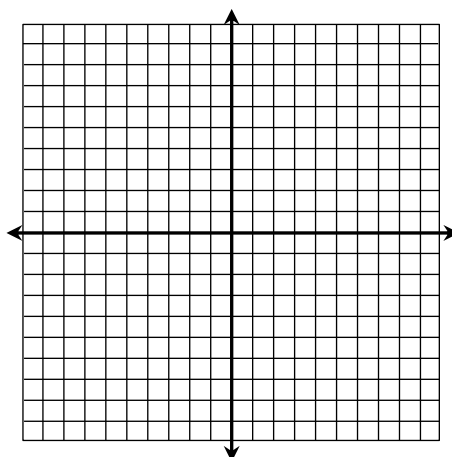
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

5. $f(x) = \log_2 x - 3$



Domain: _____

Range: _____

End Behavior:

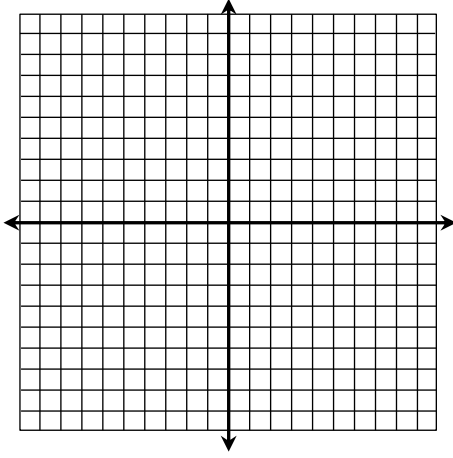
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

6. $f(x) = \log_{\frac{1}{3}}(x+2) + 1$



Domain: _____

Range: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

Topic 2: Exponential vs. Logarithmic Forms

Write in logarithmic form.			
7. $8^2 = 64$	8. $2^{x-4} = 32$	9. $10^{2x} = 54$	10. $e^6 = x - 2$
Write in exponential form.			
11. $\log_3 27 = 3$	12. $\log_x 7 = \frac{1}{2}$	13. $\log_4 90 = x$	14. $\ln x = 38$

Topic 3: Evaluating Logarithms

Evaluate each logarithm. Use the Change of Base Formula when necessary: $\log_b a =$			
15. $\log_9 81$	16. $\log_{81} 3$	17. $\log_5 \frac{1}{25}$	18. $\log_6 1$
19. $\log 63$	20. $\log_7 95$	21. $\log_2 78$	22. $\ln 42$

Topic 4: Properties of Logarithms

Product Rule	Quotient Rule	Power Rule
$\log_b(m \cdot n) =$	$\log_b\left(\frac{m}{n}\right) =$	$\log_b m^n =$

Condense each expression into a single logarithm.		
23. $3 \cdot \log 2 + \log(x - 4)$	24. $\frac{1}{2} \cdot \log_5 324 - \log_5 2$	25. $3 \cdot \ln 6 - \frac{3}{2} \cdot \ln 4$
Expand each expression.		
26. $\log_3(x^2 y^5)^3$	27. $\ln\left(\frac{2}{a^3}\right)^4$	28. $\log_4 \sqrt{p^3 q^{10}}$

Topic 5: Solving Logarithmic & Exponential Equations

Solve each equation. Check for extraneous solutions.	
29. $\log_4(5x + 7) = \log_4(2x + 31)$	30. $\ln(p^2 - p) = \ln(6p + 18)$
31. $\frac{1}{2} \cdot \log_8 36 + \log_8(3k + 7) = \log_8 132$	32. $2 \cdot \log(y + 5) = \log 20 - \log 5$

33. $\log_2(9m + 2) = 7$

34. $5 \cdot \ln(2a - 1) = 15$

35. $64^{x+7} = 4^{5x-3}$

36. $9^{w-8} = \left(\frac{1}{27}\right)^{2w}$

37. $8^{n-5} = 48$

38. $2 \cdot 3^{4y} - 11 = 61$

39. $e^{a+1} = 65$

40. $-3 \cdot e^{2m-5} - 7 = -34$

Topic 6: Applications

Use the models below to answer questions 41-44.

Exponential Growth	Exponential Decay	Compound Interest
a = initial amount r = growth rate t = time	a = initial amount r = decay rate t = time (in years)	P = initial amount r = rate n = # of times compounded/year t = time (in years)
41. The population of an endangered species of birds is decreasing at a rate of 1.5% each year. There are currently 250,000 of these birds. Write a function to model this situation, then find the number of birds in 25 years.	42. A baseball card that was valued at \$200 in 1980 has increased in value by 7% each year. Write a function to model this situation, then find the value of the card in 2016.	
43. Miles invested \$2,400 into a retirement account that earns 1.8% interest compounded bimonthly. Write a function to model this situation, then find the balance of the account after 25 years.	44. Sarah moved \$30,000 of her savings to a new investment account that earns 4% interest compounded quarterly. Write a function to model this situation, then find the amount of interest the account will earn after 12 years.	
45. Once a car tire hits a nail, the pressure in pounds per square inch, psi, of the tire can be modeled by the function $f(t) = 32e^{-0.04t}$ where t is the number of minutes since the tire hit the nail. Find the air pressure of the tire after 45 minutes.	46. Jeremiah invested \$2,000 in shares of a certain stock. The function $f(t) = 2000e^{0.18t}$ represents the value of his investment, t years after his initial investment. Find the value of his investment after 5 years.	

Topic 7: Regression

47. The table below shows the value of a stock over the course of five years. Using an **exponential model**, write an equation for the curve of best fit, then estimate the value of the stock in 2025.

Year	Value (\$)
1998	400
1999	438
2000	480
2001	525
2002	575

48. The table below shows the wind speed along with the corresponding wind chill factor. Using a **logarithmic model**, write an equation for the curve of best fit, then find the approximate wind speed that corresponds to a wind chill factor of -18°F .

Wind Speed (mi/hr)	Wind Chill Factor ($^{\circ}\text{F}$)
3	1
5	-3
14	-8
20	-11
27	-14

49. The table below shows the number of seniors at Greenville High School that are also taking college credit classes during certain years. Which model would best fit this data: linear, or quadratic? Use the model to write a best-fit equation, then estimate the number of seniors taking college classes in 2020.

Year	Students
2004	18
2006	24
2008	39
2012	48
2015	64

50. The table below shows the number of used textbooks sold each year on an online bookstore. Which model would best fit this data: cubic, quartic, or exponential? Use the model to write a best-fit equation, then estimate the number of books sold in 2014.

Year	Books Sold (thousands)
2008	1
2009	6.2
2010	9.7
2011	12.1
2012	13.3

Name: _____

Unit 7 Test

Date: _____ Per: _____

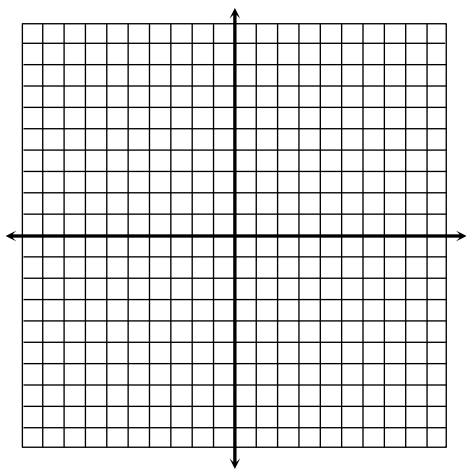
Exponential & Logarithmic Functions

1. Which functions represent an exponential growth? Check all that apply.

☐ $f(x) = 0.1 \cdot 8^x$	☐ $f(x) = \frac{4}{5} \cdot \left(\frac{9}{2}\right)^x$	☐ $f(x) = \frac{8}{5} \cdot \left(\frac{3}{4}\right)^x$
☐ $f(x) = 5 \cdot \left(\frac{1}{3}\right)^x$	☐ $f(x) = \frac{3}{8} \cdot 2^x$	☐ $f(x) = 16 \cdot 0.3^x$

Graph each function and identify its key characteristics.

2. $f(x) = \frac{1}{2} \cdot 2^{x+1} + 4$



Domain: _____

Range: _____

End Behavior:

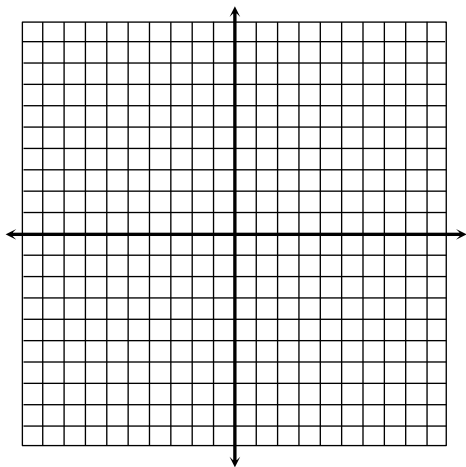
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

y-intercept: _____

Asymptote: _____

3. $f(x) = \log_3(x+7) - 1$



Domain: _____

Range: _____

End Behavior:

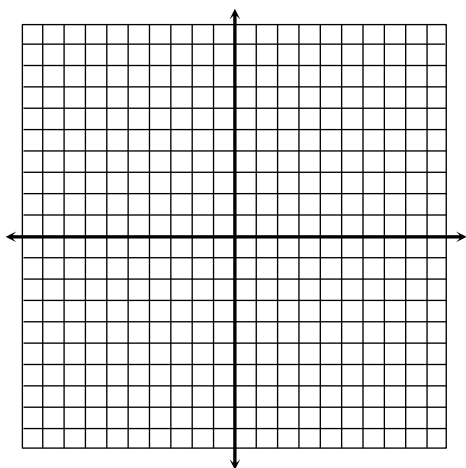
As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

4. $f(x) = \log_{\frac{1}{2}}(x+2) + 1$



Domain: _____

Range: _____

End Behavior:

As $x \rightarrow$ _____, $f(x) \rightarrow \infty$

As $x \rightarrow$ _____, $f(x) \rightarrow -\infty$

x-intercept: _____

Asymptote: _____

5. Which line represents the asymptote of the graph of the following function?

$$f(x) = 3^{x-4} + 1$$

- A. $x = -4$
- B. $x = 1$
- C. $y = -4$
- D. $y = 1$

6. What is the domain of the function given below?

$$f(x) = \log(x + 5) - 2$$

- A. $\{x \mid x > -5\}$
- B. $\{x \mid x > 5\}$
- C. $\{x \mid x > -2\}$
- D. $\{x \mid x > 2\}$

7. Which is equivalent to the equation given below?

$$7^x = 24$$

- A. $\log_7 x = 24$
- B. $\log_x 7 = 24$
- C. $\log_{24} 7 = x$
- D. $\log_7 24 = x$

8. Which is equivalent to the equation given below?

$$e^{x+3} = 60$$

- A. $\ln(x + 3) = 60$
- B. $\ln 60 = x + 3$
- C. $\log(x + 3) = 60$
- D. $\log 60 = x + 3$

9. Which is equivalent to the equation given below?

$$\log_2 18 = x - 1$$

- A. $2^{18} = x - 1$
- B. $2^{x-1} = 18$
- C. $18^2 = x - 1$
- D. $18^{x-1} = 2$

10. Which is equivalent to the equation given below?

$$\ln 5x = 4$$

- A. $e^{4x} = 5$
- B. $e^{5x} = 4$
- C. $e^5 = 4x$
- D. $e^4 = 5x$

11. The expression $\log_{36} 6$ is equivalent to which of the following?

- A. 2
- B. 6
- C. $\frac{1}{2}$
- D. $\frac{1}{6}$

12. Find the value of the expression below.

$$\log_3 3 + \log_2 64$$

- A. 5
- B. 7
- C. 9
- D. 33

13. Approximate the value of $\log_5 84$ to the nearest ten-thousandths place.

14. Which of the following have values less than 1? Check all that apply.

☐ $\log_6 39$	☐ $\log_2 50$
☐ $\log 24$	☐ $\log_{15} 1$
☐ $\log_7 3$	☐ $\ln 6$

15. Condense the expression below into a single logarithm.

$$\frac{1}{3} \cdot \log 64 + \log(x + 5)$$

16. Condense the expression below into a single logarithm. Do not evaluate.

$$7 \cdot \ln 3 - \frac{1}{2} \cdot \ln 81$$

17. Which expression is not equivalent to $\log_x 16$?

- A. $8 \cdot \log_x 2$
- B. $2 \cdot \log_x 4$
- C. $\log_x 8 + \log_x 2$
- D. $\log_x 432 - 3 \cdot \log_x 3$

18. Write a value in the box that would make the equation true.

$$\ln 9 = \ln 72 - \boxed{} \cdot \ln 2$$

19. Expand the logarithm below.

$$\log_5 (m^2 n^3)^4$$

20. Expand the logarithm below.

$$\log_2 \sqrt[3]{x^{12} y}$$

21. Which is equivalent to the expression below?

$$\log\left(\frac{b^3}{a}\right)$$

- A. $3 \cdot (\log b - \log a)$
- B. $\log 3b - \log a$
- C. $3 \cdot \log b - \log a$
- D. $3 \cdot \log b + \log a$

22. Which is equivalent to the expression below?

$$2 \cdot \log x - (3 \cdot \log y + \log y)$$

- A. $\log\left(\frac{x}{y}\right)^2$
- B. $\log\left(\frac{x^2}{y^4}\right)$
- C. $\log(xy)^2$
- D. $\log(x^2y^4)$

For questions 23-30, solve each equation. Round to the nearest ten-thousandths place when necessary.

23. $\log_2(6x - 15) = \log_2(41 - 2x)$

24. $\ln(k^2 - 1) = \ln 24 - \frac{1}{2} \cdot \ln 9$

25. $\log_3(p + 4) + \log_3(p - 2) = 3$

26. $4 \cdot \ln w + 6 = 12$

27. $\left(\frac{1}{8}\right)^{2-2n} = 16$

28. $3^{k-2} + 7 = 82$

29. $2 \cdot 10^{5-y} + 9 = 87$

30. $e^{5p-8} + 3 = 15$

31. Jack bought a new car in 2014 for \$28,000. If the value of the car decreases by 14% each year, write an exponential model for the value of the car. Then, estimate the year the car will have a value of \$5,000.

Equation:

Answer:

32. Logan inherited \$8,000 from his grandmother. He split the money evenly and placed the amounts in the two accounts below. Find the total amount of money he will have in the accounts combined after 5 years.

- **Account A:** earns 7% interest compounded quarterly
- **Account B:** earns 5.5% interest compounded annually

Answer:

33. A bacteria culture begins with 500 bacteria and is growing exponentially. The number of bacteria after t hours can be modeled by the function $f(t) = 500e^{0.9t}$. Find the number of bacteria after 4 hours.

Answer:

34. The table below shows the number of new restaurants in a fast-food chain that opened during the years of 1988 through 1992. Using an **exponential model**, write an equation for the curve of best fit, then estimate the number of new restaurants that opened in 2005.

Year	New Restaurants
1988	49
1989	81
1990	112
1991	150
1992	262

Equation:

Answer:

35. The table below shows the water temperature at certain depths of a lake. Using an **logarithmic model**, write an equation for the curve of best fit, then estimate the depth of water at a temperature of $50^{\circ} F$.

Depth (ft)	Temp ($^{\circ}F$)
2	86
12	78
14	75
45	71
68	68

Equation:

Answer:

36. The population (in thousands) of a town during certain years is given in the table below. Which model would best fit this data: linear, quadratic, or exponential? Choose the model that best fits the data, write an equation to model the data, then predict the approximate population in 2020.

Year	Population
1960	50.6
1970	55.5
1980	64.0
1990	80.3
2000	102.9

Model:

Equation:

Answer:

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

EXPONENTIAL
Parent Function

$$f(x) = b^x$$

- If $b > 1$, the function is an exponential growth and is increasing.
- If $b < 1$, the function is an exponential decay and is decreasing.

ASYMPTOTE

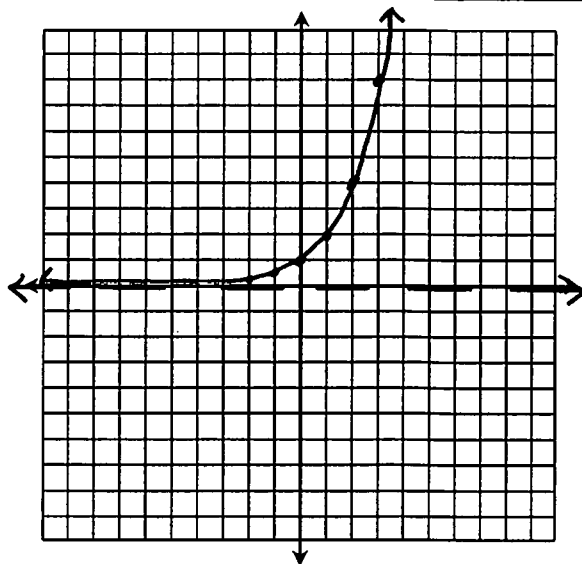
A line which the graph
Approaches.

Directions: Classify as an exponential growth or decay, graph, then identify its key characteristics.

1. $f(x) = 2^x$

X	y
-2	.25
-1	.5
0	1
1	2
2	4

growth

Domain: $\mathbb{R}$ Range: $y > 0$

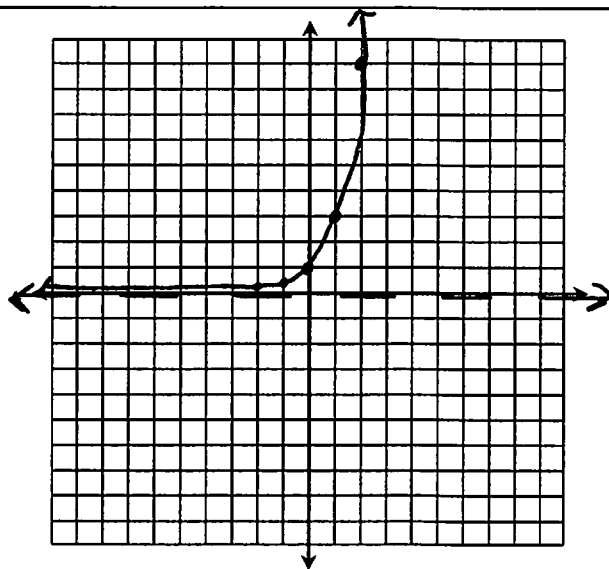
End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow 0$ y-intercept: (0, 1)Asymptote: $y = 0$

2. $f(x) = 3^x$

X	y
-2	$\frac{1}{9}$
-1	$\frac{1}{3}$
0	1
1	3
2	9

growth

Domain: $\mathbb{R}$ Range: $y > 0$

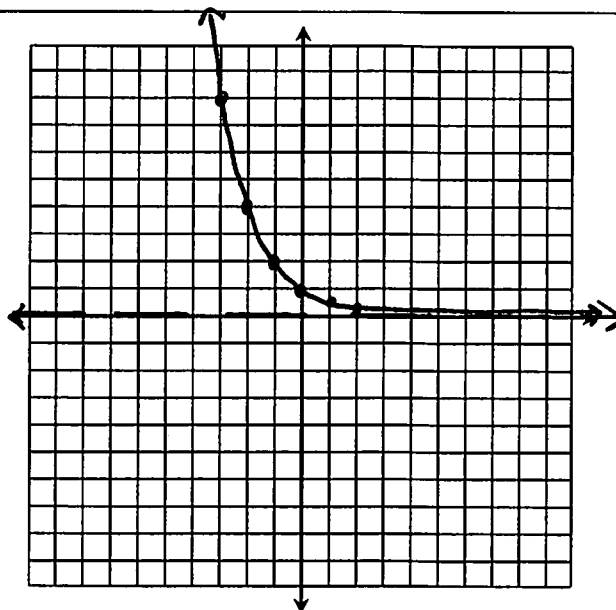
End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow 0$ y-intercept: (0, 1)Asymptote: $y = 0$

3. $f(x) = \left(\frac{1}{2}\right)^x$

X	y
-2	4
-1	2
0	1
1	.5
2	.25

decay



Domain: $\mathbb{R}$

Range: $y > 0$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow 0$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

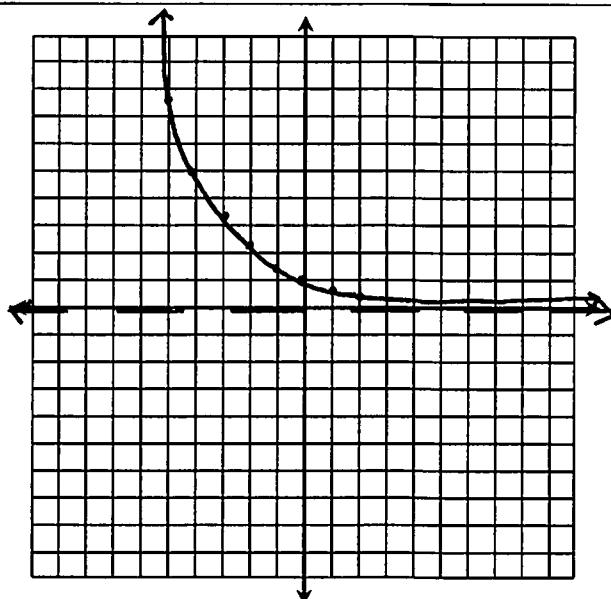
y-intercept: $(0, 1)$

Asymptote: $y = 0$

4. $f(x) = \left(\frac{2}{3}\right)^x$

X	y
-2	2.25
-1	1.5
0	1
1	0.6
2	0.4

decay



Domain: $\mathbb{R}$

Range: $y > 0$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow 0$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

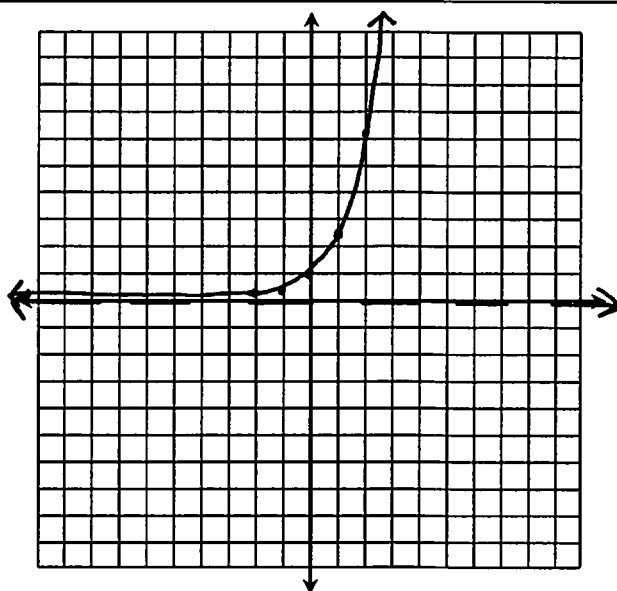
y-intercept: $(0, 1)$

Asymptote: $y = 0$

5. $f(x) = \left(\frac{5}{2}\right)^x$

X	y
-2	0.16
-1	0.4
0	1
1	2.5
2	6.25

growth



Domain: $\mathbb{R}$

Range: $y > 0$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow 0$

y-intercept: $(0, 1)$

Asymptote: $y = 0$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

TRANSFORMATIONS

of Exponential Functions

$$f(x) = a \cdot b^{x-h} + k$$

- h is the horizontal shift. (+ shifts left, - shifts right)
- k is the vertical shift. (+ shifts up, - shifts down)
- If a is negative, the function is reflection across the x-axis
- $|a| > 1$ represents a vertical stretch.
- $0 < |a| < 1$ represents a vertical compression.

Directions: (a) Identify the parent function, and (b) describe the transformations.

1. $f(x) = 3^x + 5$

a) $f(x) = 3^x$

b) shift up 5

2. $f(x) = 2 \cdot \left(\frac{1}{4}\right)^{x-1}$

a) $f(x) = \left(\frac{1}{4}\right)^x$

b) vertical stretch by 2,
shift right 1

3. $f(x) = -\left(\frac{4}{3}\right)^{x+2} + 7$

a) $f(x) = \left(\frac{4}{3}\right)^x$

b) reflect across x-axis,
shift left 2 and up 7

4. $f(x) = \frac{1}{2} \cdot 5^{x-4} - 2$

a) $f(x) = 5^x$

b) vertical compression by $\frac{1}{2}$,
shift right 4 and down 2

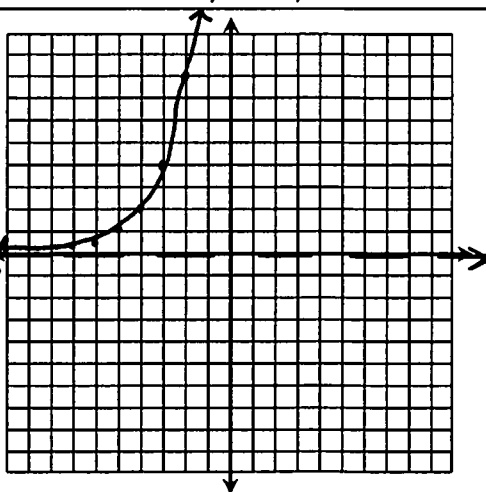
Directions: Graph each function and identify its key characteristics.

5. $f(x) = 2^{x+5}$

$f(x) = 2^x \rightarrow f(x) = 2^{x+5}$

x	y
-2	.25
-1	.5
0	1
1	2
2	4

x	y
-7	.25
-6	.5
-5	1
-4	2
-3	4

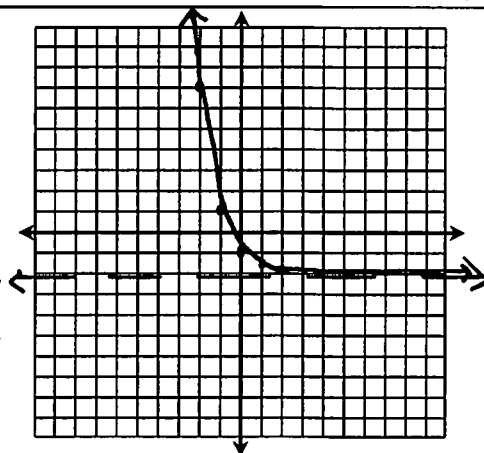
Domain: $\mathbb{R}$ Range: $y > 0$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow 0$ y-intercept: $2^{0+5} = 32 \rightarrow (0, 32)$ Asymptote: $y = 0$

6. $f(x) = \left(\frac{1}{3}\right)^x - 2$

X	y	→	X	y
-2	9		-2	7
-1	3		-1	1
0	1		0	-1
1	0.3		1	-1.6
2	0.1		2	-1.8



Domain: $\mathbb{R}$

Range: $y > -2$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -2$

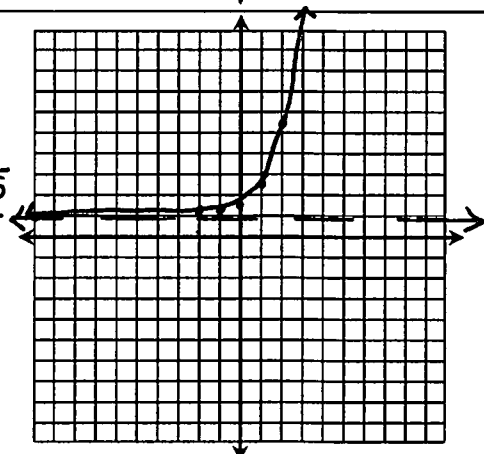
As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

y-intercept: $(0, -1)$

Asymptote: $y = -2$

7. $f(x) = \frac{1}{2} \cdot 3^x + 1$

X	y	→	X	y
-2	.1		-2	1.05
-1	.3		-1	1.16
0	1		0	1.5
1	3		1	2.5
2	9		2	5.5



Domain: $\mathbb{R}$

Range: $y > 1$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

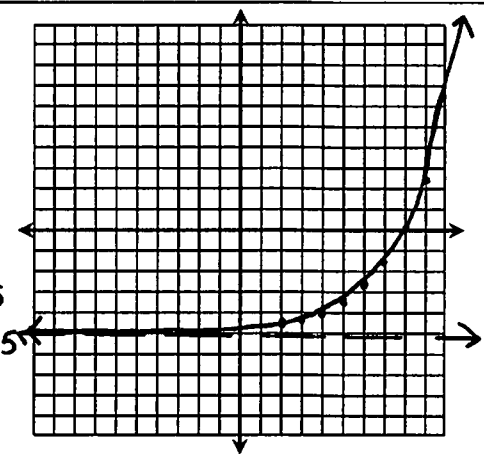
As $x \rightarrow -\infty$, $f(x) \rightarrow 1$

y-intercept: $(0, 1.5)$

Asymptote: $y = 1$

8. $f(x) = \left(\frac{3}{2}\right)^{x-4} - 5$

X	y	→	X	y
-2	0.4		2	-4.5
-1	0.6		3	-4.3
0	1		4	-4
1	1.5		5	-3.5
2	2.25		6	-2.75



Domain: $\mathbb{R}$

Range: $y > -5$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

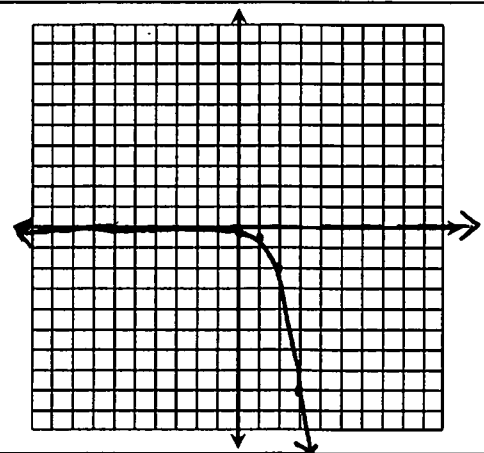
As $x \rightarrow -\infty$, $f(x) \rightarrow -5$

y-intercept: $(0, -4.802)$

Asymptote: $y = -5$

9. $f(x) = -2 \cdot 4^{x-2}$

X	y	→	X	y
-2	.06		0	-.125
-1	.25		1	-.5
0	1		2	-2
1	4		3	-8
2	16		4	-32



Domain: $\mathbb{R}$

Range: $y < 0$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow 0$

y-intercept: $(0, -0.125)$

Asymptote: $y = 0$

Name: _____

Unit 7: Exponential & Logarithmic Functions

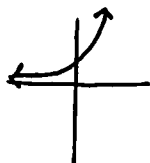
Date: _____ Per: _____

Homework 1: Graphing Exponential Functions

**** This is a 2-page document! ******Directions:** Classify as an exponential growth or an exponential decay. Then, sketch the curve.

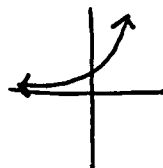
1. $f(x) = \frac{1}{2} \cdot 5^x$

growth



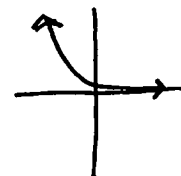
2. $f(x) = \left(\frac{6}{5}\right)^x$

growth



3. $f(x) = 4 \cdot \left(\frac{3}{8}\right)^x$

decay

**Directions:** (a) Identify the parent function and (b) describe the transformations.

4. $f(x) = \left(\frac{4}{5}\right)^{x+2}$

a) $f(x) = \left(\frac{4}{5}\right)^x$

b) shift left 2

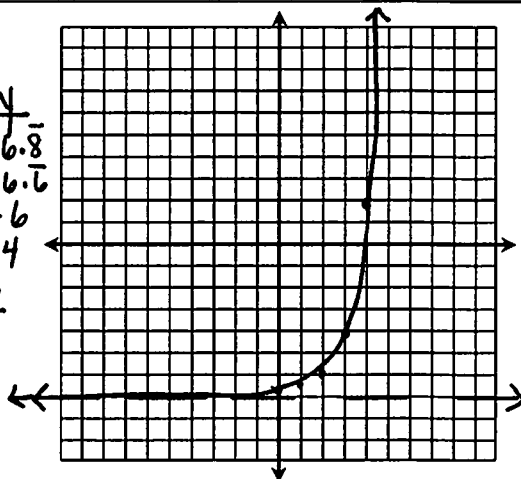
5. $f(x) = -3 \cdot 2^{x-1} + 7$

a) $f(x) = 2^x$

b) reflected across x-axis,
vertical stretch by 3,
shift up 7**Directions:** Graph each function and identify its key characteristics.

6. $f(x) = 3^{x-2} - 7$

X	Y	X	Y
-2	0.1	0	-6.8
-1	0.3	1	-6.6
0	1	2	-6
1	3	3	-4
2	9	4	2

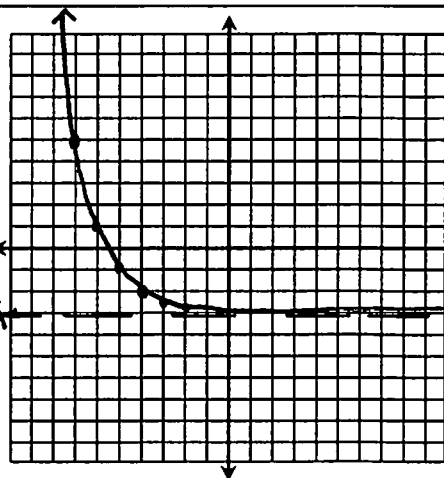
Domain: $\mathbb{R}$ Range: $y > -7$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow -7$ y-intercept: $(0, -6.8)$ Asymptote: $y = -7$

7. $f(x) = \left(\frac{1}{2}\right)^{x+4} - 3$

X	Y	X	Y
-2	4	-6	1
-1	2	-5	-1
0	1	-4	-2
1	.5	-3	-2.5
2	.25	-2	-2.75

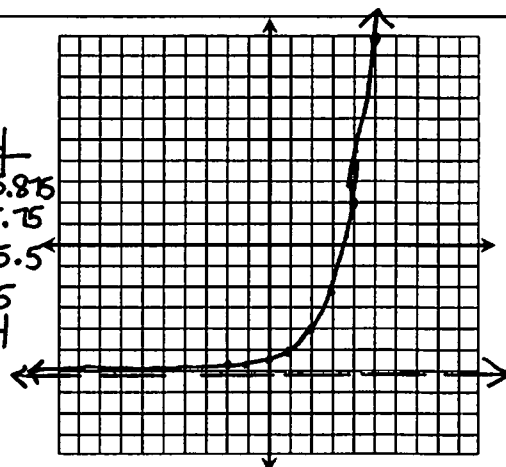
Domain: $\mathbb{R}$ Range: $y > -3$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -3$ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ y-intercept: $(0, -2.9375)$ Asymptote: $y = -3$

8. $f(x) = \frac{1}{2} \cdot 2^x - 6$

X	y	X	y
-2	.25	-2	-5.875
-1	.5	-1	-5.75
0	1	0	-5.5
1	2	1	-5
2	4	2	-4



Domain: $\mathbb{R}$

Range: $y > -6$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

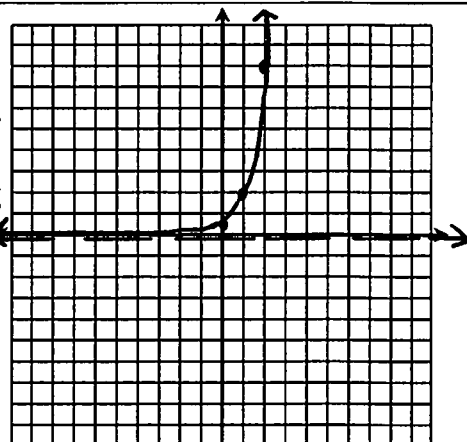
As $x \rightarrow -\infty$, $f(x) \rightarrow -6$

y-intercept: $(0, -5.5)$

Asymptote: $y = -6$

9. $f(x) = \frac{1}{8} \cdot 4^{x+1}$

X	y	X	y
-2	.06	-3	.008
-1	.25	-2	.03125
0	1	-1	.125
1	4	0	.5
2	16	1	2



Domain: $\mathbb{R}$

Range: $y > 0$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

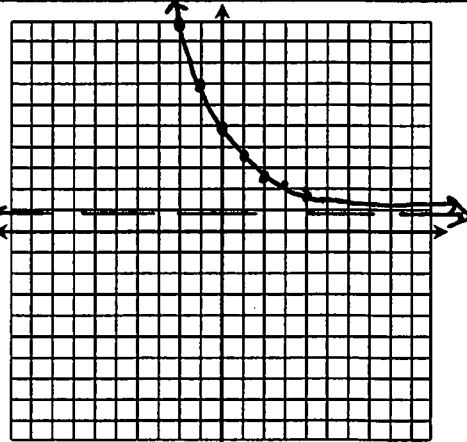
As $x \rightarrow -\infty$, $f(x) \rightarrow 0$

y-intercept: $(0, 0.5)$

Asymptote: $y = 0$

10. $f(x) = 4 \cdot \left(\frac{2}{3}\right)^x + 1$

X	y	X	y
-2	2.25	-2	10
-1	1.5	-1	7
0	1	0	5
1	0.6	1	3.6
2	0.4	2	2.7



Domain: $\mathbb{R}$

Range: $y > 1$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow 1$

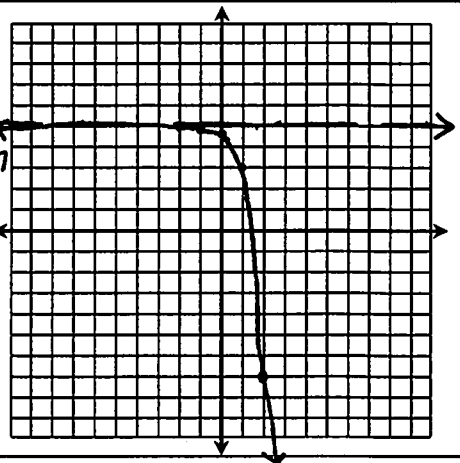
As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

y-intercept: $(0, 5)$

Asymptote: $y = 1$

11. $f(x) = -\frac{1}{3} \cdot 6^x + 5$

X	y	X	y
-2	.027	-2	4.9907
-1	.16	-1	4.94
0	1	0	4.6
1	6	1	3
2	36	2	-7



Domain: $\mathbb{R}$

Range: $y < 5$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow 5$

y-intercept: $(0, 4.6)$

Asymptote: $y = 5$

Name:		Date:	
Topic:		Class:	
Main Ideas/Questions		Notes/Examples	
Solving EXPONENTIAL EQUATIONS	①	Use the properties of exponents to SIMPLIFY each side of the equation.	
	②	Rewrite the equation so both sides have the SAME BASE .	
	③	Drop the bases and SET THE EXPONENTS EQUAL TO EACH OTHER .	
SET 1: WITH A COMMON BASE	Solve each equation.		
	1. $2^{x+1} = 2^9$ $x+1 = 9$ $x = 8$		2. $5^{4n+5} = 5^{n-7}$ $4n+5 = n-7$ $3n = -12$ $n = -4$
	3. $3^k \cdot 3^{k+2} = 3^{5k-1}$ $k+k+2 = 5k-1$ $2k+2 = 5k-1$ $3 = 3k$ $k = 1$		4. $10^{-4} \cdot 10^9 = 10^{v+4} \cdot 10^{2v-11}$ $-4+9 = v+4+2v-11$ $5 = 3v-7$ $12 = 3v$ $v = 4$
SET 2: WITHOUT A COMMON BASE	Solve each equation.		
	5. $6^{2x-10} = 36$ $6^{2x-10} = 6^2$ $2x-10 = 2$ $2x = 12$ $x = 6$		6. $2^{p-7} = 8$ $2^{p-7} = 2^3$ $p-7 = 3$ $p = 10$
	7. $7^{4x+11} = \frac{1}{7}$ $7^{4x+11} = 7^{-1}$ $4x+11 = -1$ $4x = -12$ $x = -3$		8. $32 = 2^{2m-9}$ $2^5 = 2^{2m-9}$ $5 = 2m-9$ $14 = 2m$ $m = 7$
	9. $27^{2x+6} = 3^{2x}$ $(3^3)^{2x+6} = 3^{2x}$ $3(2x+6) = 2x$ $6x+18 = 2x$ $18 = -4x$ $x = -\frac{9}{2}$		10. $4^{y+2} = 16^{y-3}$ $4^{y+2} = (4^2)^{y-3}$ $y+2 = 2(y-3)$ $y+2 = 2y-6$ $8 = y$

$$11. 125^y = 25$$

$$(5^3)^y = 5^2$$

$$3y = 2$$

$$y = \frac{2}{3}$$

$$12. 16^{3x} = 8^{x+2}$$

$$(2^4)^{3x} = (2^3)^{x+2}$$

$$12x = 3x + 6$$

$$9x = 6$$

$$x = \frac{2}{3}$$

$$13. 4^{3x} = 8^{x-1}$$

$$(2^2)^{3x} = (2^3)^{x-1}$$

$$6x = 3x - 3$$

$$3x = -3$$

$$x = -1$$

$$14. 81^{2x+5} = \left(\frac{1}{3}\right)^{2x}$$

$$(3^4)^{2x+5} = (3^{-1})^{2x}$$

$$8x + 20 = -2x$$

$$20 = -10x$$

$$x = -2$$

$$15. 8^{2a-1} = 32^{2a+1}$$

$$(2^3)^{2a-1} = (2^5)^{2a+1}$$

$$6a - 3 = 10a + 5$$

$$-8 = 4a$$

$$a = -2$$

$$16. 27^{2x} = 243^{x-2}$$

$$(3^3)^{2x} = (3^5)^{x-2}$$

$$6x = 5x - 10$$

$$x = -10$$

$$17. 64 = 4 \cdot 4^{4x}$$

$$4^3 = 4^1 \cdot 4^{4x}$$

$$3 = 1 + 4x$$

$$2 = 4x$$

$$x = \frac{1}{2}$$

$$18. 9^{2x+4} \cdot 9^{2x} = \frac{1}{81}$$

$$9^{2x+4} \cdot 9^{2x} = 9^{-2}$$

$$2x + 4 + 2x = -2$$

$$4x + 4 = -2$$

$$4x = -6$$

$$x = -\frac{3}{2}$$

$$19. \frac{1}{7} = 49^{x-5} \cdot 7^{x-9}$$

$$7^{-1} = (7^2)^{x-5} \cdot 7^{x-9}$$

$$-1 = 2x - 10 + x - 9$$

$$-1 = 3x - 19$$

$$18 = 3x$$

$$x = 6$$

$$20. 4^{2x} \cdot \frac{1}{16} = 4^{6x+18}$$

$$4^{2x} \cdot 4^{-2} = 4^{6x+18}$$

$$2x - 2 = 6x + 18$$

$$-20 = 4x$$

$$x = -5$$

Name: _____

Unit 7: Exponential & Logarithmic Functions

Date: _____ Per: _____

Homework 2: Solving Exponential Equations

**** This is a 2-page document! ******Directions:** Solve each equation using a common base.

1. $9^{3x-7} = 9^{5-x}$

$$3x-7 = 5-x$$

$$4x = 12$$

$$x = 3$$

2. $2^{w+4} \cdot 2^{4w+6} = 2^{2w+1}$

$$w+4 + 4w+6 = 2w+1$$

$$5w + 10 = 2w + 1$$

$$3w = -9$$

$$w = -3$$

3. $8^{6y+4} = 64$

$$8^{6y+4} = 8^2$$

$$6y+4 = 2$$

$$6y = -2$$

$$y = -\frac{1}{3}$$

4. $\frac{1}{5} = 5^{2c+3}$

$$5^{-1} = 5^{2c+3}$$

$$-1 = 2c+3$$

$$-4 = 2c$$

$$c = -2$$

5. $\frac{1}{27} = 3^{4m-1}$

$$3^{-3} = 3^{4m-1}$$

$$-3 = 4m-1$$

$$-2 = 4m$$

$$m = -\frac{1}{2}$$

6. $216 = 6^{2r-11}$

$$6^3 = 6^{2r-11}$$

$$3 = 2r-11$$

$$14 = 2r$$

$$r = 7$$

7. $2^{3k-1} \cdot 2^{5k-7} = 16$

$$2^{3k-1} \cdot 2^{5k-7} = 2^4$$

$$3k-1 + 5k-7 = 4$$

$$8k-8 = 4$$

$$8k = 12$$

$$k = \frac{3}{2}$$

8. $4^n \cdot 4^{2n-9} = 64$

$$4^n \cdot 4^{2n-9} = 4^3$$

$$n + 2n - 9 = 3$$

$$3n - 9 = 3$$

$$3n = 12$$

$$n = 4$$

$$9. 8^{x+2} = 4$$

$$(2^3)^{x+2} = 2^2$$

$$3x+6 = 2$$

$$3x = -4$$

$$\boxed{x = -\frac{4}{3}}$$

$$10. 125 = 25^{2h+1}$$

$$5^3 = (5^2)^{2h+1}$$

$$3 = 4h+2$$

$$1 = 4h$$

$$\boxed{h = \frac{1}{4}}$$

$$11. 49^{p+1} = 343^{2p}$$

$$(7^2)^{p+1} = (7^3)^{2p}$$

$$2p+2 = 6p$$

$$2 = 4p$$

$$\boxed{\frac{1}{2} = p}$$

$$12. 16^{r-2} = 64^{r+2}$$

$$(4^2)^{r-2} = (4^3)^{r+2}$$

$$2r-4 = 3r+6$$

$$\boxed{-10 = r}$$

$$13. 27^{3n} = 81^{2n+1}$$

$$(3^3)^{3n} = (3^4)^{2n+1}$$

$$9n = 8n+4$$

$$\boxed{n = 4}$$

$$14. \left(\frac{1}{4}\right)^{2x} = 32^{4x-2}$$

$$(2^{-2})^{2x} = (2^5)^{4x-2}$$

$$-4x = 20x-10$$

$$-24x = -10$$

$$\boxed{x = \frac{5}{12}}$$

$$15. 16 \cdot 2^{6m} = 2^{3m-8}$$

$$2^4 \cdot 2^{6m} = 2^{3m-8}$$

$$4+6m = 3m-8$$

$$3m = -12$$

$$\boxed{m = -4}$$

$$16. \left(\frac{1}{9}\right)^a \cdot \left(\frac{1}{3}\right)^a = 3^{16-a}$$

$$(3^{-2})^a \cdot (3^{-1})^a = 3^{16-a}$$

$$-2a + (-a) = 16-a$$

$$-2a = 16$$

$$\boxed{a = -8}$$

$$17. 256^y \cdot 16^{y-1} = 4^{2y-22}$$

$$(4^4)^y \cdot (4^2)^{y-1} = 4^{2y-22}$$

$$4y + 2y-2 = 2y-22$$

$$6y-2 = 2y-22$$

$$4y = -20$$

$$\boxed{y = -5}$$

$$18. 36^{n-3} \cdot 216^n = 216^{2n+1}$$

$$(6^2)^{n-3} \cdot (6^3)^n = (6^3)^{2n+1}$$

$$2n-6+3n = 6n+3$$

$$5n-6 = 6n+3$$

$$\boxed{-9 = n}$$

Name: _____ Algebra 2

Date: _____ Per: _____ Unit 7: Exponential & Logarithmic Functions

Quiz 7-1: Exponential Functions & Equations

For questions 1-3, indicate whether the function represents an exponential growth or decay.

1. $f(x) = \left(\frac{3}{4}\right)^x$

2. $f(x) = \frac{1}{6} \cdot 4^x$

3. $f(x) = \frac{1}{4} \cdot \left(\frac{5}{2}\right)^x$

1. decay

2. growth

3. growth

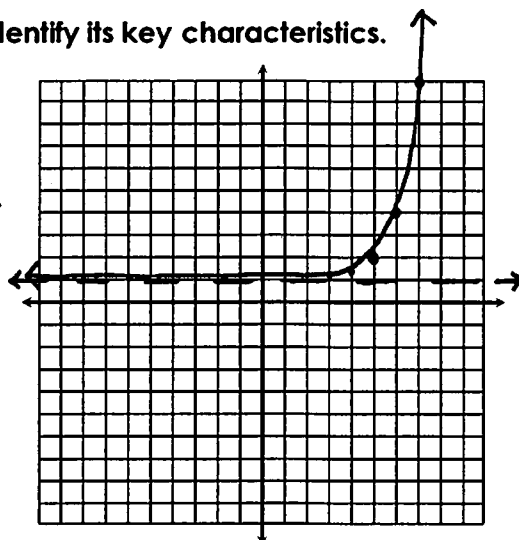
4. The parent function $f(x) = 2^x$ is reflected across the x -axis, then shifted 2 units left and 6 units down. Write an equation to represent the new function.

4. $f(x) = -(2)^{x+2} - 6$

Graph each function and identify its key characteristics.

5. $f(x) = 3^{x-5} + 1$

X	y	X	y
-2	0.1	3	1.1
-1	0.3	4	1.3
0	1	5	2
1	3	6	4
2	9	7	10



Domain: $\mathbb{R}$

Range: $y > 1$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

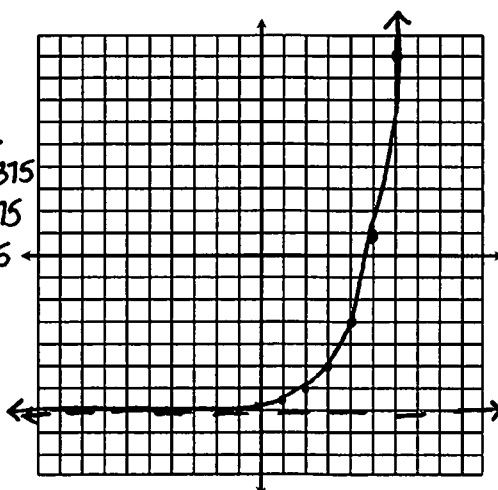
As $x \rightarrow -\infty$, $f(x) \rightarrow 1$

y-intercept: $(0, 1.0041)$

Asymptote: $y = 1$

6. $f(x) = \frac{1}{4} \cdot 2^x - 7$

X	y	X	y
-2	.25	-2	-6.9375
-1	.5	-1	-6.875
0	1	0	-6.75
1	2	1	-6.5
2	4	2	-6
		3	-5
		4	-3



Domain: $\mathbb{R}$

Range: $y > -7$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

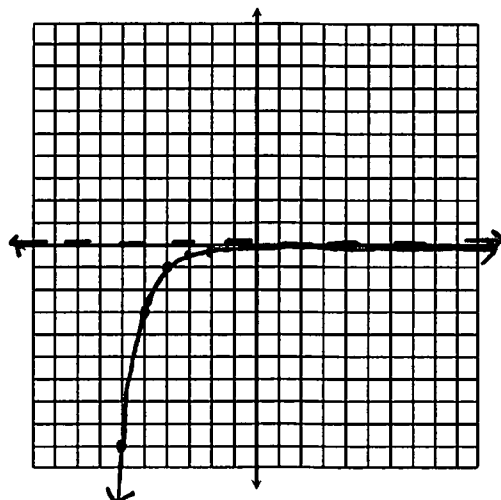
As $x \rightarrow -\infty$, $f(x) \rightarrow -7$

y-intercept: $(0, -6.75)$

Asymptote: $y = -7$

7. $f(x) = -\left(\frac{1}{3}\right)^{x+4}$

X	y
-6	-9
-5	-3
-4	-1
-3	-0.3
-2	-0.1



Domain: $\mathbb{R}$

Range: $y < 0$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow 0$

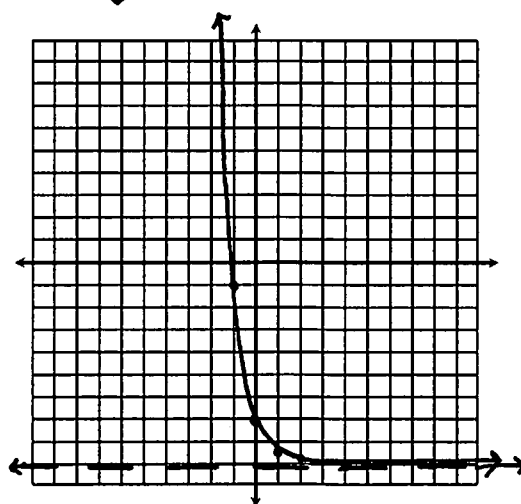
As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

y-intercept: $(0, -0.0123)$

Asymptote: $y = 0$

8. $f(x) = 2\left(\frac{1}{4}\right)^x - 9$

X	y
-2	23
-1	-1
0	-7
1	-8.5
2	-8.875



Domain: $\mathbb{R}$

Range: $y > -9$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -9$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

y-intercept: $(0, -7)$

Asymptote: $y = -9$

Solve each equation.

9. $5^{3x-1} \cdot 5^{2x-5} = 5^{x+6}$

$$3x-1 + 2x-5 = x+6$$

$$5x-6 = x+6$$

$$4x = 12$$

$$x = 3$$

10. $64^x = 8^{3x+1}$

$$(8^2)^x = 8^{3x+1}$$

$$2x = 3x+1$$

$$-x = 1$$

$$x = -1$$

9. $x = 3$

10. $x = -1$

11. $x = -5/2$

12. $x = -16$

11. $\frac{1}{216} = 36^{3x+6}$

$$6^{-3} = (6^2)^{3x+6}$$

$$-3 = 6x+12$$

$$-15 = 6x$$

$$-\frac{5}{2} = x$$

12. $32^{x+6} \cdot \frac{1}{2} = 8^{x-1}$

$$(2^5)^{x+6} \cdot 2^{-1} = (2^3)^{x-1}$$

$$5x+30 + (-1) = 3x-3$$

$$5x+29 = 3x-3$$

$$2x = -32$$

$$x = -16$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples						
What is a LOGARITHM?	A logarithm (log) is another way of writing exponents. Logarithmic Form $\log_b a = x$ → Exponential Form $b^x = a$ ↖ Read as "log base <i>b</i> of <i>a</i> equals <i>x</i>."						
Converting LOG ⇌ EXP	Directions: Write each equation in exponential form. <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px; vertical-align: top;"> 1. $\log_3 9 = 2$ $3^2 = 9$ </td> <td style="width: 50%; padding: 5px; vertical-align: top;"> 2. $\log_6 216 = 3$ $6^3 = 216$ </td> </tr> <tr> <td style="padding: 5px; vertical-align: top;"> 3. $\log_7 1 = 0$ $7^0 = 1$ </td> <td style="padding: 5px; vertical-align: top;"> 4. $\log_2 16 = 4$ $2^4 = 16$ </td> </tr> <tr> <td style="padding: 5px; vertical-align: top;"> 5. $\log_4 \frac{1}{16} = -2$ $4^{-2} = \frac{1}{16}$ </td> <td style="padding: 5px; vertical-align: top;"> 6. $\log_9 27 = \frac{3}{2}$ $9^{\frac{3}{2}} = 27$ </td> </tr> </table>	1. $\log_3 9 = 2$ $3^2 = 9$	2. $\log_6 216 = 3$ $6^3 = 216$	3. $\log_7 1 = 0$ $7^0 = 1$	4. $\log_2 16 = 4$ $2^4 = 16$	5. $\log_4 \frac{1}{16} = -2$ $4^{-2} = \frac{1}{16}$	6. $\log_9 27 = \frac{3}{2}$ $9^{\frac{3}{2}} = 27$
1. $\log_3 9 = 2$ $3^2 = 9$	2. $\log_6 216 = 3$ $6^3 = 216$						
3. $\log_7 1 = 0$ $7^0 = 1$	4. $\log_2 16 = 4$ $2^4 = 16$						
5. $\log_4 \frac{1}{16} = -2$ $4^{-2} = \frac{1}{16}$	6. $\log_9 27 = \frac{3}{2}$ $9^{\frac{3}{2}} = 27$						
Converting EXP ⇌ LOG	Directions: Write each equation in logarithmic form. <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 50%; padding: 5px; vertical-align: top;"> 7. $14^2 = 196$ $\log_{14} 196 = 2$ </td> <td style="width: 50%; padding: 5px; vertical-align: top;"> 8. $3^4 = 81$ $\log_3 81 = 4$ </td> </tr> <tr> <td style="padding: 5px; vertical-align: top;"> 9. $12^1 = 12$ $\log_{12} 12 = 1$ </td> <td style="padding: 5px; vertical-align: top;"> 10. $36^{\frac{1}{2}} = 6$ $\log_{36} 6 = \frac{1}{2}$ </td> </tr> <tr> <td style="padding: 5px; vertical-align: top;"> 11. $2^{-3} = \frac{1}{8}$ $\log_2 \frac{1}{8} = -3$ </td> <td style="padding: 5px; vertical-align: top;"> 12. $8^{\frac{4}{3}} = 16$ $\log_8 16 = \frac{4}{3}$ </td> </tr> </table>	7. $14^2 = 196$ $\log_{14} 196 = 2$	8. $3^4 = 81$ $\log_3 81 = 4$	9. $12^1 = 12$ $\log_{12} 12 = 1$	10. $36^{\frac{1}{2}} = 6$ $\log_{36} 6 = \frac{1}{2}$	11. $2^{-3} = \frac{1}{8}$ $\log_2 \frac{1}{8} = -3$	12. $8^{\frac{4}{3}} = 16$ $\log_8 16 = \frac{4}{3}$
7. $14^2 = 196$ $\log_{14} 196 = 2$	8. $3^4 = 81$ $\log_3 81 = 4$						
9. $12^1 = 12$ $\log_{12} 12 = 1$	10. $36^{\frac{1}{2}} = 6$ $\log_{36} 6 = \frac{1}{2}$						
11. $2^{-3} = \frac{1}{8}$ $\log_2 \frac{1}{8} = -3$	12. $8^{\frac{4}{3}} = 16$ $\log_8 16 = \frac{4}{3}$						

COMMON LOGARITHM	A logarithm with base 10 is called a common logarithm and can be written without the base. $\log_{10} x \rightarrow \log x$	
EVALUATING LOGARITHMS	Directions: Use your knowledge of exponents to evaluate the following logarithms.	
	13. $\log_7 49$ $7^x = 49$ $x = 2$	14. $\log_3 27$ $3^x = 27$ $x = 3$
	15. $\log_{10} 100$ $10^x = 100$ $x = 2$	16. $\log_{12} 1$ $12^x = 1$ $x = 0$
	17. $\log_2 64$ $2^x = 64$ $x = 6$	18. $\log_3 243$ $3^x = 243$ $x = 5$
	19. $\log_9 \frac{1}{81}$ $9^x = \frac{1}{81}$ $x = -2$	20. $\log_{64} 4$ $64^x = 4$ $x = \frac{1}{3}$
CHANGE OF BASE FORMULA Choose BASE 10 because there is a calculator button for it!	Some logarithms are not as easy to evaluate as those above, and will require the change of base formula. $\log_b a = \frac{\log a}{\log b}$	
	Directions: Evaluate each log using the change of base formula.	
	21. $\log_{16} 64$ $\frac{\log 64}{\log 16} = 1.5$	22. $\log_8 32$ $\frac{\log 32}{\log 8} = 1.\overline{6}$
	23. $\log_2 54$ $\frac{\log 54}{\log 2} = 5.7549$	24. $\log_{10} 294$ $\frac{\log 294}{\log 10} = 2.4683$
	25. $\log_4 136$ $\frac{\log 136}{\log 4} = 3.5437$	26. $\log_6 \frac{1}{36}$ $\frac{\log \frac{1}{36}}{\log 6} = -2$

Name: _____

Unit 7: Exponential & Logarithmic Functions

Date: _____ Per: _____

Homework 3: Intro to Logarithms



Directions: Write each equation in exponential form.		
1. $\log_2 128 = 7$ $2^7 = 128$	2. $\log_8 64 = 2$ $8^2 = 64$	3. $\log_3 \frac{1}{27} = -3$ $3^{-3} = \frac{1}{27}$
Directions: Write each equation in logarithmic form.		
4. $4^4 = 256$ $\log_4 256 = 4$	5. $8^3 = 512$ $\log_8 512 = 3$	6. $27^{\frac{2}{3}} = 9$ $\log_{27} 9 = \frac{2}{3}$
Directions: Evaluate each logarithm. Use the change of base formula when necessary.		
7. $\log_6 36$ $6^x = 36$ $x = 2$	8. $\log_2 32$ $2^x = 32$ $x = 5$	9. $\log_4 64$ $4^x = 64$ $x = 3$
10. $\log_3 81$ $3^x = 81$ $x = 4$	11. $\log_{100} 10$ $100^x = 10$ $x = \frac{1}{2}$	12. $\log_7 \frac{1}{7}$ $7^x = \frac{1}{7}$ $x = -1$
13. $\log_{18} 1$ $18^x = 1$ $x = 0$	14. $\log_2 \frac{1}{16}$ $2^x = \frac{1}{16}$ $x = -4$	15. $\log 1000$ $10^x = 1000$ $x = 3$
16. $\log_{16} 8$ $\frac{\log 8}{\log 16} = 0.75$	17. $\log_{243} 27$ $\frac{\log 27}{\log 243} = 0.6$	18. $\log_3 92$ $\frac{\log 92}{\log 3} = 4.1159$
19. $\log_7 35$ $\frac{\log 35}{\log 7} = 1.8271$	20. $\log_2 260$ $\frac{\log 260}{\log 2} = 8.0224$	21. $\log_5 38$ $\frac{\log 38}{\log 5} = 2.2602$

Name:		Date:	
Topic:		Class:	
Main Ideas/Questions			
Notes/Examples			
PRODUCT Property $\log_b(m \cdot n) = \log_b m + \log_b n$ Answers may vary →	Condense into a single logarithm. Simplify if possible.		
	1. $\log_2 7 + \log_2 4$ $\log_2 7 \cdot 4$ $= \boxed{\log_2 28}$	2. $\log 25 + \log 4$ $\log 25 \cdot 4$ $= \boxed{\log 100}$	3. $\log_4 2x + \log_4 4x^2$ $\log_4 2x \cdot 4x^2$ $= \boxed{\log_4 8x^3}$
	Expand using the product property.		
QUOTIENT Property $\log_b\left(\frac{m}{n}\right) = \log_b m - \log_b n$ Answers may vary →	4. $\log 6$ $\log 2 \cdot 3$ $= \boxed{\log 2 + \log 3}$	5. $\log_7 45$ $\log_7 5 \cdot 9$ $= \boxed{\log_7 5 + \log_7 9}$	6. $\log_2 (5x)$ $\log_2 5 \cdot x$ $= \boxed{\log_2 5 + \log_2 x}$
	Condense into a single logarithm. Simplify if possible.		
	7. $\log_3 24 - \log_3 8$ $\log_3 \frac{24}{8}$ $= \boxed{\log_3 3}$	8. $\log_2 15 - \log_2 15$ $\log_2 \frac{15}{15}$ $= \boxed{\log_2 1}$	9. $\log_4 x^9 - \log_4 x^2$ $\log_4 \frac{x^9}{x^2}$ $= \boxed{\log_4 x^7}$
POWER Property $\log_b m^n = n \cdot \log_b m$	Expand using the quotient property.		
	10. $\log_8 4$ $\log_8 \frac{8}{2}$ $= \boxed{\log_8 8 - \log_8 2}$	11. $\log_5 \frac{1}{3}$ $\log_5 \frac{15}{45}$ $= \boxed{\log_5 15 - \log_5 45}$	12. $\log\left(\frac{m}{7}\right)$ $= \boxed{\log m - \log 7}$
	Condense into a single logarithm. Simplify if possible.		
POWER Property $\log_b m^n = n \cdot \log_b m$	13. $5 \cdot \log_4 2$ $\log_4 2^5$ $= \boxed{\log_4 32}$	14. $7 \cdot \log_2 x$ $\boxed{\log_2 x^7}$	15. $\frac{1}{3} \cdot \log 8$ $\log 8^{1/3}$ $= \boxed{\log 2}$
	Expand using the power property. Simplify if possible.		
	16. $\log_2 8^7$ $\boxed{7 \cdot \log_2 8}$	17. $3 \cdot \log 4^{x-1}$ $(x-1) \cdot 3 \cdot \log 4$ $= \boxed{(3x-3) \cdot \log 4}$	18. $\log_7 \sqrt{w}$ $\log_7 w^{1/2}$ $= \boxed{\frac{1}{2} \cdot \log_7 w}$

PUTTING IT ALL TOGETHER!

CONDENSING LOGS

Rewrite as a single logarithm. Simplify if possible.

19. $2 \cdot \log_6 6 - \log_6 9$

$$\log_6 \frac{6^2}{9} = \log_6 \frac{36}{9} = \boxed{\log_6 4}$$

20. $4 \cdot \log_4 a + 2 \cdot \log_4 b$

$$\log_4 a^4 + \log_4 b^2 = \boxed{\log_4 a^4 b^2}$$

21. $7 \cdot \log_4 u - 3 \cdot \log_4 v^2$

$$\log_4 u^7 - \log_4 (v^2)^3 = \boxed{\log_4 \frac{u^7}{v^6}}$$

22. $\log_2 15 + \log_2 4 - \log_2 6$

$$\log_2 \frac{15 \cdot 4}{6} = \boxed{\log_2 10}$$

23. $\log_3 4 + \log_3 y + \frac{1}{2} \cdot \log_3 49$

$$\log_3 4 \cdot y \cdot 49^{1/2} = \boxed{\log_3 28y}$$

24. $\frac{1}{3}(\log_5 8 + \log_5 27) - \log_5 3$

$$\log_5 \frac{8^{1/3} \cdot 27^{1/3}}{3} = \log_5 \frac{2 \cdot 3}{3} = \boxed{\log_5 2}$$

25. $3 \cdot \log_2 4 - \log_2 32$

$$\log_2 \frac{4^3}{32} = \log_2 \frac{64}{32} = \boxed{\log_2 2}$$

26. $2 \cdot \log_6 6 - \frac{1}{4} \cdot \log_6 16 + \log_6 3$

$$\log_6 \frac{6^2}{16^{1/4}} + \log_6 3 = \log_6 \frac{36}{2} + \log_6 3 = \boxed{\log_6 54}$$

EXPANDING LOGS

Expand each logarithm completely.

27. $\log_6 (xyz^4)$

$$\log_6 x + \log_6 y + \log_6 z^4 = \boxed{\log_6 x + \log_6 y + 4 \cdot \log_6 z}$$

28. $\log_4 \left(\frac{a^9}{b} \right)$

$$\log_4 a^9 - \log_4 b = \boxed{9 \cdot \log_4 a - \log_4 b}$$

29. $\log_7 (q^4 r^2)^2$

$$2 \cdot (\log_7 q^4 + \log_7 r^2) = 2 \cdot (4 \cdot \log_7 q + 2 \cdot \log_7 r) = \boxed{8 \log_7 q + 4 \log_7 r}$$

30. $\log_2 \left(\frac{y}{z^5} \right)^2 = 2 \cdot (\log_2 y - \log_2 z^5)$

$$= 2 \cdot (\log_2 y - 5 \cdot \log_2 z) = \boxed{2 \log_2 y - 10 \log_2 z}$$

31. $\log \sqrt{7x^3} = \log (7x^3)^{1/2}$

$$= \frac{1}{2} \cdot (\log 7 + \log x^3) = \frac{1}{2} \cdot (\log 7 + 3 \cdot \log x) = \boxed{\frac{1}{2} \log 7 + \frac{3}{2} \log x}$$

32. $\log_3 \sqrt[4]{m^5 n^2} = (\log_3 m^5 + \log_3 n^2)^{1/4}$

$$= \frac{1}{4} \cdot (5 \cdot \log_3 m + 2 \cdot \log_3 n) = \boxed{\frac{5}{4} \log_3 m + \frac{1}{2} \log_3 n}$$

PROPERTIES OF LOGARITHMS

Graphic Organizer

Name	Rule(s)	Example 1	Example 2
BASIC LOGARITHMS	$\log_b b = 1$; $\log_b 1 = 0$	Simplify: $\log_4 4 = 1$	Simplify: $\log_3 1 = 0$
PRODUCT RULE	$\log_b (m \cdot n) = \log_b m + \log_b n$	Condense: $\log_5 6 + \log_5 7 =$ $\log_5 6 \cdot 7$ $= \boxed{\log_5 42}$	Expand: $\log_2 63 =$ $\log_2 7 \cdot 9$ $= \boxed{\log_2 7 + \log_2 9}$
QUOTIENT RULE	$\log_b \left(\frac{m}{n} \right) = \log_b m - \log_b n$	Condense: $\log_4 84 - \log_4 12 =$ $\log_4 \frac{84}{12}$ $= \boxed{\log_4 7}$	Expand: $\log 9 =$ $\log \frac{81}{9}$ $= \boxed{\log 81 - \log 9}$
POWER RULE	$\log_b m^n = n \cdot \log_b m$	Condense: $2 \cdot \log_3 8 =$ $\log_3 8^2$ $= \boxed{\log_3 64}$	Expand: $\log_2 6^{x-1} =$ $\boxed{(x-1) \cdot \log_2 6}$
CHANGE OF BASE FORMULA	$\log_b a = \frac{\log a}{\log b}$	Using a common base, evaluate the expression below. $\log_7 32 = \frac{\log 32}{\log 7} = \boxed{1.7810}$	

REMEMBER: BASE 10 LOGS ARE **COMMON LOGS** AND WRITTEN WITHOUT A BASE! ($\log x$)

Name: _____

Unit 7: Exponential & Logarithmic Functions

Date: _____ Per: _____

Homework 4: Properties of Logarithms

**** This is a 2-page document! ******Directions:** Complete each rule.**PRODUCT RULE**

$$\log_b(m \cdot n) = \log_b m + \log_b n$$

QUOTIENT RULE

$$\log_b\left(\frac{m}{n}\right) = \log_b m - \log_b n$$

POWER RULE

$$\log_b m^n = n \cdot \log_b m$$

Directions: Condense each expression into a single logarithm. Simplify if possible.

1. $\log_7 9 + \log_7 4$

$$\log_7 9 \cdot 4 = \boxed{\log_7 36}$$

2. $\log_2 80 - \log_2 5$

$$\log_2 \frac{80}{5} = \boxed{\log_2 16}$$

3. $\frac{1}{2} \cdot \log_3 81$

$$\log_3 81^{1/2} = \log_3 \sqrt{81} = \boxed{\log_3 9}$$

4. $3 \cdot \log_4 8 - 5 \cdot \log_4 2$

$$\log_4 \frac{8^3}{2^5} = \log_4 \frac{512}{32} = \boxed{\log_4 16}$$

5. $7 \cdot \log_5 x + 3 \cdot \log_5 y^4$

$$\log_5 x^7 \cdot (y^4)^3 = \boxed{\log_5 x^7 y^{12}}$$

6. $\log_3 a^7 + (\log_3 a^4 - 2 \cdot \log_3 b)$

$$\log_3 a^7 \cdot \frac{a^4}{b^2} = \boxed{\log_3 \frac{a^{11}}{b^2}}$$

7. $\log_4 x^7 - \log_4 x^2 + 4 \cdot \log_4 x^3$

$$\log_4 \frac{x^7}{x^2} \cdot (x^3)^4 = \log_4 x^5 \cdot x^{12} = \boxed{\log_4 x^{17}}$$

8. $\frac{1}{2}(\log_6 45 - \log_6 5) + \log_6 12$

$$\log_6 \left(\frac{45}{5}\right)^{1/2} \cdot 12 = \log_6 \sqrt{9} \cdot 12 = \boxed{\log_6 36}$$

Directions: Condense, then use the **change of base formula** to evaluate the logarithm.

9. $\log_9 35 - \log_9 7$

$$\log_9 \frac{35}{7}$$

$$= \log_9 5 \rightarrow \frac{\log 5}{\log 9}$$

$$= \boxed{0.7325}$$

10. $2 \cdot \log_3 8 - 4 \cdot \log_3 2$

$$\log_3 \frac{8^2}{2^4}$$

$$= \log_3 \frac{64}{16}$$

$$= \log_3 4$$

$$\frac{\log 4}{\log 3}$$

$$= \boxed{1.2619}$$

11. $\frac{1}{3} \cdot \log_4 8 + \log_4 15$

$$\log_4 8^{1/3} \cdot 15$$

$$= \log_4 2 \cdot 15$$

$$= \log_4 30$$

$$\frac{\log 30}{\log 4}$$

$$= \boxed{2.4534}$$

12. $\log_2 27 + \log_2 4 - 2 \cdot \log_2 3$

$$\log_2 \frac{27 \cdot 4}{3^2}$$

$$= \log_2 \frac{108}{9}$$

$$= \log_2 12$$

$$\frac{\log 12}{\log 2}$$

$$= \boxed{3.5850}$$

Directions: Expand each logarithm completely.

13. $\log_8(mn^2)$

$$\boxed{\log_8 m + 2 \cdot \log_8 n}$$

14. $\log\left(\frac{x^9}{y^4}\right)$

$$\boxed{9 \cdot \log x - 4 \cdot \log y}$$

15. $\log_2(a^5 b^2)^3$

$$\log_2 a^{15} b^6$$

$$= \boxed{15 \cdot \log_2 a + 6 \cdot \log_2 b}$$

16. $\log\left(\frac{p^4}{q^7}\right)^2$

$$\log\left(\frac{p^8}{q^{14}}\right)$$

$$= \boxed{8 \cdot \log p - 14 \cdot \log q}$$

17. $\log_5 \sqrt{a^7 b}$

$$\log_5 (a^7 b)^{1/2}$$

$$= \log_5 a^{7/2} + \log_5 b^{1/2}$$

$$= \boxed{\frac{7}{2} \log_5 a + \frac{1}{2} \log_5 b}$$

18. $\log_4 \sqrt[3]{c^2 d^{15}}$

$$\log_4 (c^2 d^{15})^{1/3}$$

$$= \log_4 c^{2/3} + \log_4 d^5$$

$$= \boxed{\frac{2}{3} \cdot \log_4 c + 5 \cdot \log_4 d}$$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

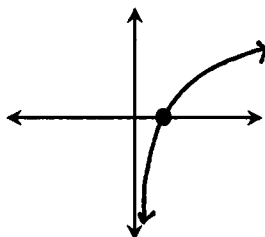
LOGARITHMIC

Parent Function

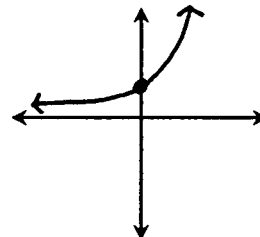
$$f(x) = \log x$$

A logarithmic function is the **inverse** of an exponential function.
Using your graphing calculator, sketch the following graphs:

$$f(x) = \log x$$



$$f(x) = 10^x$$



Because you can only graph base 10 logs on your calculator, you will need to use the inverse exponential function, then invert the values from the table to graph the logarithmic function.

Directions: Graph each function and identify its key characteristics.

1. $f(x) = \log_2 x$

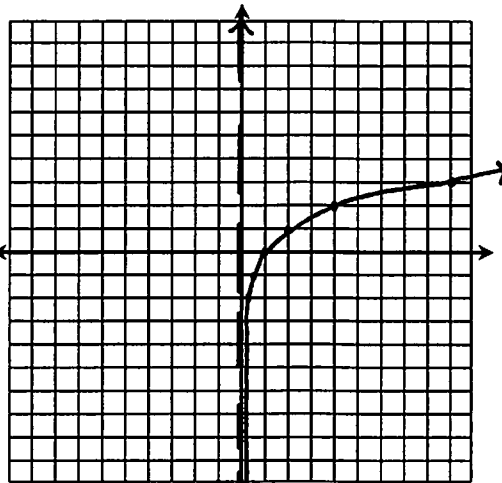
$$f(x) = 2^x$$

X	Y
-2	.25
-1	.5
0	1
1	2
2	4

$$f(x) = \log_2 x$$

X	Y
.25	-2
.5	-1
1	0
2	1
4	2

inverse



Domain: $x > 0$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 0$, $f(x) \rightarrow -\infty$

x-intercept: $(1, 0)$

Asymptote: $x = 0$

2. $f(x) = \log_{\frac{1}{3}} x$

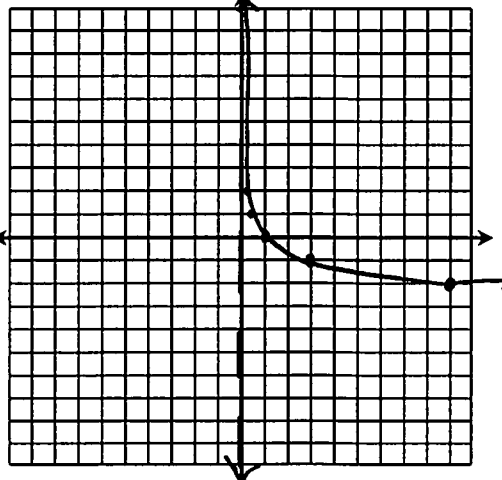
$$f(x) = \left(\frac{1}{3}\right)^x$$

X	Y
-2	9
-1	3
0	1
1	$0.\bar{3}$
2	$0.\bar{1}$

$$f(x) = \log_{\frac{1}{3}} x$$

X	Y
9	-2
3	-1
1	0
$0.\bar{3}$	1
$0.\bar{1}$	2

inverse



Domain: $x > 0$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow 0$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

x-intercept: $(1, 0)$

Asymptote: $x = 0$

3. $f(x) = \log_4(x-1)$

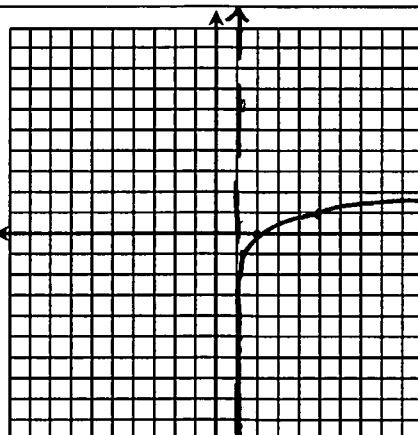
$f(x) = \log_4 x$

$f(x) = \log_4(x-1)$

X	Y
.06	-2
.25	-1
1	0
4	1
16	2

X	Y
1.06	-2
1.25	-1
2	0
5	1
17	2

right 1



Domain: $x > 1$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 1$, $f(x) \rightarrow -\infty$

x-intercept: $(2, 0)$

Asymptote: $x = 1$

4. $f(x) = \log_3 x - 2$

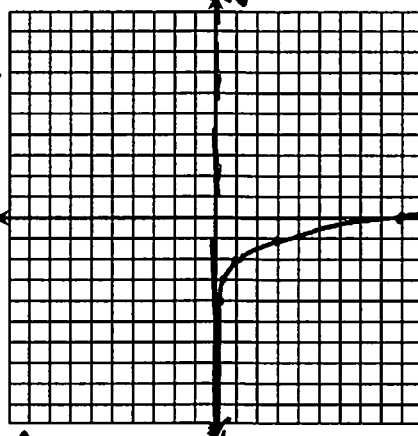
$f(x) = \log_3 x$

$f(x) = \log_3 x - 2$

X	Y
.1	-2
.3	-1
1	0
3	1
9	2

X	Y
.1	-4
.3	-3
1	-2
3	-1
9	0

down 2



Domain: $x > 0$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 0$, $f(x) \rightarrow -\infty$

x-intercept: $(9, 0)$

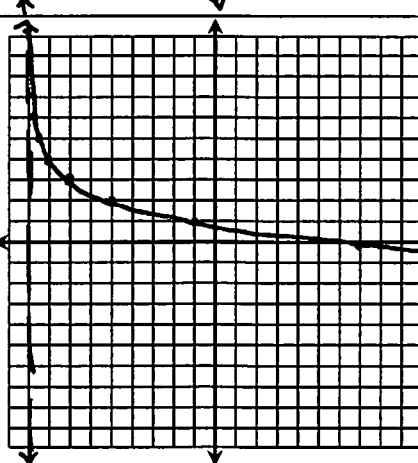
Asymptote: $x = 0$

5. $f(x) = \log_{\frac{1}{2}}(x+9) + 4$

X	Y
4	-2
2	-1
1	0
.5	1
.25	2

X	Y
-5	2
-7	3
-8	4
-8.5	5
-8.75	6

left 9, up 4



Domain: $x > -9$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow -9$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

x-intercept: $(7, 0)$

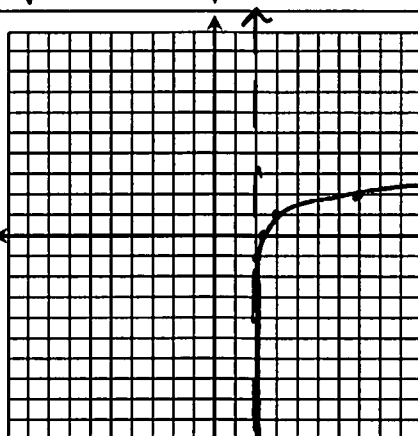
Asymptote: $x = -9$

6. $f(x) = \log_5(x-2) + 1$

X	Y
.04	-2
.2	-1
1	0
5	1
25	2

X	Y
2.04	-1
2.2	0
3	1
7	2
27	3

right 2, up 1



Domain: $x > 2$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 2$, $f(x) \rightarrow -\infty$

x-intercept: $(2.2, 0)$

Asymptote: $x = 2$

Name: _____

Unit 7: Exponential & Logarithmic Functions

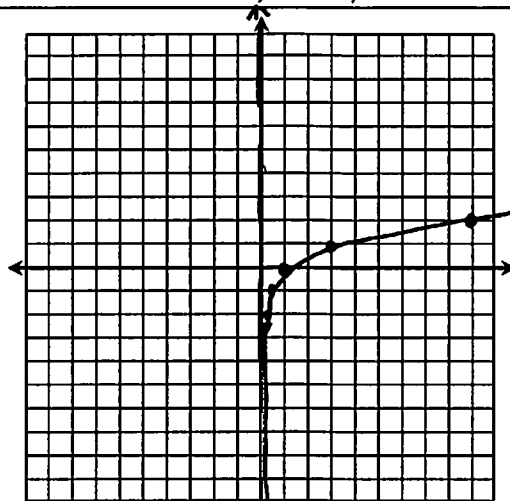
Date: _____ Per: _____

Homework 5: Graphing Logarithmic Functions

**** This is a 2-page document! ******Directions:** Graph each function and identify its key characteristics.

1. $f(x) = \log_3 x$

X	y
.1	-2
.3	-1
1	0
3	1
9	2

Domain: $x > 0$ Range: $\mathbb{R}$

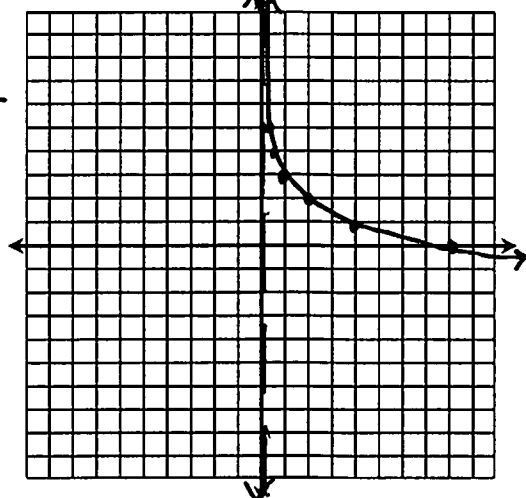
End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow 0$, $f(x) \rightarrow -\infty$ x-intercept: $(1, 0)$ Asymptote: $x = 0$

2. $f(x) = \log_{\frac{1}{2}} x + 3$

X	y	X	y
4	-2	4	1
2	-1	2	2
1	0	1	3
.5	1	.5	4
.25	2	.25	5

up 3

Domain: $x > 0$ Range: $\mathbb{R}$

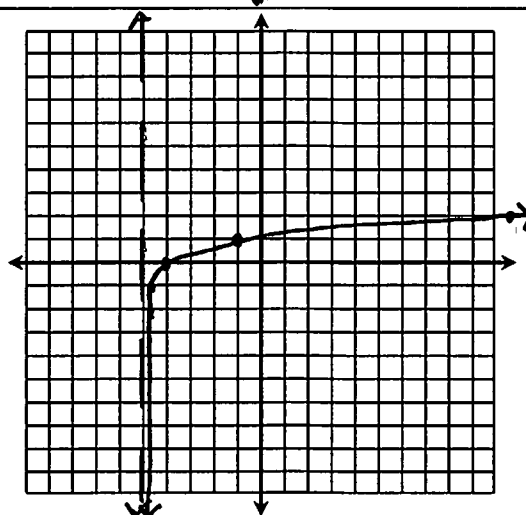
End Behavior:

As $x \rightarrow 0$, $f(x) \rightarrow \infty$ As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$ x-intercept: $(8, 0)$ Asymptote: $x = 0$

3. $f(x) = \log_4(x + 5)$

X	y	X	y
.06	-2	-4.94	-2
.25	-1	-4.75	-1
1	0	-4	0
4	1	-1	1
16	2	11	2

left 5

Domain: $x > -5$ Range: $\mathbb{R}$

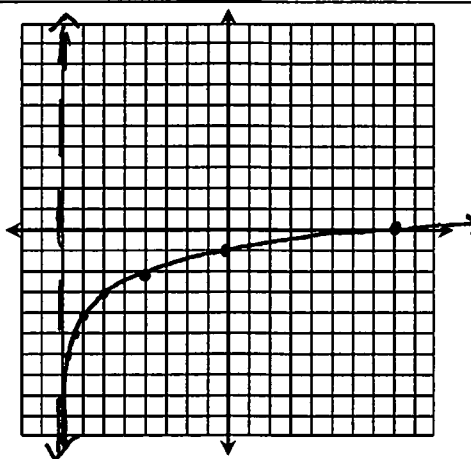
End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -5$, $f(x) \rightarrow -\infty$ x-intercept: $(-4, 0)$ Asymptote: $x = -5$

4. $f(x) = \log_2(x+8) - 4$

X	Y	X	Y
.25	-2	-1.75	-6
.5	-1	-7.5	-5
1	0	-7	-4
2	1	-6	-3
4	2	-4	-2
16	4	8	0

left 8, down 4



Domain: $x > -8$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -8$, $f(x) \rightarrow -\infty$

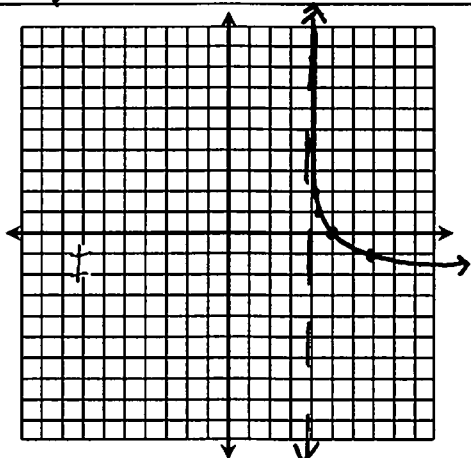
x-intercept: $(8, 0)$

Asymptote: $x = -8$

5. $f(x) = \log_{\frac{1}{3}}(x-4)$

X	Y	X	Y
9	-2	13	-2
3	-1	7	-1
1	0	5	0
.3	1	4.3	1
.7	2	4.7	2

right 4



Domain: $x > 4$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow 4$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

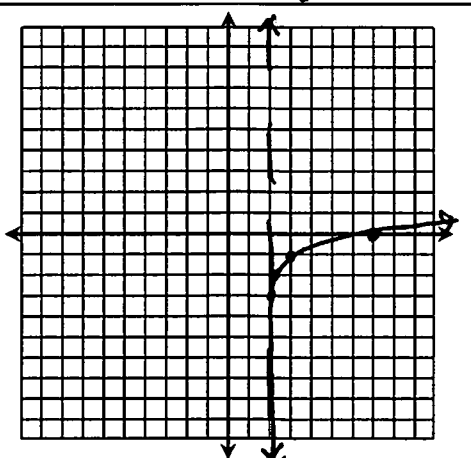
x-intercept: $(5, 0)$

Asymptote: $x = 4$

6. $f(x) = \log_5(x-2) - 1$

X	Y	X	Y
.04	-2	2.04	-3
.2	-1	2.2	-2
1	0	3	-1
5	1	7	0
25	2	27	1

right 2, down 1



Domain: $x > 2$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 2$, $f(x) \rightarrow -\infty$

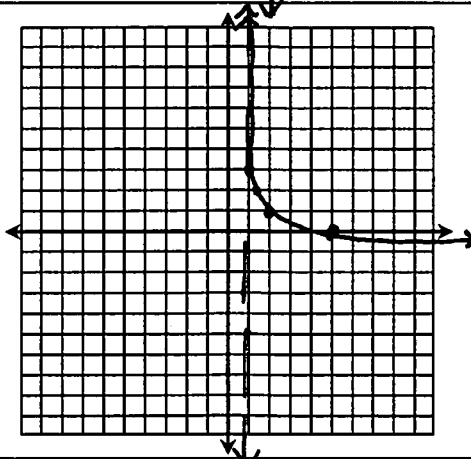
x-intercept: $(7, 0)$

Asymptote: $x = 2$

7. $f(x) = \log_{\frac{1}{4}}(x-1) + 1$

X	Y	X	Y
16	-2	17	-1
4	-1	5	0
1	0	2	1
.25	1	1.25	2
.06	2	1.06	3

right 1, up 1



Domain: $x > 1$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow 1$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

x-intercept: $(5, 0)$

Asymptote: $x = 1$

Name: _____ Algebra 2

Date: _____ Per: _____ Unit 7: Exponential & Logarithmic Functions

Quiz 7-2: Logarithmic Expressions & Functions

Write each expression in exponential form.

1. $\log_3 81 = 4$

2. $\log_x 7 = \frac{1}{2}$

Write each expression in logarithmic form.

3. $2^{-3} = \frac{1}{8}$

4. $9^x = 36$

Evaluate. Use the change of base formula when necessary.

5. $\log_7 343$
 $7^x = 343$

6. $\log_3 \frac{1}{9}$
 $3^x = \frac{1}{9}$

7. $\log_5 1$
 $5^x = 1$

8. $\log_2 64$
 $2^x = 64$

9. $\log_4 25$
 $\frac{\log 25}{\log 4}$

10. $\log_{12} 90$
 $\frac{\log 90}{\log 12}$

Condense each log into a single logarithm. DO NOT EVALUATE.

11. $\log_{12} 18 + 3 \cdot \log_{12} 2$
 $= \log_{12} 18 \cdot 2^3$
 $= \log_{12} 18 \cdot 8$

12. $\log_2 288 - \frac{1}{2} \cdot \log_2 81$
 $\log_2 \frac{288}{\sqrt{81}}$
 $= \log_2 \frac{288}{9}$

13. $5 \cdot \log x - 3 \cdot \log y$
 $\log x^5 - \log y^3$

14. $\frac{1}{4} \log_3 a^{20} + 3 \cdot \log_3 a^2$
 $\log_3 (a^{20})^{1/4} \cdot (a^2)^3$
 $= \log_3 a^5 \cdot a^6$

1. $3^4 = 81$

2. $x^{1/2} = 7$

3. $\log_2 \frac{1}{8} = -3$

4. $\log_9 36 = x$

5. $x = 3$

6. $x = -2$

7. $x = 0$

8. $x = 6$

9. $x = 2.3219$

10. $x = 1.8109$

11. $\log_{12} 144$

12. $\log_2 32$

13. $\log \frac{x^5}{y^3}$

14. $\log_3 a^{11}$

Expand each logarithm.

15. $\log_4(2w^4)^3$

$3(\log_4 2 + 4 \log_4 w)$

16. $\log\left(\frac{m^5}{n}\right)^2$

$2 \cdot (5 \log m - \log n)$

15. $3 \cdot \log_4 2 + 12 \cdot \log_4 w$

16. $10 \cdot \log m - 2 \cdot \log n$

17. $\frac{1}{2} \cdot \log_5 a + \frac{7}{2} \cdot \log_5 b$

18. $\frac{5}{4} \cdot \log_2 p + 3 \cdot \log_2 q$

17. $\log_5 \sqrt{ab^7}$

$\frac{1}{2} \cdot (\log_5 a + 7 \log_5 b)$

18. $\log_2 \sqrt[4]{p^5 q^{12}}$

$\frac{1}{4} \cdot (5 \log_2 p + 12 \cdot \log_2 q)$

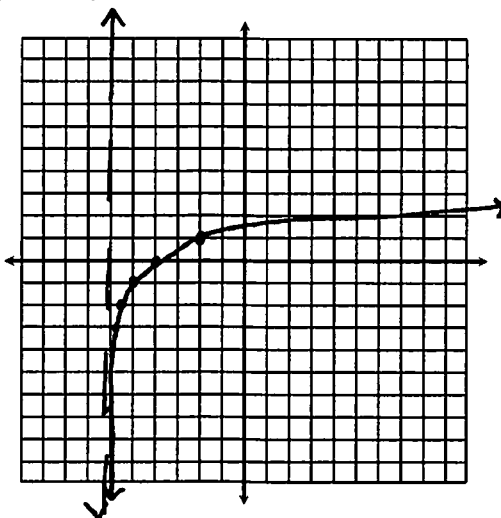
Graph each function and identify its key characteristics.

19. $f(x) = \log_2(x+6) - 1$

X	Y
.25	-2
.5	-1
1	0
2	1
4	2

X	Y
-5.75	-3
-5.5	-2
-5	-1
-4	0
-2	1

left 6, down 1



Domain: $x > -6$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -6$, $f(x) \rightarrow -\infty$

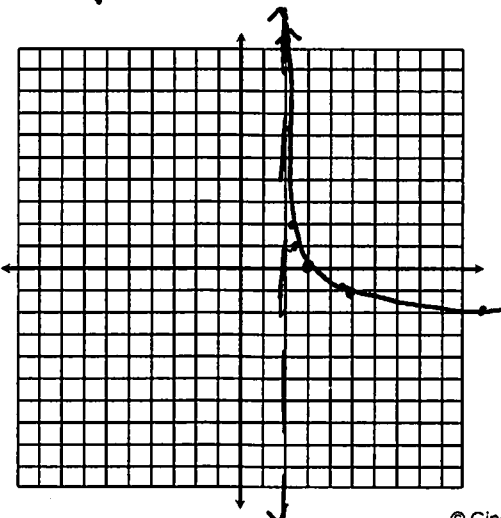
x-intercept: $(-4, 0)$

Asymptote: $x = -6$

20. $f(x) = \log_{\frac{1}{3}}(x-2)$

X	Y
9	-2
3	-1
1	0
.3	1
.1	2

X	Y
11	-2
5	-1
3	0
2.3	1
2.1	2



Domain: $x > 2$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow 2$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

x-intercept: $(3, 0)$

Asymptote: $x = 2$

Name:

Date:

Topic:

Class:

Main Ideas/Questions	Notes/Examples	
<i>Solving</i> LOGARITHMIC EQUATIONS Type 1: LOG = LOG	①	CONDENSE each logarithm.
	②	Use the One-to-One Property: If $\log_b m = \log_b n$, then $m = n$
	③	SOLVE and CHECK FOR EXTRANEIOUS SOLUTIONS.
	1. $\log_5(5x+9) = \log_5(6x)$	2. $\log_2(1-4n) = \log_2(2n+43)$
	$5x+9 = 6x$ $\boxed{9=x} \checkmark$	$1-4n = 2n+43$ $1 = 6n+43$ $\begin{array}{r} -43 \quad -43 \\ \hline -42 = 6n \\ \boxed{n=-7} \checkmark \end{array}$
	3. $\log_9(6-3w) = \log_9(-2w)$	4. $\log(y+5) + \log 4 = \log 72$
	$6-3w = -2w$ $\boxed{6=w} \text{ No Solution!}$ (Logs cannot be negative.) $\log_9(6-3 \cdot 6) = \log_9(-2 \cdot 6)$ $\log_9(-12) = \log_9(-12)$	$\log(y+5) \cdot 4 = \log 72$ $4y+20 = 72$ $4y = 52$ $\boxed{y=13} \checkmark$
	5. $3 \cdot \log_7 4 = \log_7(4a-8)$	6. $\log_4 68 - \log_4 4 = \log_4(3n+11)$
	$\log_7 4^3 = \log_7(4a-8)$ $64 = 4a-8$ $72 = 4a$ $\boxed{18=a} \checkmark$	$\log_4 \frac{68}{4} = \log_4(3n+11)$ $17 = 3n+11$ $6 = 3n$ $\boxed{n=2} \checkmark$
	7. $\frac{1}{2} \cdot \log_6 25 = \log_6(23-4w)$	8. $\log_3(2p-5) = 2 \cdot \log_3 6 - \log_3 4$
	$\log_6(25)^{1/2} = \log_6(23-4w)$ $5 = 23-4w$ $-18 = -4w$ $\boxed{w = \frac{9}{2}} \checkmark$	$\log_3(2p-5) = \log_3 \frac{6^2}{4}$ $2p-5 = 9$ $2p = 14$ $\boxed{p=7} \checkmark$

	9. $\log_4(m^2) = \log_4(18-7m)$ $m^2 = 18-7m$ $m^2 + 7m - 18 = 0$ $\frac{(m+9)(m-2)}{m+9=0 \quad m-2=0}$ $\checkmark \boxed{m=-9} \quad \boxed{m=2} \checkmark$	10. $\log 2 + \log(k^2) = \log(k^2+16)$ $\log 2(k^2) = \log(k^2+16)$ $2k^2 = k^2+16$ $k^2-16=0$ $\frac{(k+4)(k-4)}{(k+4)(k-4)=0}$ $\checkmark \boxed{k=-4} \quad \boxed{k=4} \checkmark$
Solving LOGARITHMIC EQUATIONS Type 2: LOG = NUMBER	❶ CONDENSE and ISOLATE the logarithm.	
	❷ Write the equation in EXPONENTIAL FORM.	
	❸ SOLVE and CHECK FOR EXTRANEOUS SOLUTIONS.	
	11. $\log_2(x-4) = 6$ $2^6 = x-4$ $64 = x-4$ $\boxed{68=x} \checkmark$	12. $\log_3(4x+8) - 7 = -3$ $\log_3(4x+8) = 4$ $3^4 = 4x+8$ $81 = 4x+8$ $73 = 4x$ $\boxed{x = \frac{73}{4}} \checkmark$
	13. $\log(2x) + \log(x-5) = 2$ $\log 2x(x-5) = 2$ $10^2 = 2x^2 - 10x$ $0 = 2x^2 - 10x - 100$ $0 = 2(x^2 - 5x - 50)$ $0 = 2(x-10)(x+5)$ $\checkmark \boxed{x=10} \quad \cancel{x=-5}$	14. $2 \cdot \log x - \log 4 = 2$ $\log \frac{x^2}{4} = 2$ $10^2 = \frac{x^2}{4}$ $400 = x^2$ $0 = x^2 - 400$ $0 = (x+20)(x-20)$ $\cancel{x=-20} \quad \boxed{x=20} \checkmark$
15. $\log_6(x+9) + \log_6 x = 2$ $\log_6 (x+9)(x) = 2$ $6^2 = x^2 + 9x$ $0 = x^2 + 9x - 36$ $0 = (x+12)(x-3)$ $\cancel{x=-12} \quad \boxed{x=3} \checkmark$	16. $\log(x-3) + \log x = 1$ $\log (x-3)(x) = 1$ $10^1 = x^2 - 3x$ $0 = x^2 - 3x - 10$ $0 = (x-5)(x+2)$ $\checkmark \boxed{x=5} \quad \cancel{x=-2}$	

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Unit 7: Exponential & Logarithmic Functions

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Homework 6: Solving Logarithmic Equations

**** This is a 2-page document! ******Directions:** Solve each equation. Check for extraneous solutions.

1. $\log_3(3x-11) = \log_3(25-x)$

$$3x-11 = 25-x$$

$$4x-11 = 25$$

$$4x = 36$$

$$\boxed{x=9} \checkmark$$

2. $\log_7(4n-7) = \log_7(-3n)$

$$4n-7 = -3n$$

$$-7 = -7n$$

$$1 = n$$

$$\boxed{\text{No solution!}}$$

3. $\log_2 75 = \log_2 3 + \log_2(2y-1)$

$$\log_2 75 = \log_2 3(2y-1)$$

$$75 = 6y-3$$

$$78 = 6y$$

$$\boxed{y=13} \checkmark$$

4. $2 \cdot \log m = \log 36$

$$\log m^2 = \log 36$$

$$m^2 = 36$$

$$m^2 - 36 = 0$$

$$\frac{(m+6)(m-6)=0}{m=-6 \quad \boxed{m=6} \checkmark}$$

5. $\log_4 108 - \log_4 9 = \log_4(7a-9)$

$$\log_4 \frac{108}{9} = \log_4(7a-9)$$

$$12 = 7a-9$$

$$21 = 7a$$

$$\boxed{a=3} \checkmark$$

6. $\frac{1}{3} \cdot \log_5 64 = \log_5 8 + \log_5 p$

$$\log_5 (64)^{\frac{1}{3}} = \log_5 8 \cdot p$$

$$4 = 8p$$

$$\boxed{\frac{1}{2} = p} \checkmark$$

7. $\log(w^2 + 21) = \log(10w)$

$$w^2 + 21 = 10w$$

$$w^2 - 10w + 21 = 0$$

$$\frac{(w-7)(w-3)=0}{\checkmark \quad \boxed{w=7} \quad \boxed{w=3} \checkmark}$$

8. $\log_2(2x) + \log_2(x-7) = \log_2(4x)$

$$\log_2 2x(x-7) = \log_2 4x$$

$$2x^2 - 14x = 4x$$

$$2x^2 - 18x = 0$$

$$\frac{2x(x-9)=0}{x \neq 0 \quad \boxed{x=9} \checkmark}$$

9. $\log_4(2m^3 - 14m^2) - \log_4(2m) = \log_4 8$ $\log_4 \frac{2m^3 - 14m^2}{2m} = \log_4 8$ $m^2 - 7m = 8$ $m^2 - 7m - 8 = 0$ $(m-8)(m+1) = 0$ $\checkmark \boxed{m=8} \quad m=-1$	10. $2 \cdot \log(x-3) = \log 25$ $\log(x-3)^2 = \log 25$ $x^2 - 6x + 9 = 25$ $x^2 - 6x - 16 = 0$ $(x-8)(x+2) = 0$ $\checkmark \boxed{x=8} \quad x=-2$
11. $\log_3(2x-7) = 4$ $3^4 = 2x-7$ $81 = 2x-7$ $88 = 2x$ $\boxed{x=44} \checkmark$	12. $\log_8(28k-20) + 15 = 18$ $\log_8(28k-20) = 3$ $8^3 = 28k-20$ $512 = 28k-20$ $532 = 28k$ $\boxed{k=19} \checkmark$
12. $\log_9(15-4n) = \frac{1}{2}$ $9^{1/2} = 15-4n$ $3 = 15-4n$ $-12 = -4n$ $\boxed{n=3} \checkmark$	14. $\log_2 4 + \log_2(c-9) = 5$ $\log_2 4(c-9) = 5$ $2^5 = 4c-36$ $32 = 4c-36$ $68 = 4c$ $\boxed{17=c} \checkmark$
15. $2 \cdot \log_4 k = 4$ $\log_4 k^2 = 4$ $4^4 = k^2$ $256 = k^2$ $0 = k^2 - 256$ $0 = (k+16)(k-16)$ $k=-16 \quad \boxed{k=16} \checkmark$	16. $\log_8(p^2+15) = 2$ $8^2 = p^2+15$ $64 = p^2+15$ $0 = p^2-49$ $0 = (p+7)(p-7)$ $\checkmark \boxed{p=-7} \quad \boxed{p=7} \checkmark$

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Main Ideas/Questions	Notes/Examples			
REVIEW: Using a common base to solve an exponential equation.	Directions: Solve the equations below using a common base.			
	1. $5^{n+10} = 25$ $5^{n+10} = 5^2$ $n+10 = 2$ $n = -8$	2. $9^{a+2} = 27^{4a-2}$ $(3^2)^{a+2} = (3^3)^{4a-2}$ $2a+4 = 12a-6$ $4 = 10a-6$ $10 = 10a$ $1 = a$		
What if a common base is NOT POSSIBLE? There are a couple ways to solve problems like these. Solve using both methods, then choose what works best for you!	METHOD 1	Take the Log of both sides and solve.	METHOD 2	Write in Logarithmic Form and Solve.
	$2^x = 61$ $\log 2^x = \log 61$ $\frac{x \cdot \log 2}{\log 2} = \frac{\log 61}{\log 2}$ $x = 5.9307$		$2^x = 61$ $\log_2 61 = x$ $\frac{\log 61}{\log 2} = x$ $x = 5.9307$	
Solving Exponential Equations: USING LOGS	Solve each equation. Round to the nearest ten-thousandth.			
	3. $12^x = 8$ $\log 12^x = \log 8$ $\frac{x \cdot \log 12}{\log 12} = \frac{\log 8}{\log 12}$ $x = 0.8368$		4. $8^{m-7} = 92$ $\log_8 92 = m-7$ $\frac{\log 92}{\log 8} = m-7$ $2.1745 = m-7$ $9.1745 = m$	
	5. $4 \cdot 7^n = 148$ $7^n = 37$ $\log 7^n = \log 37$ $n \cdot \log 7 = \log 37$ $n = 1.8556$		6. $4^{3w} - 5 = 3$ $4^{3w} = 8$ $\log_4 8 = 3w$ $1.5 = 3w$ $0.5 = w$	
	Empty space for additional work		Empty space for additional work	

$$7. 7 - 4^{x+1} = 18$$

$$-4^{x+1} = 11$$

$$4^{x+1} = -11$$

$$\log_4(-11) = x+1$$



No Solution!

Logs cannot be negative!

$$8. \frac{10 \cdot 5^{3k-3}}{10} = \frac{40}{10}$$

$$5^{3k-3} = 4$$

$$\log_5 4 = 3k-3$$

$$\frac{\log 4}{\log 5} = 3k-3$$

$$0.8614 = 3k-3$$

$$3.8614 = 3k$$

$$\boxed{k = 1.2871}$$

$$9. 4 \cdot 3^n + 15 = 359$$

$$\frac{4 \cdot 3^n}{4} = \frac{344}{4}$$

$$3^n = 86$$

$$\log_3 86 = n$$

$$\frac{\log 86}{\log 3} = n$$

$$\boxed{n = 4.0545}$$

$$10. -2 \cdot 5^p + 7 = -63$$

$$\frac{-2 \cdot 5^p}{-2} = \frac{-70}{-2}$$

$$5^p = 35$$

$$\log_5 35 = p$$

$$\frac{\log 35}{\log 5} = p$$

$$\boxed{p = 2.2091}$$

$$11. 5 \cdot 9^{v-1} + 1 = 181$$

$$5 \cdot 9^{v-1} = 180$$

$$9^{v-1} = 36$$

$$\log_9 36 = v-1$$

$$\frac{\log 36}{\log 9} = v-1$$

$$1.6309 = v-1$$

$$\boxed{2.6309 = v}$$

$$12. 8 \cdot 11^{7k} - 3 = 213$$

$$8 \cdot 11^{7k} = 216$$

$$11^{7k} = 27$$

$$\log_{11} 27 = 7k$$

$$\frac{\log 27}{\log 11} = 7k$$

$$1.3745 = 7k$$

$$\boxed{k = .1964}$$

$$13. 6 \cdot 16^{7y+2} - 2 = 82$$

$$6 \cdot 16^{7y+2} = 84$$

$$16^{7y+2} = 14$$

$$\log_{16} 14 = 7y+2$$

$$\frac{\log 14}{\log 16} = 7y+2$$

$$0.9518 = 7y+2$$

$$-1.0482 = 7y$$

$$\boxed{y = -.1497}$$

$$14. 3 \cdot 8^{3-7n} + 10 = 94$$

$$3 \cdot 8^{3-7n} = 84$$

$$8^{3-7n} = 28$$

$$\log_8 28 = 3-7n$$

$$\frac{\log 28}{\log 8} = 3-7n$$

$$1.6025 = 3-7n$$

$$-1.3975 = -7n$$

$$\boxed{n = .1996}$$

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Unit 7: Exponential & Logarithmic Functions

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Homework 7: Solving Exponential Equations
(using logs)**** This is a 2-page document! ******Directions:** Solve each exponential equation using logarithms.

1. $3^x = 18$

$$\log_3 18 = x$$

$$\frac{\log 18}{\log 3} = x$$

$$\boxed{2.6309 = x}$$

2. $7^y = 24$

$$\log_7 24 = y$$

$$\frac{\log 24}{\log 7} = y$$

$$\boxed{1.6332 = y}$$

3. $12^{n-3} = 60$

$$\log_{12} 60 = n-3$$

$$\frac{\log 60}{\log 12} = n-3$$

$$1.6477 = n-3$$

$$\boxed{n = 4.6477}$$

4. $2^{3a} = 142$

$$\log_2 142 = 3a$$

$$\frac{\log 142}{\log 2} = 3a$$

$$7.1497 = 3a$$

$$\boxed{a = 2.3832}$$

5. $15^{3v-5} = 87$

$$\log_{15} 87 = 3v-5$$

$$\frac{\log 87}{\log 15} = 3v-5$$

$$1.6491 = 3v-5$$

$$6.6491 = 3v$$

$$\boxed{v = 2.2164}$$

6. $4^{8n-2} = 84$

$$\log_4 84 = 8n-2$$

$$\frac{\log 84}{\log 4} = 8n-2$$

$$3.1962 = 8n-2$$

$$5.1962 = 8n$$

$$\boxed{n = 0.6495}$$

7. $4 \cdot 10^k = 60$

$$10^k = 15$$

$$\log_{10} 15 = k$$

$$\frac{\log 15}{\log 10} = k$$

$$\boxed{k = 1.1761}$$

8. $16^n - 6 = 45$

$$16^n = 51$$

$$\log_{16} 51 = n$$

$$\frac{\log 51}{\log 16} = n$$

$$\boxed{1.4181 = n}$$

9. $13^{c-8} - 9 = 17$ $13^{c-8} = 26$ $\log_{13} 26 = c-8$ $\frac{\log 26}{\log 13} = c-8$ $1.2702 = c-8$ $9.2702 = c$	10. $2 \cdot 8^{5r} = 28$ $8^{5r} = 14$ $\log_8 14 = 5r$ $\frac{\log 14}{\log 8} = 5r$ $1.2691 = 5r$ $r = 0.2538$
11. $10^{2x-7} - 3 = 57$ $10^{2x-7} = 60$ $\log_{10} 60 = 2x-7$ $\frac{\log 60}{\log 10} = 2x-7$ $1.7782 = 2x-7$ $8.7782 = 2x$ $x = 4.3891$	12. $8^{6-4x} + 6 = 22$ $8^{6-4x} = 16$ $\log_8 16 = 6-4x$ $\frac{\log 16}{\log 8} = 6-4x$ $1.\bar{3} = 6-4x$ $-4.\bar{6} = -4x$ $1.\bar{6} = x$
13. $6 \cdot 4^m - 14 = 88$ $6 \cdot 4^m = 102$ $4^m = 17$ $\log_4 17 = m$ $\frac{\log 17}{\log 4} = m$ $2.0437 = m$	14. $9 \cdot 12^{r+4} - 8 = 127$ $9 \cdot 12^{r+4} = 135$ $12^{r+4} = 15$ $\log_{12} 15 = r+4$ $\frac{\log 15}{\log 12} = r+4$ $1.09 = r+4$ $r = -2.91$
15. $-5 \cdot 4^{6x} + 5 = -30$ $-5 \cdot 4^{6x} = -35$ $4^{6x} = 7$ $\log_4 7 = 6x$ $\frac{\log 7}{\log 4} = 6x$ $1.4037 = 6x$ $0.2339 = x$	16. $8 \cdot 11^{3p-9} + 10 = 194$ $8 \cdot 11^{3p-9} = 184$ $11^{3p-9} = 23$ $\log_{11} 23 = 3p-9$ $\frac{\log 23}{\log 11} = 3p-9$ $1.3076 = 3p-9$ $10.3076 = 3p$ $p = 3.4359$

REVIEW: Logarithmic & Exponential Equations

LOGARITHMIC EQUATIONS

Solve. Check for extraneous solutions.

1. $\log_7(9x-4) = \log_7(x+20)$

$$9x-4 = x+20$$

$$8x-4 = 20$$

$$8x = 24$$

$$\boxed{x=3} \checkmark$$

2. $\log_5(m^2-12) = \log_5 m$

$$m^2-12 = m$$

$$m^2-m-12=0$$

$$(m-4)(m+3)=0$$

$$\boxed{m=4} \quad | \quad m \neq -3$$

3. $\log_3 4 + \log_3(a+5) = \log_3 56$

$$\log_3 4(a+5) = \log_3 56$$

$$4a+20 = 56$$

$$4a = 36$$

$$\boxed{a=9} \checkmark$$

4. $\log(2y-10) = 7 \cdot \log 2 - \log 8$

$$\log(2y-10) = \log \frac{2^7}{8}$$

$$2y-10 = 16$$

$$2y = 26$$

$$\boxed{y=13} \checkmark$$

5. $\log_4(5m+9) = 3$

$$4^3 = 5m+9$$

$$64 = 5m+9$$

$$55 = 5m$$

$$\boxed{m=11}$$

6. $\log_{36}(20-4p) = \frac{1}{2}$

$$36^{1/2} = 20-4p$$

$$6 = 20-4p$$

$$-14 = -4p$$

$$\boxed{\frac{7}{2} = p}$$

7. $\log_6(7k-1) = 3$

$$6^3 = 7k-1$$

$$216 = 7k-1$$

$$217 = 7k$$

$$\boxed{k=31}$$

8. $\log(n+8) + \log 4 = 2$

$$\log 4(n+8) = 2$$

$$10^2 = 4n+32$$

$$100 = 4n+32$$

$$68 = 4n$$

$$\boxed{n=17}$$

EXPONENTIAL EQUATIONS

Solve. Check for extraneous solutions. Round to the nearest ten-thousandth when necessary.

9. $25^{v-2} = 625$

$$(5^2)^{v-2} = 5^4$$

$$2v-4 = 4$$

$$2v = 8$$

$$\boxed{v=4}$$

10. $\frac{1}{16} = 8^{4x-2}$

$$2^{-4} = (2^3)^{4x-2}$$

$$-4 = 12x - 6$$

$$2 = 12x$$

$$\boxed{\frac{1}{6} = x}$$

11. $8^k = 78$

$$\log_8 78 = k$$

$$\frac{\log 78}{\log 8} = k$$

$$\boxed{k = 2.0951}$$

12. $9^{m-6} = 78$

$$\log_9 78 = m-6$$

$$\frac{\log 78}{\log 9} = m-6$$

$$1.9828 = m-6$$

$$\boxed{m = 7.9828}$$

13. $15^{3a} + 7 = 67$

$$15^{3a} = 60$$

$$\log_{15} 60 = 3a$$

$$\frac{\log 60}{\log 15} = 3a$$

$$1.5119 = 3a$$

$$\boxed{a = 0.5040}$$

14. $14^{3-8x} + 9 = 77$

$$14^{3-8x} = 68$$

$$\log_{14} 68 = 3-8x$$

$$\frac{\log 68}{\log 14} = 3-8x$$

$$1.5989 = 3-8x$$

$$-1.4011 = -8x$$

$$\boxed{x = 0.1751}$$

15. $8 \cdot 3^{n-1} - 21 = 51$

$$8 \cdot 3^{n-1} = 72$$

$$3^{n-1} = 9$$

$$\log_3 9 = n-1$$

$$\frac{\log 9}{\log 3} = n-1$$

$$2 = n-1$$

$$\boxed{n=3}$$

16. $2 \cdot 18^{10r-3} - 1 = 73$

$$2 \cdot 18^{10r-3} = 74$$

$$18^{10r-3} = 37$$

$$\log_{18} 37 = 10r-3$$

$$\frac{\log 37}{\log 18} = 10r-3$$

$$1.2493 = 10r-3$$

$$4.2493 = 10r$$

$$\boxed{0.4249=r}$$

Name: _____

Unit 7: Exponential & Logarithmic Functions

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Homework 8: Logarithmic & Exponential Equations

**** This is a 2-page document! ******Directions:** Solve each equation. Check all answers for extraneous solutions.

1. $\log_4(25 - 2x) = \log_4(6x + 1)$

$$25 - 2x = 6x + 1$$

$$24 - 2x = 6x$$

$$24 = 8x$$

$$\boxed{x = 3} \checkmark$$

2. $\log_9(8y - 9) = \log_9 108 - \log_9 4$

$$\log_9(8y - 9) = \log_9 \frac{108}{4}$$

$$8y - 9 = 27$$

$$8y = 36$$

$$\boxed{y = \frac{9}{2}} \checkmark$$

3. $6 \cdot \log_2 2 = \log_2 8 + \log_2(a - 2)$

$$\log_2 2^6 = \log_2 8(a - 2)$$

$$64 = 8a - 16$$

$$80 = 8a$$

$$\boxed{10 = a} \checkmark$$

4. $\log_6(5w + 14) = 2 \cdot \log_6 w$

$$\log_6(5w + 14) = \log_6 w^2$$

$$5w + 14 = w^2$$

$$0 = w^2 - 5w - 14$$

$$0 = (w - 7)(w + 2)$$

$$\checkmark \boxed{w = 7} \quad w \neq -2$$

5. $\log_7(3x + 5) = 2$

$$7^2 = 3x + 5$$

$$49 = 3x + 5$$

$$44 = 3x$$

$$\boxed{\frac{44}{3} = x}$$

6. $\log_{27}(11 - 2k) = \frac{1}{3}$

$$27^{\frac{1}{3}} = 11 - 2k$$

$$3 = 11 - 2k$$

$$-8 = -2k$$

$$\boxed{4 = k}$$

7. $\log(24x + 64) = 3$

$$10^3 = 24x + 64$$

$$1000 = 24x + 64$$

$$936 = 24x$$

$$\boxed{39 = x}$$

8. $5 = \log_3 8 + \log_3(r + 6)$

$$5 = \log_3 8(r + 6)$$

$$3^5 = 8r + 48$$

$$243 = 8r + 48$$

$$195 = 8r$$

$$\boxed{24.375 = r}$$

$$9. 5^{x-4} = 25^{x-6}$$

$$5^{x-4} = (5^2)^{x-6}$$

$$x-4 = 2x-12$$

$$-4 = x-12$$

$$\boxed{8=x}$$

$$10. 36^{5v+2} = \left(\frac{1}{6}\right)^{11-v}$$

$$(6^2)^{5v+2} = (6^{-1})^{11-v}$$

$$10v+4 = -11+v$$

$$9v+4 = -11$$

$$9v = -15$$

$$\boxed{v = -\frac{5}{3}}$$

$$11. 5^m = 220$$

$$\log_5 220 = m$$

$$\frac{\log 220}{\log 5} = m$$

$$\boxed{3.3512=m}$$

$$12. 14^{p-8} = 62$$

$$\log_{14} 62 = p-8$$

$$\frac{\log 62}{\log 14} = p-8$$

$$1.5639 = p-8$$

$$\boxed{9.5639=p}$$

$$13. 3 \cdot 4^{n+2} = 78$$

$$4^{n+2} = 26$$

$$\log_4 26 = n+2$$

$$\frac{\log 26}{\log 4} = n+2$$

$$2.3502 = n+2$$

$$\boxed{0.3502=n}$$

$$14. 5^{8-2y} - 10 = 45$$

$$5^{8-2y} = 55$$

$$\log_5 55 = 8-2y$$

$$\frac{\log 55}{\log 5} = 8-2y$$

$$2.4899 = 8-2y$$

$$-5.5101 = -2y$$

$$\boxed{2.7551=y}$$

$$15. 2 \cdot 10^{6c} + 9 = 17$$

$$2 \cdot 10^{6c} = 8$$

$$10^{6c} = 4$$

$$\log_{10} 4 = 6c$$

$$\frac{\log 4}{\log 10} = 6c$$

$$0.6021 = 6c$$

$$\boxed{c = 0.1003}$$

$$16. 9 \cdot 14^{5a+9} + 8 = 107$$

$$9 \cdot 14^{5a+9} = 99$$

$$14^{5a+9} = 11$$

$$\log_{14} 11 = 5a+9$$

$$\frac{\log 11}{\log 14} = 5a+9$$

$$0.9086 = 5a+9$$

$$-8.0914 = 5a$$

$$\boxed{a = -1.6183}$$

Name: _____ Algebra 2

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Quiz 7-3: Logarithmic & Exponential Equations

Part I: Logarithmic Equations

1. $\log_2(3x-10) = \log_2(14-x)$

$$3x-10 = 14-x$$

$$4x-10 = 14$$

$$4x = 24$$

$$\boxed{x=6}$$

2. $\log 12 + \log 5 = \log(4m+8)$

$$\log 12 \cdot 5 = \log(4m+8)$$

$$60 = 4m+8$$

$$52 = 4m$$

$$\boxed{m=13}$$

3. $\log_7(9p-5) = 4 \cdot \log_7 2$

$$\log_7(9p-5) = \log_7 2^4$$

$$9p-5 = 16$$

$$9p = 21$$

$$\boxed{p = \frac{7}{3}}$$

4. $\log_4(k^2+2k) = \log_4 60 - \log_4 4$

$$\log_4(k^2+2k) = \log_4 \frac{60}{4}$$

$$k^2+2k = 15$$

$$k^2+2k-15 = 0$$

$$(k+5)(k-3) = 0$$

$$\boxed{k=-5} \quad \boxed{k=3}$$

5. $\log_3(2a-15) = 2$

$$3^2 = 2a-15$$

$$9 = 2a-15$$

$$24 = 2a$$

$$\boxed{a=12}$$

6. $\frac{1}{4} = \log_{16}(17-5n)$

$$16^{1/4} = 17-5n$$

$$2 = 17-5n$$

$$-15 = -5n$$

$$\boxed{n=3}$$

7. $\log_3 9 + \log_3(c-2) = 3$

$$\log_3 9(c-2) = 3$$

$$3^3 = 9c-18$$

$$27 = 9c-18$$

$$45 = 9c$$

$$\boxed{5=c}$$

8. $\frac{1}{2} \cdot \log_4 36 = \log_2(7w+1)$

$$\log_2 36^{1/2} = \log_2(7w+1)$$

$$6 = 7w+1$$

$$5 = 7w$$

$$\boxed{\frac{5}{7} = w}$$

1. $\underline{X=6}$

2. $\underline{m=13}$

3. $\underline{p = \frac{7}{3}}$

4. $\underline{k = -5, k = 3}$

5. $\underline{a=12}$

6. $\underline{n=3}$

7. $\underline{c=5}$

8. $\underline{w = \frac{5}{7}}$

Part II: Exponential Equations

$$9. \left(\frac{1}{8}\right)^{x+3} = 64$$

$$(2^{-3})^{x+3} = 2^6$$

$$-3x - 9 = 6$$

$$-3x = 15$$

$$\boxed{x = -5}$$

$$11. 4^w = 10$$

$$\log_4 10 = w$$

$$\frac{\log 10}{\log 4} = w$$

$$\boxed{1.6610 = w}$$

$$13. 7^{10n} + 2 = 81$$

$$7^{10n} = 79$$

$$\log_7 79 = 10n$$

$$\frac{\log 79}{\log 7} = 10n$$

$$2.2455 = 10n$$

$$\boxed{n = .2245}$$

$$15. -2 \cdot 10^x - 1 = -33$$

$$-2 \cdot 10^x = -32$$

$$10^x = 16$$

$$\log_{10} 16 = x$$

$$\boxed{1.2041 = x}$$

$$10. 36^{2m} = 216$$

$$(6^2)^{2m} = 6^3$$

$$4m = 3$$

$$\boxed{m = \frac{3}{4}}$$

$$12. 9^{a-5} = 12$$

$$\log_9 12 = a - 5$$

$$\frac{\log 12}{\log 9} = a - 5$$

$$1.1309 = a - 5$$

$$\boxed{6.1309 = a}$$

$$14. 12^{3p+8} + 10 = 97$$

$$12^{3p+8} = 87$$

$$\log_{12} 87 = 3p + 8$$

$$\frac{\log 87}{\log 12} = 3p + 8$$

$$1.7972 = 3p + 8$$

$$-6.2028 = 3p$$

$$\boxed{p = -2.0676}$$

$$16. 4 \cdot 18^{k-8} + 3 = 59$$

$$4 \cdot 18^{k-8} = 56$$

$$18^{k-8} = 14$$

$$\log_{18} 14 = k - 8$$

$$\frac{\log 14}{\log 18} = k - 8$$

$$0.9131 = k - 8$$

$$\boxed{k = 8.9131}$$

$$9. \quad \underline{x = -5}$$

$$10. \quad \underline{m = \frac{3}{4}}$$

$$11. \quad \underline{w = 1.6610}$$

$$12. \quad \underline{a = 6.1309}$$

$$13. \quad \underline{n = .2245}$$

$$14. \quad \underline{p = -2.0676}$$

$$15. \quad \underline{x = 1.2041}$$

$$16. \quad \underline{k = 8.9131}$$

Name:		Date:	
Topic:		Class:	

Main Ideas/Questions	Notes/Examples		
EXPONENTIAL <i>Growth</i>	Exponential growth occurs when a quantity exponentially increases by a fixed percent each year (or other other time period).		
	Formula: $f(t) = a(1+r)^t$	$a =$ <u>initial amount</u> $r =$ <u>growth rate</u> $t =$ <u>time</u>	
EXAMPLES	1. The original value of an investment is \$1,800. If the value has increased by 7% each year, write an exponential function to model the situation. Then, find the value of the investment after 15 years. $f(t) = 1800(1.07)^t$ $f(15) = 1800(1.07)^{15}$ = \$4,966.26		2. In 2002, there were 972 students enrolled at Oakview High School. Since then, the number of students has increased by 1.5% each year. Write an exponential function to model the situation, then find the number of students enrolled in 2014. $f(t) = 972(1.015)^t$ $f(12) = 972(1.015)^{12}$ = 1162 students
EXPONENTIAL <i>Decay</i>	Exponential decay occurs when a quantity exponentially decreases by a fixed percent each year (or other other time period).		
	Formula: $f(t) = a(1-r)^t$	$a =$ <u>initial amount</u> $r =$ <u>decay rate</u> $t =$ <u>time</u>	
EXAMPLES	3. An investment of \$12,000 is losing value at a rate of 4% each year. Write an exponential function to model the situation, then find the value of the investment after 9 years. $f(t) = 12000(0.96)^t$ $f(9) = 12000(0.96)^9$ = \$8,310.41		4. Mark bought a brand new car for \$35,000 in 2008. If the value of the car depreciates approximately 8% each year, write an exponential function to model the situation. Then, find the value of the car in 2015. $f(t) = 35000(0.92)^t$ $f(7) = 35000(0.92)^7$ = \$19,524.63

COMPOUND Interest	Compound interest occurs when interest is calculated on both the principal amount AND the accrued interest thus far.	
	Formula: $A = P \left(1 + \frac{r}{n}\right)^{nt}$	$P =$ <u>principle/initial amount</u> $r =$ <u>rate</u> $n =$ <u># times compounded (yearly)</u> $t =$ <u>time</u>
EXAMPLES	5. Laura deposited \$12,000 into an account that earns 8% interest. How much money will she have in 5 years if the interest is compounded <u>quarterly</u> ? $A = 12000 \left(1 + \frac{0.08}{4}\right)^{4 \cdot 5} \quad n=4$ $A = 12000 (1.02)^{20}$ $A = \\$17,831.37$	
	6. Jack moved \$25,000 of his savings to a new investment account that earns 4.5% interest compounded <u>monthly</u> . Find the total amount in the account after 6 years. $A = 25000 \left(1 + \frac{0.045}{12}\right)^{12 \cdot 6} \quad n=12$ $A = 25000 (1.00375)^{72}$ $A = \\$32,732.58$	
	7. An investment account pays 3.9% interest compounded <u>semi-annually</u> . If \$4,000 is invested in this account, what will the balance be after 12 years? $A = 4000 \left(1 + \frac{0.039}{2}\right)^{2 \cdot 12} \quad n=2$ $A = 4000 (1.0195)^{24}$ $A = \\$6,358.48$	
	8. A savings account offers 0.8% interest compounded <u>bimonthly</u> . If Bob deposited \$300 into this account, <u>how much interest</u> will he earn after 10 years? $A = 300 \left(1 + \frac{0.008}{24}\right)^{24 \cdot 10} \quad n=24$ $A = 300 (1.00033)^{240}$ $A = \\$324.98$ $\begin{array}{r} 324.98 \\ - 300.00 \\ \hline \\$24.98 \end{array}$	
	9. Suppose you invest \$750 into an account that pays 3% interest compounded <u>weekly</u> . <u>How much interest</u> will you have earned after 20 years? $A = 750 \left(1 + \frac{0.03}{52}\right)^{52 \cdot 20} \quad n=52$ $A = 750 (1.000576923)^{1040}$ $A = \$1,366.35$ $\text{Interest} = \\$616.35$	

Name: _____

Unit 7: Exponential & Logarithmic Functions

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Homework 9: Applications of Exponential Functions

**** This is a 2-page document! ******Exponential Growth & Decay**

1. Vanessa invested \$2,500 into an account that will increase in value by 3.5% each year. Write an exponential function to model this situation, then find the value of the investment after 20 years.

$$f(t) = 2500 (1.035)^t$$

$$f(20) = 2500 (1.035)^{20}$$

$$f(20) = \$4974.47$$

2. The average price of a movie ticket in 1990 was \$4.22. Since then, the price has increased by approximately 3.1% each year. Write an exponential function to model this situation, then find the price of a ticket in 2016.

$$f(t) = 4.22 (1.031)^t$$

$$f(26) = 4.22 (1.031)^{26}$$

$$f(26) = \$9.33$$

3. A virus has infected 400 people in the town and is spreading to 25% more people each day. Write an exponential function to model this situation, then find the number of people that will be infected in 10 days.

$$f(t) = 400 (1.25)^t$$

$$f(10) = 400 (1.25)^{10}$$

$$f(10) = 3,725.29$$

↓

3,725 people

4. The population of a small town was 10,800 in 2002. Since then, the population has decreased at a rate of 2.5% each year. Write an exponential function to model the situation, then find the population of the town in 2020.

$$f(t) = 10800 (.975)^t$$

$$f(18) = 10800 (.975)^{18}$$

$$f(18) = 6847.10$$

↓

6847 people

5. Manny bought a brand-new car in 2012 for \$28,750. If the car depreciates by 12% each year, write an exponential function to model the situation, then find the value of the car in 2018.

$$f(t) = 28750 (.88)^t$$

$$f(6) = 28750 (.88)^6$$

$$f(6) = \$13,351.62$$

Compound Interest

6. Anisha invested \$8,000 in an account that earns 10% interest. How much money will she have in 15 years if the interest is compounded quarterly? $n=4$

$$A = 8000 \left(1 + \frac{.10}{4}\right)^{4 \cdot 15}$$

$$A = 8000 (1.025)^{60}$$

$$A = \$35,198.32$$

7. A realtor took \$32,500 made on the sale of a home and placed it in a new account that earns 6% compounded annually. Find the total amount in the account after 5 years. $n=1$

$$A = 32500 \left(1 + \frac{.06}{1}\right)^{1 \cdot 5}$$

$$A = 32500 (1.06)^5$$

$$A = \$43,492.33$$

8. Scott invested \$1,600 into a retirement account that earns 2.4% interest compounded monthly. What will the balance of the account be after 30 years? $n=12$

$$A = 1600 \left(1 + \frac{.024}{12}\right)^{12 \cdot 30}$$

$$A = 1600 (1.002)^{360}$$

$$A = \$3,284.73$$

9. Kaylee used her graduation money to set up a savings account that earns 3.4% interest compounded weekly. If the original amount deposited was \$500, how much interest will she have earned after 10 years? $n=52$

$$A = 500 \left(1 + \frac{.034}{52}\right)^{52 \cdot 10}$$

$$A = 500 (1.006538462)^{520}$$

$$A = \$702.40$$

$$\begin{array}{r} 702.40 \\ - 500.00 \\ \hline \end{array}$$

$$\$202.40$$

10. Ryan inherited land from his grandfather. He decided to sell the land for \$160,000, then place the money in a new account that earns 1.2% interest compounded bimonthly. How much interest will he have earned after 30 years? $n=24$

$$A = 160000 \left(1 + \frac{.012}{24}\right)^{24 \cdot 30}$$

$$A = 160000 (1.0005)^{720}$$

$$A = \$229,312.07$$

$$\$69,312.07$$

Name:

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Main Ideas/Questions

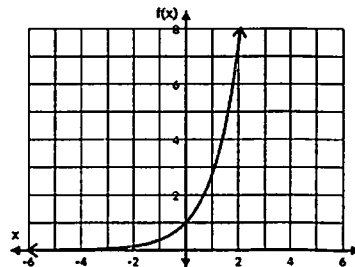
Notes/Examples

What is the
NATURAL BASE,
"e"?

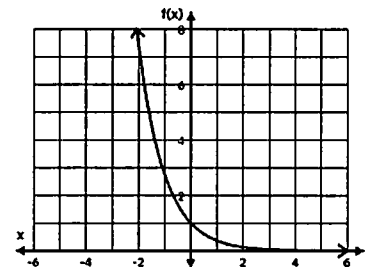
- Like π , e is special number discovered in mathematics and occurs in real-world applications of continuous exponential growth and decay.
- e is an irrational number with an approximate value of 2.71828...

Natural Base
EXPONENTIAL
Functions

The function $f(x) = e^x$
is used to model continuous
exponential growth.

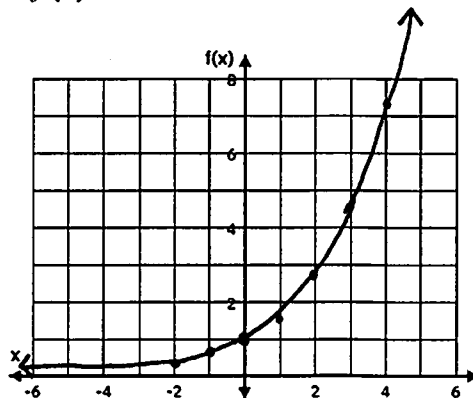


The function $f(x) = e^{-x}$
is used to model continuous
exponential decay.

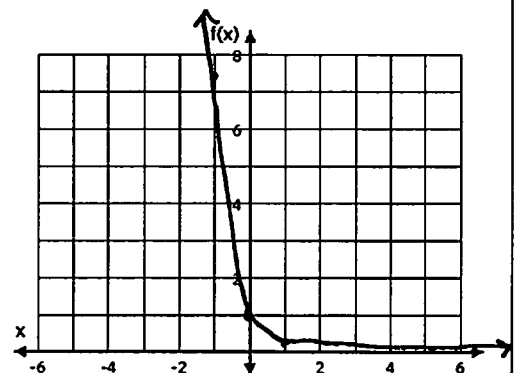


Classify as an exponential growth or decay, then graph the function.

1. $f(x) = e^{0.5x}$



2. $f(x) = e^{-2x}$



APPLICATIONS

3. A person is given 800 milligrams of a certain medicine. The amount of medicine, in milligrams, in their system after t hours is modeled by the function $f(t) = 800e^{-0.6t}$. How many milligrams of the medicine remains in the system after 2 hours?

$$\begin{aligned} f(2) &= 800e^{-0.6(2)} \\ &= 800e^{-1.2} \\ &= 240.96 \text{ mg} \end{aligned}$$

4. The balance in an investment account is modeled by the function $f(t) = 2000e^{0.05t}$ where t is the years since the initial investment. Find the balance in the account after 15 years.

$$\begin{aligned} f(15) &= 2000e^{0.05(15)} \\ &= 2000e^{0.75} \\ &= \$4234.00 \end{aligned}$$

Natural LOGARITHMS	• A <u>natural logarithm</u> is a logarithm with base e and written as <u>$\log_e X$</u> , but usually as <u>$\ln x$</u> .		
CONVERTING <i>Between Forms</i>	Write each equation in logarithmic form.		
	5. $e^x = 24$ $\log_e 24 = x$ $\boxed{\ln 24 = x}$	6. $e^9 = x$ $\log_e x = 9$ $\boxed{\ln x = 9}$	7. $e^{x+5} = 72$ $\log_e 72 = x+5$ $\boxed{\ln 72 = x+5}$
	Write each equation in exponential form.		
	8. $\ln x = 58$ $\log_e x = 58$ $\boxed{e^{58} = x}$	9. $\ln 6 = x$ $\log_e 6 = x$ $\boxed{e^x = 6}$	10. $\ln (x-9) = 32$ $\log_e (x-9) = 32$ $\boxed{e^{32} = x-9}$
<i>Simplifying</i> EXPRESSIONS	Condense each expression into a single logarithm.		
	11. $\ln 3 + \ln 16$ $\ln 3 \cdot 16$ $\boxed{= \ln 48}$	12. $\ln 63 - 2 \cdot \ln 3$ $\ln \frac{63}{3^2}$ $= \ln \frac{63}{9}$ $\boxed{= \ln 7}$	13. $\frac{1}{3} \cdot \ln 64 + 2 \cdot \ln x$ $\ln 64^{1/3} \cdot x^2$ $\boxed{= \ln 4x^2}$
	Expand each logarithm.		
	14. $\ln 5x$ $\boxed{\ln 5 + \ln x}$	15. $\ln \left(\frac{a^3}{b} \right)^2$ $\ln \frac{a^6}{b^2}$ $\boxed{= 6 \cdot \ln a - 2 \cdot \ln b}$	16. $\ln \sqrt[3]{m^2 n}$ $\ln m^{2/3} \cdot n^{1/3}$ $\boxed{= \frac{2}{3} \cdot \ln m + \frac{1}{3} \cdot \ln n}$
<i>Solving</i> EQUATIONS	Solve each equation. Check for extraneous solutions.		
	17. $\ln(4x-27) = \ln(15-2x)$ $4x-27 = 15-2x$ $6x-27 = 15$ $6x = 42$ $\boxed{x=7} \checkmark$	18. $2 \cdot \ln k = \ln(2k+15)$ $\ln k^2 = \ln(2k+15)$ $k^2 = 2k+15$ $k^2-2k-15=0$ $(k-5)(k+3)=0$ $\checkmark \boxed{k=5} \quad k=-3$	

$$\begin{aligned}
 19. \ln 72 - \ln 4 &= \ln 6 + \ln(a-2) \\
 \ln \frac{72}{4} &= \ln 6 \cdot (a-2) \\
 \ln 18 &= \ln 6a - 12 \\
 18 &= 6a - 12 \\
 30 &= 6a \\
 \checkmark \boxed{5=a}
 \end{aligned}$$

$$\begin{aligned}
 20. 2 \cdot \ln(m+4) &= \ln 4 \\
 \ln(m+4)^2 &= \ln 4 \\
 (m+4)^2 &= 4 \\
 m^2 + 8m + 16 &= 4 \\
 m^2 + 8m + 12 &= 0 \\
 (m+2)(m+6) &= 0 \\
 \checkmark \boxed{m=-2} \quad | \quad \cancel{m=-6}
 \end{aligned}$$

$$\begin{aligned}
 21. \ln 8x &= 2 \\
 \log_e 8x &= 2 \\
 e^2 &= 8x \\
 7.3891 &= 8x \\
 \boxed{0.9236=x}
 \end{aligned}$$

$$\begin{aligned}
 22. \ln x - \ln 9 &= 7 \\
 \ln \frac{x}{9} &= 7 \\
 \log_e \frac{x}{9} &= 7 \\
 e^7 &= \frac{x}{9} \\
 1096.6332 &= \frac{x}{9} \\
 \boxed{x=9869.6984}
 \end{aligned}$$

$$\begin{aligned}
 23. e^x &= 57 \\
 \log_e 57 &= x \\
 \ln 57 &= x \\
 \boxed{4.0431=x}
 \end{aligned}$$

$$\begin{aligned}
 24. e^{y+3} - 6 &= 24 \\
 e^{y+3} &= 30 \\
 \log_e 30 &= y+3 \\
 \ln 30 &= y+3 \\
 3.4012 &= y+3 \\
 \boxed{0.4012=y}
 \end{aligned}$$

$$\begin{aligned}
 25. 5e^{4n} &= 95 \\
 e^{4n} &= 19 \\
 \log_e 19 &= 4n \\
 \ln 19 &= 4n \\
 2.9444 &= 4n \\
 \boxed{0.7361=n}
 \end{aligned}$$

$$\begin{aligned}
 26. 2e^{c-9} + 3 &= 87 \\
 2e^{c-9} &= 84 \\
 e^{c-9} &= 42 \\
 \log_e 42 &= c-9 \\
 \ln 42 &= c-9 \\
 3.7377 &= c-9 \\
 \boxed{12.7377=c}
 \end{aligned}$$

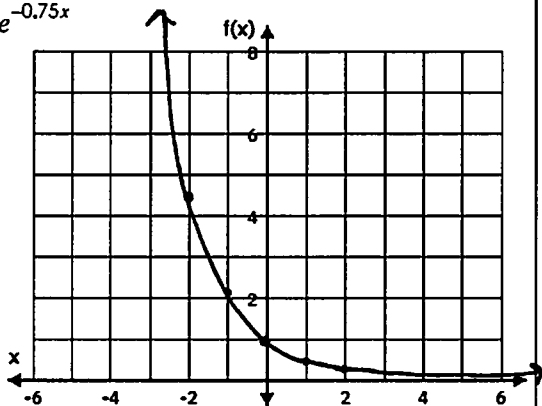
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Unit 7: Exponential & Logarithmic Functions

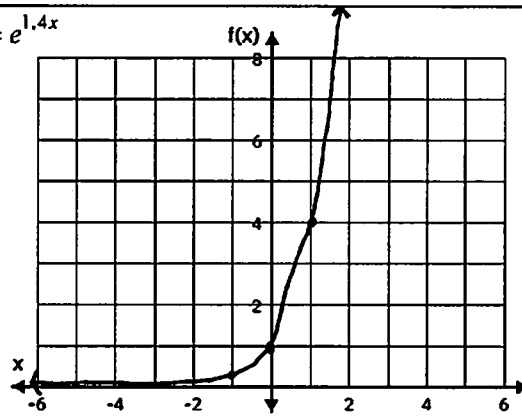
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Homework 10: Base e and Natural Logarithms**** This is a 2-page document! ******Classify as an exponential growth or decay, then graph.**

1. $f(x) = e^{-0.75x}$



2. $f(x) = e^{1.4x}$



3. The amount of a bacteria in a culture can be modeled by the function $f(t) = 400e^{0.15t}$, where t is the number of hours since 7 AM. Find the bacteria count at 3:30 PM on the same day.

$t = 8.5 \text{ hr}$

$$f(8.5) = 400e^{0.15(8.5)}$$

$$= 400e^{1.275}$$

$$= 1431.48$$

4. The value of a certain boat depreciates according to the model $f(t) = 30000e^{-0.2t}$, where t is the number of years after the boat was purchased new. Find the value of the boat in 18 months. $\rightarrow t = 1.5 \text{ yr}$

$$f(1.5) = 30000e^{-0.2(1.5)}$$

$$= \$22,224.55$$

Write each equation in logarithmic form.

5. $e^3 = x$

$\log_e x = 3$

$\ln x = 3$

6. $e^x = 36$

$\log_e 36 = x$

$\ln 36 = x$

7. $e^{x-9} = 74$

$\log_e 74 = x-9$

$\ln 74 = x-9$

Write each equation in exponential form.

8. $\ln 53 = x$

$\log_e 53 = x$

$e^x = 53$

9. $\ln x = 18$

$\log_e x = 18$

$e^{18} = x$

10. $\ln 87 = x+4$

$\log_e 87 = x+4$

$e^{x+4} = 87$

Condense each expression as a single logarithm.

11. $\ln 4 + \ln 3x$

$\ln 4 \cdot 3x$

$= \ln 12x$

12. $\frac{1}{2} \cdot \ln 256 - 3 \cdot \ln 2$

$\ln \frac{\sqrt{256}}{2^3}$

$= \ln \frac{16}{8} = \ln 2$

13. $7 \cdot \ln a - 4 \cdot \ln b$

$\ln \frac{a^7}{b^4}$

Expand each logarithmic expression.

14. $\ln(2m^8)$

$$= \ln 2 + \ln m^8$$

$$= \boxed{\ln 2 + 8 \ln m}$$

15. $\ln\left(\frac{m^5}{n^2}\right)^3 = \ln \frac{m^{15}}{n^6}$

$$= \boxed{15 \ln m - 6 \ln n}$$

16. $\ln \sqrt[8]{r^8 s^5}$

$$= \ln(r^8 s^5)^{1/2}$$

$$= \ln r^4 s^{5/2}$$

$$= \boxed{4 \ln r + \frac{5}{2} \ln s}$$

Solve each equation. Be sure to check for extraneous solutions.

17. $\ln(2x^2 - 15) = \ln(x^2 + 34)$

$$2x^2 - 15 = x^2 + 34$$

$$x^2 - 15 = 34$$

$$x^2 = 49$$

$$x^2 - 49 = 0$$

$$(x+7)(x-7) = 0$$

$$\boxed{x = -7} \quad \boxed{x = 7}$$

18. $\ln 8 + \ln(n-9) = 5 \ln 2$

$$\ln 8(n-9) = \ln 2^5$$

$$8n - 72 = 32$$

$$8n = 104$$

$$\boxed{n = 13}$$

19. $\ln(4w+9) = 5$

$$\log_e(4w+9) = 5$$

$$e^5 = 4w+9$$

$$148.4132 = 4w+9$$

$$139.4132 = 4w$$

$$\boxed{w = 34.8533}$$

20. $\ln k - \ln 14 = 2$

$$\ln \frac{k}{14} = 2$$

$$\log_e \frac{k}{14} = 2$$

$$e^2 = \frac{k}{14}$$

$$7.3891 = \frac{k}{14}$$

$$\boxed{k = 103.4468}$$

21. $e^x = 21$

$$\log_e 21 = x$$

$$\ln 21 = x$$

$$\boxed{x = 3.0445}$$

22. $e^{y-1} - 27 = 54$

$$e^{y-1} = 81$$

$$\log_e 81 = y-1$$

$$\ln 81 = y-1$$

$$4.3944 = y-1$$

$$\boxed{5.3944 = y}$$

23. $4e^{3k} + 1 = 85$

$$4e^{3k} = 84$$

$$e^{3k} = 21$$

$$\log_e 21 = 3k$$

$$\ln 21 = 3k$$

$$3.0445 = 3k$$

$$\boxed{k = 1.0148}$$

24. $3e^{4m-7} - 8 = 106$

$$3e^{4m-7} = 114$$

$$e^{4m-7} = 38$$

$$\log_e 38 = 4m-7$$

$$\ln 38 = 4m-7$$

$$3.6376 = 4m-7$$

$$10.6376 = 4m$$

$$\boxed{m = 2.6594}$$

Name:

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Main Ideas/Questions

Notes/Examples

Year	Population
1900	3.1
1920	4.6
1940	6.4
1960	9.5
1980	14.2

1. The table to the left shows the year and population (in millions) of Texas. Using an **exponential model**, write an equation for the curve of best fit, then estimate the population of Texas in 2020. $\rightarrow X = 120$

$$f(x) = 3.10 \cdot 1.02^x$$

$$f(120) = 3.10 \cdot 1.02^{120}$$

$$= 33.37$$

Year	Closing Price (\$)
1992	633
1994	793
1996	1,100
1998	1,771
2000	4,696

2. The table to the left shows the closing prices for the NASDAQ Stock Index at the end of February during certain years. Using an **exponential model**, write an equation for the curve of best fit, then estimate the year the closing price will reach \$10,000. $y = 10000$

$$f(x) = 518.18 \cdot 1.27^x$$

$$10000 = 518.18 \cdot 1.27^x$$

$$19.2983 = 1.27^x$$

$$\log_{1.27} 19.2983 = x$$

$$2005$$

$$x = 12.38$$

Year	Population
1970	30.0
1980	35.3
1985	37.9
1990	37.2
2006	41.5

3. The table to the left gives the percent of all workers who are female during selected years from 1970 to 2006. Using a **logarithmic model**, write an equation for the curve of best fit, then estimate the percent of female workers in in 2016. $x = 47$

$$f(x) = 29.50 + 2.91 \cdot \ln x$$

$$f(47) = 29.50 + 2.91 \cdot \ln 47$$

$$= 40.7$$

Age	Height
1	4
2	6.3
3	7.9
4	9.2
5	10.3

4. The table to the left shows the height (in feet) of a pine tree in the park. When it was originally planted, it was one year old and had a height of 4 feet. Using a **logarithmic model**, write an equation for the curve of best fit, then determine the age of the tree at 20 feet. $y = 20$

$$f(x) = 3.82 + 3.89 \cdot \ln x$$

$$20 = 3.82 + 3.89 \cdot \ln x$$

$$16.18 = 3.89 \cdot \ln x$$

$$4.16 = \ln x$$

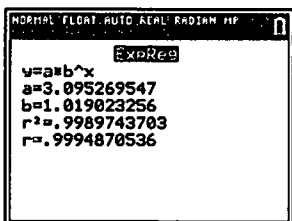
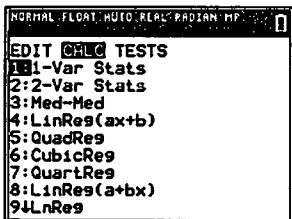
$$4.16 = \log_e x$$

$$e^{4.16} = x$$

$$64.03 = x$$

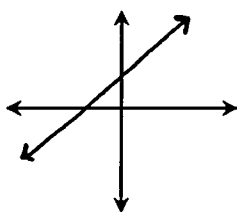
$$64 \text{ years old}$$

Choosing the BEST MODEL

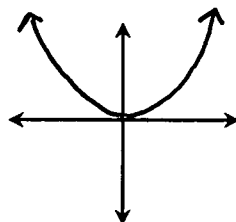


Sketch each parent function.

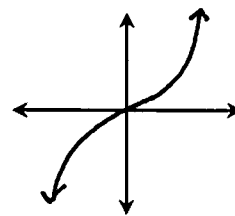
Linear (4:LinReg)



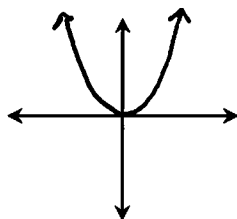
Quadratic (5:QuadReg)



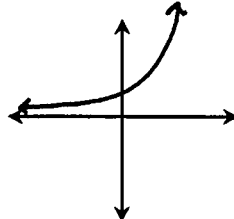
Cubic (6:CubicReg)



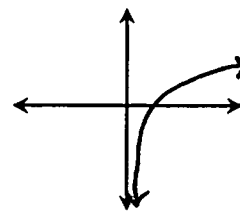
Quartic (7:QuartReg)



Exponential (0:ExpReg)



Logarithmic (9:LnReg)



To determine the best model, use the correlation coefficient, r .
The closer r is to -1 or 1, the better the fit. Make sure your diagnostics are turned ON to see this value!

5. The table below shows the operating costs (in thousands of dollars) of a small business from 2000 to 2005. Which model would fit this data best: quadratic, quartic, or logarithmic? Using the model, write a best-fit equation, then estimate the operation costs in 2008. $n=9$

Quartic!

	Year	Costs
1	2000	2.3
2	2001	2.6
3	2002	3.1
4	2003	3.4
5	2004	4
6	2005	5.6

$$f(x) = .02x^4 - .27x^3 + 1.12x^2 - 1.49x + 2.92$$

$$f(9) = 14.62 \rightarrow \boxed{\$14,620}$$

6. The table below shows the balance of Marissa's savings account at the end of the year. Which model would fit this data best: linear, cubic, or exponential? Using the model, write a best-fit equation, then estimate the year the balance will reach \$1,500.

	Year	Balance (\$)
0	2009	500
1	2010	540
2	2011	583.20
3	2012	629.86
4	2013	680.24
5	2014	734.66

Exponential!

$$f(x) = 500 \cdot 1.08^x$$

$$1500 = 500 \cdot 1.08^x$$

$$3 = 1.08^x$$

$$\log_{1.08} 3 = x$$

$$x = 14.27 \rightarrow \boxed{2024}$$

Name: _____ Unit 7: Exponential & Logarithmic Functions

Date: _____ Per: _____ Homework 11: Exponential & Logarithmic Regression;
Choosing the Best Model

**** This is a 2-page document! ****

1. The table below shows the value of a car during certain years. Using an **exponential model**, write an equation for the curve of best fit, then estimate the value of the car in 2016.

	Year	Value (\$)
0	2008	27,500
1	2009	24,200
2	2010	21,296
3	2011	18,740
4	2012	16,492

$$f(x) = 27499.87 \cdot 0.88^x \quad x=8$$

$$f(8) = 27499.87 \cdot 0.88^8$$

$$= \$9,889.90$$

2. The table below shows the number of students enrolled at Glennview High School during certain years. Using an **exponential model**, write an equation for the curve of best fit, then estimate the year in which enrollment reaches 2,000 students.

Year	Students
2004	1,250
2007	1,350
2010	1,458
2013	1,574

$$f(x) = 1250.10 \cdot 1.03^x$$

$$2000 = 1250.10 \cdot 1.03^x$$

$$1.5999 = 1.03^x$$

$$\log_{1.03} 1.5999 = x$$

$$x = 15.8985 \rightarrow$$

2019

3. The table below shows the weights of five females of the same height along with their body mass index, BMI. Using a **logarithmic model**, write an equation for the curve of best fit, then estimate the BMI of a female that weighs 130 pounds.

Weight	BMI
112	17.5
125	19.1
139	22.7
156	26.4
173	30.2

$$f(x) = -124.72 + 29.96 \cdot \ln x$$

$$f(130) = -124.72 + 29.96 \cdot \ln(130)$$

$$= 21.11$$

4. The table below shows the airfare of a one-way ticket from New York City to Orlando as advertised by an airline for the first five days of April. Using a **logarithmic model**, write an equation for the curve of best fit. If Scott does not want to pay more than \$140, what day must he purchase his ticket by?

Date	Price
1	89
2	105
3	109
4	114
5	120

$$f(x) = 89.90 + 18.27 \cdot \ln x$$

$$140 = 89.90 + 18.27 \ln x$$

$$50.10 = 18.27 \ln x$$

$$2.74 = \ln x$$

$$e^{2.74} = x$$

$$x = 15.52$$

Apr 15th

5. The table below shows the height, in feet, of a rock thrown from the top of a building each second after it was thrown. Which model would fit this data best: linear, quadratic, or cubic? Use the model to write a best-fit equation, then estimate the height of the rock at 5 seconds.

Time	Height
0	250
1	280
2	284
3	249
4	202

Cubic!

$$f(x) = 1.17x^3 - 20.79x^2 + 52.48x + 249.43$$

$$f(5) = \boxed{138.33 \text{ ft}}$$

6. When the Morrison's adopted their six week old puppy, he was 8 pounds. The table below shows the weight of the puppy each week thereafter. Which model would best fit this data: exponential or logarithmic? Use the model to write a best-fit equation, then find the week at which the puppy will weigh 50 pounds.

Week	Weight
6	8
7	12
8	14
9	17
10	20

Logarithmic!

$$f(x) = -32.66 + 22.71 \cdot \ln x$$

$$50 = -32.66 + 22.71 \cdot \ln x$$

$$82.66 = 22.71 \cdot \ln x$$

$$3.64 = \ln x$$

$$e^{3.64} = x$$

$$38.09 = x$$

$$\boxed{38 \text{ weeks}}$$

7. The table below shows the value of an investment during certain years. Which model would best fit this data: quadratic, cubic, or quartic? Use the model to write a best-fit equation, then estimate the value of the investment in 2022.

	Year	Value (\$)
0	2002	1,200
3	2005	1,895
6	2008	2,399
9	2011	2,781
12	2014	3,500

Quartic!

$$f(x) = .20x^4 - 3.19x^3 + 5.42x^2 + 238.67x + 1200$$

$$f(20) = \boxed{\$14,621.40}$$

8. The total yearly revenue of a small company, in thousands of dollars, is given in the table below. Which model would best fit this data: quadratic, cubic, or exponential? Use the model to write a best-fit equation, then estimate the revenue of the company in 2020.

	Year	Revenue
0	2010	4.3
1	2011	8.3
2	2012	10.5
3	2013	11.4
4	2014	11.5

Cubic!

$$f(x) = .0833x^3 - 1.15x^2 + 5.0667x + 4.3$$

$$f(10) = 23.267 \rightarrow \boxed{\$23,267}$$

Unit 7 Test Study Guide

(Exponential & Logarithmic Functions)

Name: _____

Date: _____ Per: _____

Topic 1: Graphing Exponential & Logarithmic Functions

Describe as an exponential growth or decay.

1. $f(x) = 5\left(\frac{2}{3}\right)^x$
 $\hookrightarrow < 1$

decay

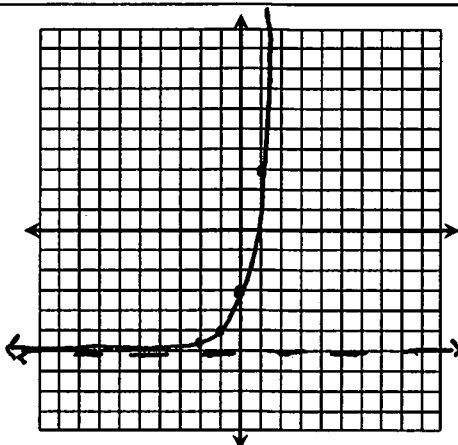
2. $f(x) = \frac{1}{3}\left(\frac{6}{5}\right)^x$
 $\hookrightarrow > 1$

growth

Graph each function, then identify its key characteristics.

3. $f(x) = 3^{x+1} - 6$

X	y
-2	-5.6
-1	-5
0	-3
1	3
2	21



Domain: $\mathbb{R}$

Range: $y > -6$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

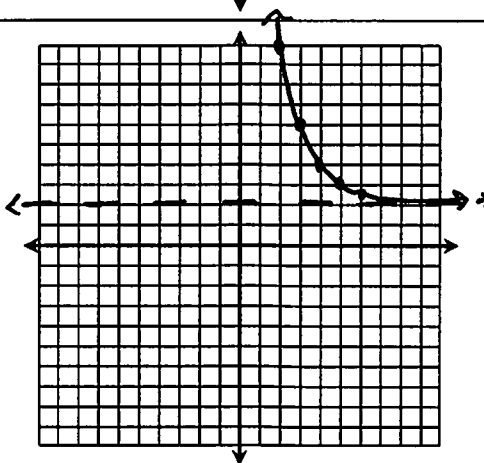
As $x \rightarrow -\infty$, $f(x) \rightarrow -6$

y-intercept: $(0, -3)$

Asymptote: $y = -6$

4. $f(x) = \left(\frac{1}{2}\right)^{x-5} + 2$

X	y
2	10
3	6
4	4
5	3
6	2.5



Domain: $\mathbb{R}$

Range: $y > 2$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow 2$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

y-intercept: $(0, 34)$

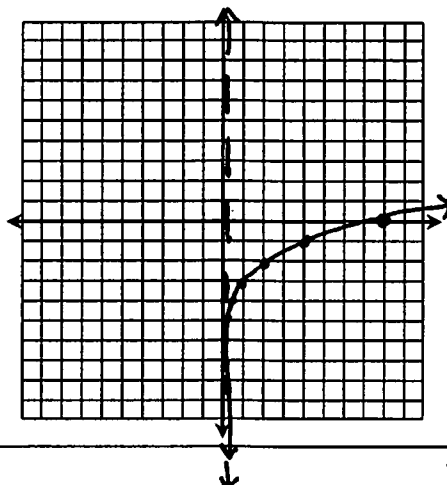
Asymptote: $y = 2$

5. $f(x) = \log_2 x - 3$

X	y
.25	-2
.5	-1
1	0
2	1
4	2
8	3

X	y
.25	-5
.5	-4
1	-3
2	-2
4	-1
8	0

down 3



Domain: $x > 0$

Range: $\mathbb{R}$

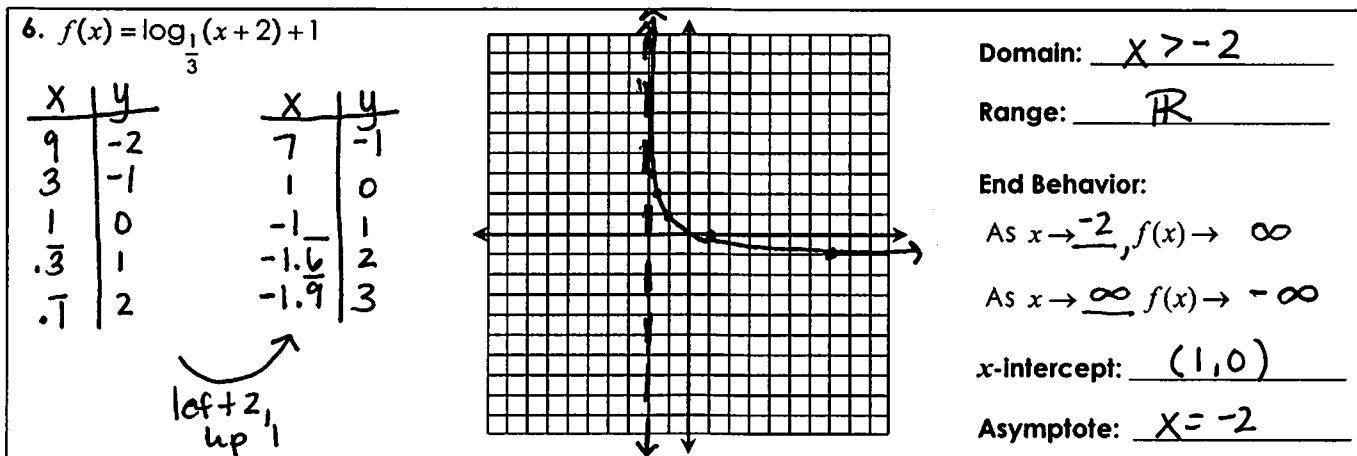
End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow 0$, $f(x) \rightarrow -\infty$

x-intercept: $(8, 0)$

Asymptote: $x = 0$



Topic 2: Exponential vs. Logarithmic Forms

Write in logarithmic form.

7. $8^2 = 64$ $\log_8 64 = 2$	8. $2^{x-4} = 32$ $\log_2 32 = x - 4$	9. $10^{2x} = 54$ $\log_{10} 54 = 2x$ $\log 54 = 2x$	10. $e^6 = x - 2$ $\log_e (x - 2) = 6$ $\ln (x - 2) = 6$
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Write in exponential form.

11. $\log_3 27 = 3$ $3^3 = 27$	12. $\log_x 7 = \frac{1}{2}$ $x^{1/2} = 7$	13. $\log_4 90 = x$ $4^x = 90$	14. $\ln x = 38$ $\log_e x = 38$ $e^{38} = x$
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Topic 3: Evaluating Logarithms

Evaluate each logarithm. Use the Change of Base Formula when necessary: $\log_b a =$

15. $\log_9 81$ $9^x = 81$ $x = 2$	16. $\log_{81} 3$ $81^x = 3$ $x = \frac{1}{4}$	17. $\log_5 \frac{1}{25}$ $5^x = \frac{1}{25}$ $x = -2$	18. $\log_6 1$ $6^x = 1$ $x = 0$
19. $\log 63$ $\frac{\log 63}{\log 10}$ $= 1.7993$	20. $\log_7 95$ $\frac{\log 95}{\log 7}$ $= 2.3402$	21. $\log_2 78$ $\frac{\log 78}{\log 2}$ $= 6.2854$	22. $\ln 42$ $= 3.7377$

Topic 4: Properties of Logarithms

Product Rule $\log_b (m \cdot n) =$ $\log_b m + \log_b n$	Quotient Rule $\log_b \left(\frac{m}{n} \right) =$ $\log_b m - \log_b n$	Power Rule $\log_b m^n =$ $n \cdot \log_b m$
--	--	---

Condense each expression into a single logarithm.		
23. $3 \cdot \log 2 + \log(x-4)$ $= \log 2^3 + \log(x-4)$ $= \log 8(x-4)$ $= \log(8x \cdot 32)$	24. $\frac{1}{2} \cdot \log_5 324 - \log_5 2$ $= \log_5 \frac{324^{1/2}}{2}$ $= \log_5 9$	25. $3 \cdot \ln 6 - \frac{3}{2} \cdot \ln 4$ $= \ln \frac{6^3}{4^{3/2}}$ $= \ln 27$
Expand each expression.		
26. $\log_3(x^2 y^5)^3$ $= \log_3 x^6 y^{15}$ $= 6 \cdot \log_3 x + 15 \cdot \log_3 y$	27. $\ln \left(\frac{2}{a^3} \right)^4$ $= \ln \frac{2^4}{a^{12}}$ $= 4 \cdot \ln 2 - 12 \cdot \ln a$	28. $\log_4 \sqrt{p^3 q^{10}}$ $= \log_4 p^{3/2} q^5$ $= \frac{3}{2} \cdot \log_4 p + 5 \cdot \log_4 q$

Topic 5: Solving Logarithmic & Exponential Equations

Solve each equation. Check for extraneous solutions.	
29. $\log_4(5x+7) = \log_4(2x+31)$ $5x+7 = 2x+31$ $3x+7 = 31$ $3x = 24$ $x = 8 \checkmark$	30. $\ln(p^2 - p) = \ln(6p+18)$ $p^2 - p = 6p + 18$ $p^2 - 7p - 18 = 0$ $(p-9)(p+2) = 0$ $\checkmark \boxed{p=9} \quad \boxed{p=-2} \checkmark$
31. $\frac{1}{2} \cdot \log_8 36 + \log_8(3k+7) = \log_8 132$ $\log_8 36^{1/2} \cdot (3k+7) = \log_8 132$ $6(3k+7) = 132$ $18k+42 = 132$ $18k = 90$ $\boxed{k=5} \checkmark$	32. $2 \cdot \log(y+5) = \log 20 - \log 5$ $\log(y+5)^2 = \log \frac{20}{5}$ $y^2 + 10y + 25 = 4$ $y^2 + 10y + 21 = 0$ $(y+7)(y+3) = 0$ $\cancel{y=-7} \quad \boxed{y=-3} \checkmark$

$$33. \log_2(9m+2)=7$$

$$2^7 = 9m+2$$

$$128 = 9m+2$$

$$126 = 9m$$

$$\boxed{m=14}$$

$$34. 5 \cdot \ln(2a-1)=15$$

$$\ln(2a-1)=3$$

$$e^3 = 2a-1$$

$$20.09 = 2a-1$$

$$21.09 = 2a$$

$$\boxed{a=10.5428}$$

$$35. 64^{x+7} = 4^{5x-3}$$

$$(4^3)^{x+7} = 4^{5x-3}$$

$$3x+21 = 5x-3$$

$$21 = 2x-3$$

$$24 = 2x$$

$$\boxed{x=12}$$

$$36. 9^{w-8} = \left(\frac{1}{27}\right)^{2w}$$

$$(3^2)^{w-8} = (3^{-3})^{2w}$$

$$2w-16 = -6w$$

$$-16 = -8w$$

$$\boxed{2=w}$$

$$37. 8^{n-5} = 48$$

$$\log_8 48 = n-5$$

$$\frac{\log 48}{\log 8} = n-5$$

$$1.8617 = n-5$$

$$\boxed{6.8617=n}$$

$$38. 2 \cdot 3^{4y} - 11 = 61$$

$$2 \cdot 3^{4y} = 72$$

$$3^{4y} = 36$$

$$\log_3 36 = 4y$$

$$\frac{\log 36}{\log 3} = 4y$$

$$3.2619 = 4y$$

$$\boxed{y=0.8155}$$

$$39. e^{a+1} = 65$$

$$\log_e 65 = a+1$$

$$\ln 65 = a+1$$

$$4.1744 = a+1$$

$$\boxed{a=3.1744}$$

$$40. -3 \cdot e^{2m-5} - 7 = -34$$

$$-3 \cdot e^{2m-5} = -27$$

$$e^{2m-5} = 9$$

$$\log_e 9 = 2m-5$$

$$\ln 9 = 2m-5$$

$$2.1972 = 2m-5$$

$$7.1972 = 2m$$

$$\boxed{m=3.5986}$$

Topic 6: Applications

Use the models below to answer questions 41-44.

Exponential Growth	Exponential Decay	Compound Interest
$A = a(1+r)^t$	$A = a(1-r)^t$	$A = P(1 + \frac{r}{n})^{nt}$
a = initial amount r = growth rate t = time	a = initial amount r = decay rate t = time (in years)	P = initial amount r = rate n = # of times compounded/year t = time (in years)
41. The population of an endangered species of birds is decreasing at a rate of 1.5% each year. There are currently 250,000 of these birds. Write a function to model this situation, then find the number of birds in 25 years. $f(x) = 250000(.985)^t$ $f(25) = 250000(.985)^{25}$ $= 171334.872$ 171,334 birds	42. A baseball card that was valued at \$200 in 1980 has increased in value by 7% each year. Write a function to model this situation, then find the value of the card in 2016. $f(x) = 200(1.07)^x$ $f(36) = 200(1.07)^{36}$ = \$2,284.79	43. Miles invested \$2,400 into a retirement account that earns 1.8% interest compounded bimonthly. Write a function to model this situation, then find the balance of the account after 25 years. $f(x) = 2400(1 + \frac{.018}{24})^{24x}$ $f(25) = 2400(1 + \frac{.018}{24})^{24 \cdot 25}$ $= 2400(1.00075)^{600}$ = \$3,763.31
45. Once a car tire hits a nail, the pressure in pounds per square inch, psi, of the tire can be modeled by the function $f(t) = 32e^{-0.04t}$ where t is the number of minutes since the tire hit the nail. Find the air pressure of the tire after 45 minutes. $f(45) = 32e^{-0.04(45)}$ = 5.2896 psi	44. Sarah moved \$30,000 of her savings to a new investment account that earns 4% interest compounded quarterly. Write a function to model this situation, then find the amount of interest the account will earn after 12 years. $f(x) = 30000(1 + \frac{.04}{4})^{4x}$ $f(12) = 30000(1 + \frac{.04}{4})^{4 \cdot 12}$ $= 30000(1.01)^{48}$ $= 48,366.78 \rightarrow \text{Interest} = \$18,366.78$	46. Jeremiah invested \$2,000 in shares of a certain stock. The function $f(t) = 2000e^{0.18t}$ represents the value of his investment, t years after his initial investment. Find the value of his investment after 5 years. $f(5) = 2000e^{.18(5)}$ = \$4919.21

Topic 7: Regression

47. The table below shows the value of a stock over the course of five years. Using an **exponential model**, write an equation for the curve of best fit, then estimate the value of the stock in 2025.

	Year	Value (\$)
0	1998	400
1	1999	438
2	2000	480
3	2001	525
4	2002	575

$$f(x) = 400.07 \cdot 1.09^x$$

$$f(27) = 400.07 \cdot 1.09^{27}$$

$$= \$4098.75$$

48. The table below shows the wind speed along with the corresponding wind chill factor. Using a **logarithmic model**, write an equation for the curve of best fit, then find the approximate wind speed that corresponds to a wind chill factor of -18°F .

Wind Speed (mi/hr)	Wind Chill Factor ($^\circ\text{F}$)
3	1
5	-3
14	-8
20	-11
27	-14

$$f(x) = 7.91 - 6.41 \cdot \ln x$$

$$-18 = 7.91 - 6.41 \cdot \ln x$$

$$-25.91 = -6.41 \ln x$$

$$4.04 = \ln x$$

$$e^{4.04} = x$$

$$x = 56.95 \rightarrow 57 \text{ mph}$$

49. The table below shows the number of seniors at Greenville High School that are also taking college credit classes during certain years. Which model would best fit this data: linear, or quadratic? Use the model to write a best-fit equation, then estimate the number of seniors taking college classes in 2020.

	Year	Students
0	2004	18
2	2006	24
4	2008	39
8	2012	48
11	2015	64

Linear!

$$f(x) = 4.09x + 18.16$$

$$f(16) = 4.09(16) + 18.16$$

$$= 83.6$$

↓

84 Seniors

50. The table below shows the number of used textbooks sold each year on an online bookstore. Which model would best fit this data: cubic, quartic, or exponential? Use the model to write a best-fit equation, then estimate the number of books sold in 2014.

	Year	Books Sold (thousands)
0	2008	1
1	2009	6.2
2	2010	9.7
3	2011	12.1
4	2012	13.3

Quartic!

$$f(x) = -.03x^4 + .28x^3 - 1.47x^2 + 6.43x + 1$$

$$f(6) = -.03(6)^4 + .28(6)^3 - 1.47(6)^2 + 6.43(6) + 1$$

$$= 8.26 \rightarrow 8,260$$

Name: _____

Unit 7 Test

Date: _____ Per: _____

Exponential & Logarithmic Functions

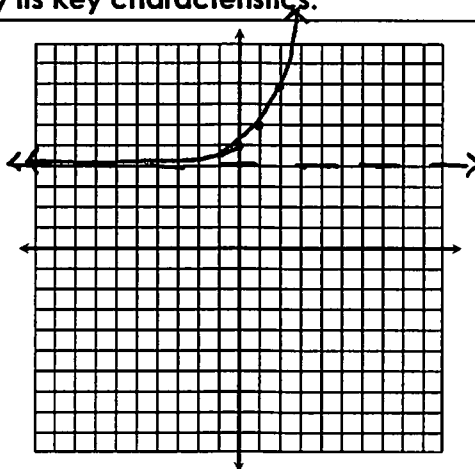
1. Which functions represent an exponential growth? Check all that apply.

☒ $f(x) = 0.1 \cdot 8^x$	☒ $f(x) = \frac{4}{5} \cdot \left(\frac{9}{2}\right)^x$	☐ $f(x) = \frac{8}{5} \cdot \left(\frac{3}{4}\right)^x$
☐ $f(x) = 5 \cdot \left(\frac{1}{3}\right)^x$	☒ $f(x) = \frac{3}{8} \cdot 2^x$	☐ $f(x) = 16 \cdot 0.3^x$

Graph each function and identify its key characteristics.

2. $f(x) = \frac{1}{2} \cdot 2^{x+1} + 4$

X	Y
-1	4.5
0	5
1	6
2	8
3	12



Domain: $\mathbb{R}$

Range: $y > 4$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow 4$

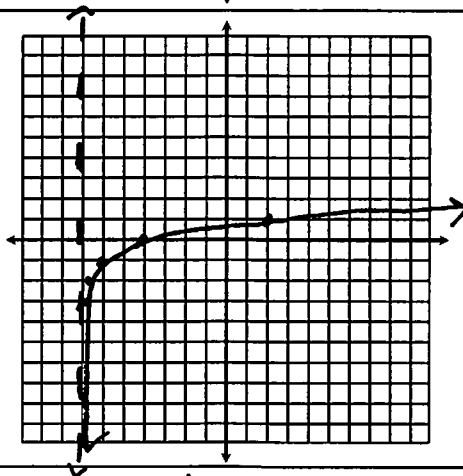
y-intercept: $(0, 5)$

Asymptote: $y = 4$

3. $f(x) = \log_3(x+7) - 1$

X	Y
.1	-2
.3	-1
1	0
3	1
9	2

X	Y
-6.9	-3
-6.6	-2
-6	-1
-4	0
2	1



Domain: $x > -7$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -7$, $f(x) \rightarrow -\infty$

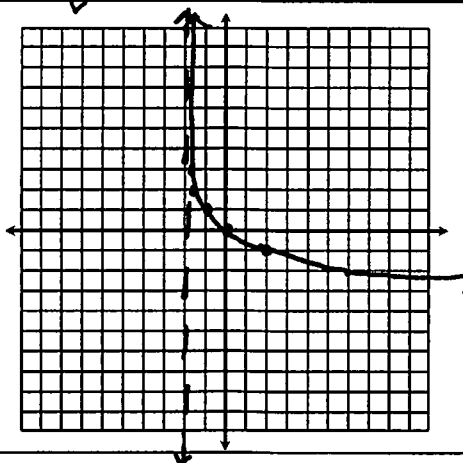
x-intercept: $(-4, 0)$

Asymptote: $x = -7$

4. $f(x) = \log_{\frac{1}{2}}(x+2) + 1$

X	Y
4	-2
2	-1
1	0
.5	1
.25	2

X	Y
2	-1
0	0
-1	1
-1.5	2
-1.75	3



Domain: $x > -2$

Range: $\mathbb{R}$

End Behavior:

As $x \rightarrow -2$, $f(x) \rightarrow \infty$

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

x-intercept: $(0, 0)$

Asymptote: $x = -2$

5. Which line represents the asymptote of the graph of the following function? $f(x) = 3^{x-4} + 1$ A. $x = -4$ B. $x = 1$ C. $y = -4$ D. $y = 1$ D	6. What is the domain of the function given below? $f(x) = \log(x + 5) - 2$ A. $\{x \mid x > -5\}$ B. $\{x \mid x > 5\}$ C. $\{x \mid x > -2\}$ D. $\{x \mid x > 2\}$ A
7. Which is equivalent to the equation given below? $7^x = 24$ A. $\log_7 x = 24$ B. $\log_x 7 = 24$ C. $\log_{24} 7 = x$ D. $\log_7 24 = x$ D	8. Which is equivalent to the equation given below? $e^{x+3} = 60$ A. $\ln(x + 3) = 60$ B. $\ln 60 = x + 3$ C. $\log(x + 3) = 60$ D. $\log 60 = x + 3$ B
9. Which is equivalent to the equation given below? $\log_2 18 = x - 1$ A. $2^{18} = x - 1$ B. $2^{x-1} = 18$ C. $18^2 = x - 1$ D. $18^{x-1} = 2$ B	10. Which is equivalent to the equation given below? $\ln 5x = 4$ A. $e^{4x} = 5$ B. $e^{5x} = 4$ C. $e^5 = 4x$ D. $e^4 = 5x$ D
11. The expression $\log_{36} 6$ is equivalent to which of the following? A. 2 B. 6 C. $\frac{1}{2}$ D. $\frac{1}{6}$ C $36^x = 6$ $36^{1/2} = 6$ $x = \frac{1}{2}$	12. Find the value of the expression below. $\log_3 3 + \log_2 64$ $1 + 6 = 7$ A. 5 B. 7 C. 9 D. 33 B

13. Approximate the value of $\log_5 84$ to the nearest ten-thousandths place.

$$\frac{\log 84}{\log 5}$$

2.7530

14. Which of the following have values less than 1? Check all that apply.

☐ $\log_6 39$	☐ $\log_2 50$
☐ $\log 24$	☒ $\log_{15} 1$
☒ $\log_7 3$	☐ $\ln 6$

15. Condense the expression below into a single logarithm.

$$\frac{1}{3} \cdot \log 64 + \log(x+5)$$

$$\log 64^{1/3} \cdot (x+5)$$

$$\log 4 \cdot (x+5)$$

$\log(4x+20)$

16. Condense the expression below into a single logarithm. Do not evaluate.

$$7 \cdot \ln 3 - \frac{1}{2} \cdot \ln 81$$

$$\ln \frac{3^7}{81^{1/2}} = \ln \frac{2187}{9}$$

$\ln 243$

17. Which expression is not equivalent to $\log_x 16$?

- A. $8 \cdot \log_x 2$
- B. $2 \cdot \log_x 4$
- C. $\log_x 8 + \log_x 2$
- D. $\log_x 432 - 3 \cdot \log_x 3$

A

18. Write a value in the box that would make the equation true.

$$\ln 9 = \ln 72 - \boxed{3} \cdot \ln 2$$

19. Expand the logarithm below.

$$\log_5 (m^2 n^3)^4$$

$$\log_5 (m^8 n^{12})$$

$8 \cdot \log_5 m + 12 \cdot \log_5 n$

20. Expand the logarithm below.

$$\log_2 \sqrt[3]{x^{12} y}$$

$$\log_2 x^4 y^{1/3}$$

$4 \cdot \log_2 x + \frac{1}{3} \cdot \log_2 y$

21. Which is equivalent to the expression below?

$$\log\left(\frac{b^3}{a}\right)$$

- A. $3 \cdot (\log b - \log a)$
- B. $\log 3b - \log a$
- C. $3 \cdot \log b - \log a$
- D. $3 \cdot \log b + \log a$

C

22. Which is equivalent to the expression below?

$$2 \cdot \log x - (3 \cdot \log y + \log y)$$

- A. $\log\left(\frac{x}{y}\right)^2$
- B. $\log\left(\frac{x^2}{y^4}\right)$
- C. $\log(xy)^2$
- D. $\log(x^2y^4)$

B

For questions 23-30, solve each equation. Round to the nearest ten-thousandths place when necessary.

23. $\log_2(6x - 15) = \log_2(41 - 2x)$

$$\begin{aligned} 6x - 15 &= 41 - 2x \\ 8x - 15 &= 41 \\ 8x &= 56 \\ x &= 7 \end{aligned}$$

$x = 7$

24. $\ln(k^2 - 1) = \ln 24 - \frac{1}{2} \cdot \ln 9$

$$\ln(k^2 - 1) = \ln \frac{24}{9^{1/2}}$$

$$k^2 - 1 = 8$$

$$k^2 - 9 = 0$$

$$(k+3)(k-3) = 0$$

$$k = -3 \quad k = 3$$

$k = \pm 3$

25. $\log_3(p + 4) + \log_3(p - 2) = 3$

$$\log_3(p + 4)(p - 2) = 3$$

$$p^2 + 2p - 8 = 27$$

$$p^2 + 2p - 35 = 0$$

$$(p+7)(p-5) = 0$$

$$p = -7 \quad p = 5$$

$p = 5$

26. $4 \cdot \ln w + 6 = 12$

$$4 \cdot \ln w = 6$$

$$\ln w = \frac{3}{2}$$

$$\log_e w = \frac{3}{2}$$

$$e^{\frac{3}{2}} = w$$

$$w = 4.4817$$

$w = 4.4817$

27. $\left(\frac{1}{8}\right)^{2-2n} = 16$

$$(2^{-3})^{2-2n} = 2^4$$

$$-6 + 6n = 4$$

$$6n = 10$$

$$n = \frac{5}{3}$$

$n = \frac{5}{3}$

28. $3^{k-2} + 7 = 82$

$$3^{k-2} = 75$$

$$\log_3 75 = k - 2$$

$$\frac{\log 75}{\log 3} = k - 2$$

$$3.9299 = k - 2$$

$$k = 5.9299$$

$k = 5.9299$

29. $2 \cdot 10^{5-y} + 9 = 87$

$$\begin{aligned} 2 \cdot 10^{5-y} &= 78 \\ 10^{5-y} &= 39 \\ \log_{10} 39 &= 5-y \\ \frac{\log 39}{\log 10} &= 5-y \\ 1.5911 &= 5-y \\ -3.4089 &= -y \end{aligned}$$

$$y = 3.4089$$

30. $e^{5p-8} + 3 = 15$

$$\begin{aligned} e^{5p-8} &= 12 \\ \log_e 12 &= 5p-8 \\ \ln 12 &= 5p-8 \\ 2.4849 &= 5p-8 \\ 10.4849 &= 5p \\ p &= 2.0970 \end{aligned}$$

$$p = 2.0970$$

31. Jack bought a new car in 2014 for \$28,000. If the value of the car decreases by 14% each year, write an exponential model for the value of the car. Then, estimate the year the car will have a value of \$5,000.

$$\begin{aligned} f(x) &= 28000(0.86)^x \\ 5000 &= 28000(0.86)^x \\ .1786 &= 0.86^x \\ \log_{.86} .1786 &= x \\ x &= 11.42 \rightarrow 11 \text{ yrs} \end{aligned}$$

Equation:

$$f(x) = 28000(0.86)^x$$

Answer:

2025

32. Logan inherited \$8,000 from his grandmother. He split the money and placed the amounts in the two accounts below. Find the total amount of money he will have in the accounts combined after 5 years.

- **Account A:** earns 7% interest compounded quarterly
- **Account B:** earns 5.5% interest compounded annually

$$\begin{aligned} A: A &= 4000 \left(1 + \frac{0.07}{4}\right)^{4 \cdot 5} \\ &= 4000 (1.0175)^{20} \\ &= \$5659.11 \end{aligned}$$

$$\begin{aligned} B: A &= 4000 \left(1 + \frac{0.055}{1}\right)^{1 \cdot 5} \\ &= 4000 (1.055)^5 \\ &= \$5227.84 \end{aligned}$$

Answer:

\$10,886.95

33. A bacteria culture begins with 500 bacteria and is growing exponentially. The number of bacteria after t hours can be modeled by the function $f(t) = 500e^{0.9t}$. Find the number of bacteria after 4 hours.

$$\begin{aligned} f(4) &= 500e^{0.9(4)} \\ &= 18299.12 \end{aligned}$$

Answer:

18,299 bacteria

34. The table below shows the number of new restaurants in a fast-food chain that opened during the years of 1988 through 1992. Using an **exponential model**, write an equation for the curve of best fit, then estimate the number of new restaurants that opened in 2005.

$$f(x) = 50.55 \cdot 1.49^x \quad x=17$$

	Year	New Restaurants
0	1988	49
1	1989	81
2	1990	112
3	1991	150
4	1992	262

$$f(17) = 50.55 \cdot 1.49^{17} \\ = 44,451.63$$

Equation:

$$f(x) = 50.55 \cdot 1.49^x$$

Answer:

44,452

35. The table below shows the water temperature at certain depths of a lake. Using an **logarithmic model**, write an equation for the curve of best fit, then estimate the depth of water at a temperature of $50^\circ F$.

Depth (ft)	Temp ($^\circ F$)
2	86
12	78
14	75
45	71
68	68

$$f(x) = 89.46 - 5.01 \ln x$$

$$50 = 89.46 - 5.01 \ln x$$

$$-7.8762 = -5.01 \ln x$$

$$e^{-7.8762 / -5.01} = x$$

$$x = 2633.97$$

Equation:

$$f(x) = 89.46 - 5.01 \ln x$$

Answer:

2634 ft

36. The population (in thousands) of a town during certain years is given in the table below. Which model would best fit this data: linear, quadratic, or exponential? Choose the model that best fits the data, write an equation to model the data, then predict the approximate population in 2020. $x=60$

	Year	Population
0	1960	50.6
10	1970	55.5
20	1980	64.0
30	1990	80.3
40	2000	102.9

Model:

Quadratic

Equation:

$$f(x) = 0.03x^2 + 0.06x + 50.95$$

Answer:

162,550

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Many thanks to these
talented artists!